Time Series Analysis

Lecture notes, Chapters 1–8

Based on Sangyeol Lee, *Time Series Analysis: Theory and Practice with R*

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Chapter 1. Time Series Data

1.1 Time series data with dependence

Motivation. Observations recorded over time are correlated with their own past, so the i.i.d. toolkit must be replaced by models for dependence.

1.1.1 Trend and moving-average smoothing

For observations $x_1, \dots, x_n$ (e.g. monthly stock index values), the *moving average* of span m is

$$y_t = \frac{x_{t-m+1} + \cdots + x_t}{m}.$$

Larger m gives a smoother curve (*smoothing*): short spans show short-run movement, long spans the trend.

Under the signal-plus-noise model

$$x_t = f(t) + z_t, \qquad t = 1, \dots, n, \tag{1.1}$$

with f deterministic and z_t i.i.d. $(0, \sigma^2)$,

$$y_t = \frac{f(t-m+1) + \cdots + f(t)}{m} + \frac{z_{t-m+1} + \cdots + z_t}{m}. \tag{1.2}$$

The second term has variance σ^2/m and is $\approx Ez_1 = 0$ by the LLN, so $y_t \approx f_m(t) := m^{-1} \sum_{j=0}^{m-1} f(t-j)$.

1.1.2 Exponential smoothing

Discounting the past geometrically, for $0 < w < 1$,

$$S_t = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j}.$$

Given S_0^*, define recursively

$$S_t^* = w x_t + (1-w) S_{t-1}^*. \tag{1.3}$$

Iterating,

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^* = S_t + (1-w)^t S_0^*, \tag{1.4}$$

so $S_t^* \approx S_t$ for large t. Take $S_0^* = x_1$ or $\bar{x}$, and choose w to minimize

$$SSE(w) = \sum_{t=2}^{n} (x_t - S_{t-1}^*)^2. \tag{1.5}$$

1.1.3 Trend, seasonality and noise

With trend $T_t = f(t)$, the decomposition $x_t = T_t + S_t + z_t$ has *seasonal component* $S_t = S_{t+s}$ ($s = 4$ quarterly, $s = 12$ monthly). Real errors z_t are usually correlated, which motivates stationarity.

1.2 Stationary time series

Motivation. Inference from one realization needs a probabilistic structure that is stable over time; its dependence is described by the autocovariance function.

Definition 1.2.1 (Time series)

If T is the set of integers $\mathbb{Z}$ or of positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$, a collection of random variables $\{X_t : t \in T\}$ is called a *time series*.

Definition 1.2.2 (Strictly stationary time series)

$\{X_t\}$ is *strictly stationary* if for all $k \geq 1$, all integers $t_1, \dots, t_k$ and h, and all real $x_1, \dots, x_k$,

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k).$$

Definition 1.2.3 (Weakly stationary time series)

$\{X_t\}$ is *(weakly) stationary* if
(1) $EX_t = \mu$ for all t;
(2) $EX_t^2 < \infty$ for all t;
(3) $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_0, X_h)$ for all t, h.

By (3) the covariance depends only on the lag; in particular the variance is constant. "Stationary" means weakly stationary.

Example 1.2.4 (Strict versus weak stationarity)

(1) An i.i.d. sequence $\{X_t\}$ is strictly stationary.
(2) If moreover $EX_1^2 < \infty$, it is also weakly stationary. An i.i.d. Cauchy sequence is strictly but not weakly stationary.
(3) If ϵ_t are i.i.d. with mean 0, variance 1, then $X_t = \epsilon_t + \epsilon_{t-1}$ is strictly and weakly stationary.

A strictly stationary series with finite variance is weakly stationary (shift invariance of the law of (X_t, X_{t+h}) gives (1) and (3)); finite variance is assumed throughout.

Definition 1.2.5 (Autocovariance function)

For a stationary $\{X_t\}$ the real function

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is called the *autocovariance function* (ACVF).

Theorem 1.2.6 (Basic properties of the ACVF)

The autocovariance function satisfies
(1) $\gamma(0) \geq 0$;
(2) $\gamma(-h) = \gamma(h)$;
(3) $|\gamma(h)| \leq \gamma(0)$.

Proof. [Supplement]

Step 1. $\gamma(0) = \mathrm{Var}(X_t) \geq 0$.

Step 2. $\gamma(-h) = \mathrm{Cov}(X_t, X_{t-h}) = \mathrm{Cov}(X_s, X_{s+h}) = \gamma(h)$ with $s = t - h$.

Step 3. Cauchy–Schwarz: $|\mathrm{Cov}(X_t, X_{t+h})| \leq \sqrt{\mathrm{Var}\, X_t \ \mathrm{Var}\, X_{t+h}} = \gamma(0)$. ■

Theorem 1.2.7 (Nonnegative definiteness)

For any reals $x_1, \ldots, x_k$ and integers $t_1, \ldots, t_k$,

$$\sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j \geq 0.$$

Proof. [Supplement] Let $Y = \sum_{i=1}^{k} x_i X_{t_i}$. Since $\mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_j - t_i) = \gamma(t_i - t_j)$ by Theorem 1.2.6(2),

$$0 \leq \mathrm{Var}(Y) = \sum_{i,j=1}^{k} x_i x_j \,\mathrm{Cov}(X_{t_i}, X_{t_j}) = \sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j.$$

■

Thus $\Gamma_k = [\gamma(t_i - t_j)]_{i,j=1}^{k}$ is nonnegative definite; conversely, every even nonnegative definite function on $\mathbb{Z}$ is the ACVF of a stationary Gaussian series.

Definition 1.2.8 (White noise)

A stationary series $\{X_t\}$ whose ACVF satisfies $\gamma(h) = 0$ for $h \neq 0$ is called *white noise*, written $\{X_t\} \sim WN(0, \sigma^2)$ with $\sigma^2 = \gamma(0)$ (mean zero is understood). If in addition the X_t are i.i.d., we write $\{X_t\} \sim IID(0, \sigma^2)$.

Definition 1.2.9 (Autocorrelation function)

For a stationary $\{X_t\}$,

$$\rho(h) = \gamma(h)/\gamma(0)$$

is the *autocorrelation function* (ACF); for fixed h, $\rho(h)$ is the autocorrelation coefficient at lag h.

By Theorem 1.2.6, $\rho(0) = 1$, $\rho(-h) = \rho(h)$ and $|\rho(h)| \leq 1$.

Definition 1.2.10 (Multivariate normal distribution)

Let $\mu = (\mu_1, \ldots, \mu_k)^T \in \mathbb{R}^k$ and let $\Sigma = (\sigma_{ij})_{i,j=1}^{k}$ be symmetric positive definite. If the random vector $Z = (Z_1, \ldots, Z_k)^T$ has joint density

$$f(z) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} \exp\Big\{-\tfrac{1}{2}(z-\mu)^T \Sigma^{-1} (z-\mu)\Big\}, \qquad z \in \mathbb{R}^k,$$

then Z follows the *multivariate normal distribution* with mean μ and covariance Σ; here $\mu_i = EZ_i$ and $\sigma_{ij} = \mathrm{Cov}(Z_i, Z_j)$.

Definition 1.2.11 (Gaussian time series)

$\{X_t\}$ is a *Gaussian time series* if $(X_{t_1}, \ldots, X_{t_k})^T$ is multivariate normal for every k and all $t_1, \ldots, t_k$.

A normal law is determined by its mean and covariance, so a weakly stationary Gaussian series is strictly stationary. For a zero-mean stationary Gaussian series the best predictor is linear:

$$E(X_{k+1} \mid X_1, \dots, X_k) = \sum_{i=1}^{k} a_i X_i,$$

with a_i estimable by minimizing $\sum_{t=k+1}^{n} (X_t - a_1 X_{t-1} - \cdots - a_k X_{t-k})^2$.

[Supplement] Proof of linearity of the conditional mean. Let $\mathbf{X} = (X_1, \dots, X_k)^T$, $\Gamma_k = \mathrm{Cov}(\mathbf{X})$ (invertible), $\gamma_k = \mathrm{Cov}(\mathbf{X}, X_{k+1})$, $a = \Gamma_k^{-1}\gamma_k$, $U = X_{k+1} - a^T\mathbf{X}$. Then $\mathrm{Cov}(\mathbf{X}, U) = \gamma_k - \Gamma_k a = 0$; as $(\mathbf{X}, U)$ is jointly normal, $U \perp \mathbf{X}$, so $E(U \mid \mathbf{X}) = 0$ and $E(X_{k+1} \mid \mathbf{X}) = a^T\mathbf{X}$. ■

Nonstationary data can often be reduced to stationarity by transformations such as differencing (Chapter 5).

1.3 Selected exercises

Motivation. Exercises on exponential smoothing, strict versus weak stationarity, Gaussian conditional means and sums of stationary series.

Exercise 1.1 (Exponential smoothing recursion)

Let $0 < w < 1$ and let S_t^* be defined by (1.3) from an initial value S_0^*. Show (1.4), i.e.

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^*, \qquad t \geq 1,$$

and deduce that $S_t^* - S_t \to 0$ as $t \to \infty$.

Solution. Induction on t.

Step 1. $t = 1$: (1.3) gives $S_1^* = wx_1 + (1-w)S_0^*$.

Step 2. $t \geq 2$: substituting the formula for $t-1$ into (1.3) and putting $k = j+1$,

$$\begin{aligned} S_t^* &= wx_t + (1-w)\Big\{ w \sum_{j=0}^{t-2} (1-w)^j x_{t-1-j} + (1-w)^{t-1} S_0^* \Big\} \\ &= wx_t + w \sum_{k=1}^{t-1} (1-w)^k x_{t-k} + (1-w)^t S_0^* = w \sum_{k=0}^{t-1} (1-w)^k x_{t-k} + (1-w)^t S_0^*. \end{aligned}$$

Step 3. Hence $S_t^* - S_t = (1-w)^t S_0^* \to 0$ since $0 < 1 - w < 1$. ■

Exercise 1.3 (Strict versus weak stationarity without independence)

Using examples that are *not* i.i.d., show that
(a) a strictly stationary series need not be weakly stationary;
(b) a weakly stationary series need not be strictly stationary.

Solution.

(a) Let Z be standard Cauchy and $X_t = Z$ for all t. Then

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(Z \leq \min_i x_i) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k),$$

so $\{X_t\}$ is strictly stationary; it is not i.i.d. ($X_1 = X_2$, non-degenerate), and $EX_t^2 = \infty$ violates (2) of Definition 1.2.3.

(b) Let $\{X_t\}$ be independent with $X_t \sim N(0,1)$ for odd t and $P(X_t = \pm 1) = \frac{1}{2}$ for even t. Then $EX_t = 0$, $\mathrm{Cov}(X_t, X_{t+h}) = \mathbb{1}\{h = 0\}$, so $\{X_t\} \sim WN(0,1)$, but the marginals differ:

$$P(X_1 \le \tfrac{1}{2}) = \Phi(\tfrac{1}{2}) \approx 0.691 \neq \tfrac{1}{2} = P(X_2 \le \tfrac{1}{2}),$$

so it is neither i.i.d. nor strictly stationary. ■

Exercise 1.6 (Gaussian one-step conditional mean)

(a) (Exercise 1.5.) Let (X, Y) be bivariate normal with $\mathrm{Var}(Y) > 0$. Show that $E(X \mid Y) = aY + b$ with $a = \mathrm{Cov}(X,Y)/\mathrm{Var}(Y)$ and $b = EX - aEY$.
(b) Let $\{X_t\}$ be a stationary Gaussian series with mean μ and ACVF γ, $\gamma(0) > 0$. Express $E(X_t \mid X_{t-1})$ and its mean squared error through μ and γ.

Solution.

(a) Put $U = X - aY - b$.

Step 1. Every $sU + tY$ is a linear combination of X, Y, hence normal, so $Ee^{i(sU+tY)} = \exp\{i(sEU + tEY) - \frac{1}{2}\mathrm{Var}(sU + tY)\}$.

Step 2. $EU = 0$ and $\mathrm{Cov}(U, Y) = \mathrm{Cov}(X, Y) - a\,\mathrm{Var}(Y) = 0$, so $\mathrm{Var}(sU + tY) = s^2\,\mathrm{Var}\,U + t^2\,\mathrm{Var}\,Y$ and the characteristic function factorizes: $U \perp Y$.

Step 3. Hence $E(X \mid Y) = aY + b + E(U \mid Y) = aY + b$.

(b) (X_t, X_{t-1}) is bivariate normal with $\mathrm{Var}\,X_{t-1} = \gamma(0)$, $\mathrm{Cov}(X_t, X_{t-1}) = \gamma(1)$; (a) gives $a = \rho(1)$, $b = \mu - \rho(1)\mu$:

$$E(X_t \mid X_{t-1}) = \mu + \frac{\gamma(1)}{\gamma(0)}\,(X_{t-1} - \mu).$$

With $U = X_t - \mu - \rho(1)(X_{t-1} - \mu)$,

$$EU^2 = \gamma(0) - 2\rho(1)\gamma(1) + \rho(1)^2\gamma(0) = \gamma(0)(1 - \rho(1)^2) = \gamma(0) - \frac{\gamma(1)^2}{\gamma(0)}.$$

■

Exercise 1.7 (Sum of independent stationary series)

Let $\{X_t\}$ and $\{Y_t\}$ be independent stationary series with ACVFs γ_X, γ_Y. Show that $\{X_t + Y_t\}$ is stationary and find its ACVF.

Solution. Let $Z_t = X_t + Y_t$; check Definition 1.2.3.

Step 1. $EZ_t = \mu_X + \mu_Y$.

Step 2. $EZ_t^2 \le 2EX_t^2 + 2EY_t^2 < \infty$.

Step 3. By bilinearity,

$$\mathrm{Cov}(Z_t, Z_{t+h}) = \mathrm{Cov}(X_t, X_{t+h}) + \mathrm{Cov}(X_t, Y_{t+h}) + \mathrm{Cov}(Y_t, X_{t+h}) + \mathrm{Cov}(Y_t, Y_{t+h}).$$

Independence gives $\mathrm{Cov}(X_s, Y_u) = 0$, so

$$\gamma_Z(h) = \gamma_X(h) + \gamma_Y(h),$$

free of t (only uncorrelatedness was used), and

$$\rho_Z(h) = \frac{\gamma_X(0)\rho_X(h) + \gamma_Y(0)\rho_Y(h)}{\gamma_X(0) + \gamma_Y(0)}.$$

■

Chapter 2. ARMA Models

2.1 Introduction

Motivation. ARMA models build stationary dependence from i.i.d. noise by summing neighbouring noise terms (MA) and regressing on the past (AR).

Let $\epsilon_1, \epsilon_2, \dots$ be i.i.d. with mean 0 and variance σ^2.

2.1.1 Summing noise: MA(1)

For a real θ put

$$X_t = \epsilon_t + \theta\epsilon_{t-1}. \tag{2.1}$$

Then $EX_t = 0$ and, since $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$,

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1},\, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2, \end{cases}$$

free of t: the stationary *moving average model of order* 1, MA(1).

2.1.2 Regressing on the past: AR(1)

Consider

$$X_t = \phi X_{t-1} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2). \tag{2.2}$$

Case $\phi = 1$ (random walk). $X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t$, so $\mathrm{Var}(X_t - X_0) = t\sigma^2$. A stationary solution would give

$$\mathrm{Var}(X_t - X_0) = 2\gamma(0) - 2\gamma(t), \tag{2.3}$$

bounded by $4\gamma(0)$ (Theorem 1.2.6(3)), while $t\sigma^2 \to \infty$. Hence no stationary solution exists. For $\phi = -1$, $X_{2t} - X_0 = \sum_{j=1}^{2t} (-1)^j \epsilon_j$ has variance $2t\sigma^2$: same contradiction.

Case $|\phi| < 1$. If a stationary solution exists, iterating (2.2),

$$X_t = \phi X_{t-1} + \epsilon_t = \phi(\phi X_{t-2} + \epsilon_{t-1}) + \epsilon_t = \cdots = \sum_{j=0}^{n-1} \phi^j \epsilon_{t-j} + \phi^n X_{t-n}. \tag{2.4}$$

As $E(\phi^n X_{t-n})^2 = \phi^{2n} EX_0^2 \to 0$, the only candidate is

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-i} \tag{2.5}$$

(convergent in L^2 and a.s.). It is a solution,

$$\phi X_{t-1} + \epsilon_t = \sum_{i=0}^{\infty} \phi^{i+1} \epsilon_{t-1-i} + \epsilon_t = \sum_{j=1}^{\infty} \phi^j \epsilon_{t-j} + \epsilon_t = X_t,$$

and stationary: $EX_t = 0$ and

$$\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}\Big(\sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}, \sum_{k=0}^{\infty} \phi^k \epsilon_{t+h-k}\Big) = \sum_{j=0}^{\infty} \phi^j \phi^{j+h} \sigma^2 = \phi^h \frac{\sigma^2}{1-\phi^2}, \quad h \geq 0.$$

This is the *AR(1) model*.

2.2 Autoregressive models

Motivation. AR(p) regresses X_t on p lags; stationarity is decided by the roots of a polynomial.

Definition 2.2.1 (Autoregressive model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.6}$$

then $\{X_t\}$ follows an *autoregressive model of order* p, written $\{X_t\} \sim AR(p)$.

The *characteristic polynomial* is

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{2.7}$$

If every root of ϕ satisfies $|z| > 1$, then on $|z| \le 1$

$$\frac{1}{\phi(z)} = \sum_{i=0}^{\infty} a_i z^i, \tag{2.8}$$

and the stationary solution of (2.6) is

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}. \tag{2.9}$$

X_t depends only on ϵ_s, $s \le t$, hence is independent of ϵ_s, $s > t$: a *causal* representation (Proposition 2.2.B).

2.2.1 The backshift operator

$BX_t = X_{t-1}$, $Bf(t) = f(t-1)$, $B^j X_t = X_{t-j}$. For $a(z) = \sum_{i\ge 0} a_i z^i$, $b(z) = \sum_{j\ge 0} b_j z^j$ with $\sum |a_i|, \sum |b_j| < \infty$,

$$c(z) = (ab)(z) = a(z)b(z) = \sum_{k=0}^{\infty} c_k z^k, \qquad c_k = \sum_{i=0}^{k} a_i b_{k-i}.$$

For stationary $\{X_t\}$,

$$a(B)\big(b(B)X_t\big) = (ab)(B)X_t = b(B)\big(a(B)X_t\big), \tag{2.10}$$

all stationary. Thus (2.6) is $\phi(B)X_t = \epsilon_t$ and (2.9) is

$$X_t = \frac{1}{\phi(B)}\epsilon_t, \qquad \frac{1}{\phi(B)} = \sum_{i=0}^{\infty} a_i B^i. \tag{2.11}$$

Lemma 2.2.A (Operator calculus; supplement)

Let $\{X_t\}$ satisfy $\sup_t E|X_t| < \infty$ (e.g. stationary), and let $\sum |a_i| < \infty$, $\sum |b_j| < \infty$. Then $b(B)X_t = \sum_j b_j X_{t-j}$ converges absolutely a.s. and in L^1, (2.10) holds a.s., and if $\{X_t\}$ is stationary so is $\{b(B)X_t\}$, with ACVF $\sum_{j,k} b_j b_k \gamma_X(h + j - k)$.

Proof. [Supplement]

Step 1. With $M = \sup_t E|X_t|$, $E\sum_j |b_j||X_{t-j}| \le M \sum_j |b_j| < \infty$ (monotone convergence): absolute a.s. and L^1 convergence. If stationary, Minkowski gives $\|\sum_j |b_j||X_{t-j}|\|_2 \le \sum_j |b_j| \, \|X_0\|_2$: L^2 convergence.

Step 2. $\sum_{i,j}|a_i||b_j|E|X_{t-i-j}| \le M\sum_i|a_i|\sum_j|b_j| < \infty$, so $\sum_{i,j}a_ib_jX_{t-i-j}$ converges absolutely a.s. and can be summed in any order: over j first gives $a(B)(b(B)X_t)$, by $k = i+j$ gives $(ab)(B)X_t$; $ab = ba$ gives the third form.

Step 3. $E\,b(B)X_t = \mu\sum b_j$, and by L^2 convergence $\mathrm{Cov}(b(B)X_t, b(B)X_{t+h}) = \sum_{j,k}b_jb_k\gamma_X(h+j-k)$, free of t. ■

Proposition 2.2.B (Causal stationary solution; supplement)

Suppose $\phi(z) \neq 0$ for all $|z| \le 1$. Then:
(1) there are $r > 1$ and $C < \infty$ with $1/\phi(z) = \sum_{i\ge 0}a_iz^i$ for $|z| < r$ and $|a_i| \le Cr_0^{-i}$ for any fixed $1 < r_0 < r$;
(2) $X_t = \sum_i a_i\epsilon_{t-i}$ is a stationary solution of (2.6);
(3) it is the only stationary solution.

Proof. [Supplement]

Step 1. Let $r > 1$ be the smallest root modulus of ϕ. Then $1/\phi$ is analytic on $|z| < r$, and Cauchy's estimate on $|z| = r_0$ gives $|a_i| \le r_0^{-i}\max_{|z|=r_0}|1/\phi(z)| = Cr_0^{-i}$; so $\sum|a_i| < \infty$.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is stationary and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \epsilon_t$, as $\phi a = 1$ on $|z| \le 1$.

Step 3. If $\phi(B)Y_t = \epsilon_t$ with $\{Y_t\}$ stationary, then by the Lemma

$$a(B)\epsilon_t = a(B)(\phi(B)Y_t) = (a\phi)(B)Y_t = Y_t,$$

so $Y_t = X_t$ a.s. ■

For $p = 1$, the root $1/\phi$ lies outside the unit circle iff $|\phi| < 1$, and $a_i = \phi^i$ recovers (2.5).

Remark 2.2.2 (Nonzero mean)

A stationary AR(p) model with mean μ is written

$$X_t - \mu - \phi_1(X_{t-1}-\mu) - \phi_2(X_{t-2}-\mu) - \cdots - \phi_p(X_{t-p}-\mu) = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0,\sigma^2).$$

Example 2.2.3 (Variance of a stationary AR(1))

Let $X_t = \phi X_{t-1} + \epsilon_t$ with $|\phi| < 1$. As $\mathrm{Cov}(X_{t-1}, \epsilon_t) = 0$,

$$\gamma(0) = \mathrm{Var}(\phi X_{t-1} + \epsilon_t) = \phi^2\gamma(0) + \sigma^2, \qquad \text{hence} \qquad \gamma(0) = \frac{\sigma^2}{1-\phi^2}.$$

2.3 Moving average models

Motivation. MA(q) models are always stationary, and their ACF vanishes beyond lag q.

Definition 2.3.1 (Moving average model)

If $\{X_t\}$ satisfies

$$X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \cdots + \theta_q\epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0,\sigma^2), \tag{2.12}$$

then $\{X_t\}$ follows a *moving average model of order* q, written $\{X_t\} \sim MA(q)$.

With the backshift operator,

$$X_t = \theta(B)\epsilon_t, \qquad \theta(z) = \sum_{i=0}^{q} \theta_i z^i, \quad \theta_0 = 1. \tag{2.13}$$

For MA(1) with $\{\epsilon_t\} \sim WN(0, \sigma^2)$ (§2.1),

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1}, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2. \end{cases} \tag{2.14}$$

In general, for $h \geq 0$,

$$\gamma(h) = \sum_{i=0}^{q}\sum_{j=0}^{q} \theta_i\theta_j \,\mathrm{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_{i=0}^{q-h} \theta_i\theta_{i+h}$$

(empty for $h > q$), so

$$\gamma(h) = \begin{cases} 0, & |h| \geq q+1, \\ (\theta_{|h|} + \theta_1\theta_{1+|h|} + \cdots + \theta_{q-|h|}\theta_q)\sigma^2, & |h| \leq q. \end{cases} \tag{2.15}$$

An MA(q) is stationary with variance $(1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2$; a sample ACF vanishing beyond some lag suggests an MA model.

2.4 Autoregressive moving average models

Motivation. ARMA models describe rich dependence with few parameters; causality (roots of ϕ) gives stationarity and invertibility (roots of θ) recovers the noise.

Definition 2.4.1 (ARMA model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t + \theta_1\epsilon_{t-1} + \cdots + \theta_q\epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.16}$$

then $\{X_t\}$ follows an *autoregressive moving average model of orders p and q*, written $\{X_t\} \sim ARMA(p, q)$. With $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$, (2.16) reads

$$\phi(B)X_t = \theta(B)\epsilon_t. \tag{2.17}$$

If $\{X_t\}$ is stationary with mean $\mu \neq 0$, X_t is replaced by $X_t - \mu$ in (2.17).

Pure AR fits often need a large order; by the *principle of parsimony* (fewer parameters for equal fit) ARMA models are preferred.

If $\phi(z) \neq 0$ for all $|z| \leq 1$, $\{X_t\}$ is stationary and

$$X_t = \sum_{i=0}^{\infty} a_i\epsilon_{t-i}, \qquad \sum_{i=0}^{\infty} a_i z^i = \frac{\theta(z)}{\phi(z)}, \tag{2.18}$$

i.e. by (2.10)

$$X_t = \frac{\theta(B)}{\phi(B)}\epsilon_t = \Big(\frac{\theta}{\phi}\Big)(B)\epsilon_t. \tag{2.19}$$

Its ACVF decays geometrically: ARMA processes are *short-memory processes*.

Proposition 2.4.A (Causal ARMA and geometric decay; supplement)

If $\phi(z) \neq 0$ for $|z| \leq 1$, then (2.18) is the unique stationary solution of (2.17), with $|a_i| \leq C r_0^{-i}$ for some $C < \infty$, $r_0 > 1$, and

$$\gamma(h) = \sigma^2 \sum_{i=0}^{\infty} a_i a_{i+|h|}, \qquad |\gamma(h)| \leq C' r_0^{-|h|}.$$

Proof. [Supplement]

Step 1. θ/ϕ is analytic on $|z| < r$, so $|a_i| \leq C r_0^{-i}$ as in Proposition 2.2.B.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is stationary and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \theta(B)\epsilon_t$. Any stationary solution satisfies $Y_t = (\phi^{-1}\phi)(B)Y_t = \phi^{-1}(B)\theta(B)\epsilon_t = a(B)\epsilon_t$, $\phi^{-1} = 1/\phi$ having summable coefficients.

Step 3. For $h \geq 0$, $\gamma(h) = \sum_{i,j} a_i a_j \operatorname{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_i a_i a_{i+h}$, and

$$|\gamma(h)| \leq \sigma^2 C^2 \sum_{i \geq 0} r_0^{-i} r_0^{-i-h} = \frac{\sigma^2 C^2}{1 - r_0^{-2}} r_0^{-h}.$$

■

2.4.1 Invertibility

If $\{X_t\}$ is stationary ARMA and $\theta(z) \neq 0$ for all $|z| \leq 1$, then

$$\sum_{j=0}^{\infty} \pi_j X_{t-j} = \epsilon_t, \qquad \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad \pi_0 = 1, \tag{2.20}$$

and $\{X_t\}$ is *invertible*: the noise is a linear combination of past observations.

Proposition 2.4.B (Invertibility; supplement)

Let $\{X_t\}$ be a stationary solution of $\phi(B)X_t = \theta(B)\epsilon_t$ and suppose $\theta(z) \neq 0$ for $|z| \leq 1$. Then (2.20) holds a.s., with $|\pi_j|$ decaying geometrically.

Proof. [Supplement] As in Proposition 2.2.B, $1/\theta = \sum_j \eta_j z^j$ and $\pi = \phi\eta$ have geometrically decaying coefficients. By the Lemma,

$$\pi(B)X_t = \eta(B)(\phi(B)X_t) = \eta(B)(\theta(B)\epsilon_t) = (\eta\theta)(B)\epsilon_t = \epsilon_t.$$

■

Example 2.4.2 (One ACF, two MA(1) models)

The models

$$X_t = \epsilon_t + 2\epsilon_{t-1} \ \text{(not invertible)}, \qquad X_t = \epsilon_t + \tfrac{1}{2}\epsilon_{t-1} \ \text{(invertible)} \tag{2.21}$$

both have $\rho(1) = \theta/(1+\theta^2) = 2/5$ and $\rho(h) = 0$ for $|h| \geq 2$. Generally θ and $1/\theta$ give the same ACF; the ACVFs agree with noise variance σ^2 for $\theta = 2$ and $4\sigma^2$ for $\theta = 1/2$.

The invertible model is preferred because (2.20) gives the predictor

$$\hat{X}_{n+1} = -\pi_1 X_n - \pi_2 X_{n-1} - \cdots - \pi_n X_1.$$

Orders p, q are chosen from data: too many parameters violate parsimony, too few cause *model bias*; AIC and BIC are used (Chapter 4). For an AR(p) with $n > p$, by causality

$$E(X_{n+1} \mid X_1, \ldots, X_n) = E(\phi_1 X_n + \cdots + \phi_p X_{n-p+1} + \epsilon_{n+1} \mid X_1, \ldots, X_n) = \phi_1 X_n + \cdots + \phi_p X_{n-p+1},$$

so a wrong p gives a wrong predictor.

Example 2.4.3 (ACVF of a causal ARMA(1,1))

Let $X_t = \phi X_{t-1} + \epsilon_t + \theta \epsilon_{t-1}$, $|\phi| < 1$. For $h \geq 2$, X_{t-h} is uncorrelated with $\epsilon_t, \epsilon_{t-1}$, so

$$\gamma(h) = \mathrm{Cov}(\phi X_{t-1} + \epsilon_t + \theta \epsilon_{t-1}, X_{t-h}) = \phi \gamma(h-1), \quad \text{so} \quad \gamma(h) = \phi^{h-1} \gamma(1), \ h \geq 1.$$

With $\mathrm{Cov}(\epsilon_{t-1}, X_{t-1}) = \mathrm{Cov}(\epsilon_t, X_t) = \sigma^2$, $\mathrm{Cov}(\epsilon_{t-1}, X_t) = (\phi + \theta)\sigma^2$:

$$\begin{aligned} \gamma(1) &= \phi \gamma(0) + \theta \sigma^2, \\ \gamma(0) &= \phi^2 \gamma(0) + \sigma^2 + \theta^2 \sigma^2 + 2\phi\theta\sigma^2. \end{aligned}$$

Hence

$$\gamma(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad \gamma(1) = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}.$$

2.5 The partial autocorrelation function

Motivation. The PACF removes the effect of intermediate observations and cuts off after lag p for an AR(p).

Let $\{X_t\}$ be stationary with mean 0,

$$F = \Big\{ \sum_{i=2}^{k} a_i X_i : a_i \in \mathbb{R} \Big\},$$

and $P_F Y$ the *projection* of Y onto F: $E\big[(Y - P_F Y) X_i\big] = 0$, $i = 2, \ldots, k$ (the best linear predictor; $F = \{0\}$ for $k = 1$).

Definition 2.5.1 (Partial autocorrelation function)

For $k \geq 1$,

$$\phi_{kk} = \mathrm{Corr}\big(X_{k+1} - P_F X_{k+1},\ X_1 - P_F X_1\big)$$

is the *partial autocorrelation function* (PACF); for fixed k, ϕ_{kk} is the partial autocorrelation coefficient at lag k.

2.5.1 Computing the PACF

ϕ_{hh} is the last component of the solution of

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(h-1) \\ \rho(1) & \rho(0) & \cdots & \rho(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(h-1) & \rho(h-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{h1} \\ \phi_{h2} \\ \vdots \\ \phi_{hh} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(h) \end{pmatrix}. \tag{2.22}$$

By Cramer's rule, with $R_h = [\rho(i-j)]_{i,j=1}^h$ and R_h^* its last column replaced by $(\rho(1),\dots,\rho(h))^T$,

$$\phi_{hh} = \frac{\det R_h^*}{\det R_h}.$$

Proposition 2.5.A (PACF via the prediction equations; supplement)

Let $\{X_t\}$ be stationary, mean 0, with $\Gamma_h = [\gamma(i-j)]_{i,j=1}^h$ nonsingular. Then the best linear predictor of X_{h+1} from $X_1,\dots,X_h$ is $\sum_{j=1}^h \phi_{hj} X_{h+1-j}$ with coefficients solving (2.22), and its coefficient ϕ_{hh} on X_1 equals the PACF of Definition 2.5.1.

Proof. [Supplement]

Step 1. $\hat{X} = \sum_j c_j X_{h+1-j}$ is the projection iff $E[(X_{h+1} - \hat{X})X_{h+1-i}] = 0$, $i = 1,\dots,h$, i.e. $\sum_j c_j \gamma(i-j) = \gamma(i)$; divide by $\gamma(0)$ to get (2.22), uniquely solvable.

Step 2. Let $F = \operatorname{span}(X_2,\dots,X_h)$ and $U = X_1 - P_F X_1 \perp F$ ($\perp$: uncorrelated). Since $\operatorname{span}(X_1,\dots,X_h) = F \oplus \operatorname{span}(U)$, the projection of X_{h+1} is

$$P_F X_{h+1} + \frac{E(X_{h+1}U)}{EU^2}\,U = P_F X_{h+1} + \frac{E(X_{h+1}U)}{EU^2}\,(X_1 - P_F X_1),$$

and as $P_F X_{h+1}, P_F X_1 \in F$, the coefficient of X_1 is $\phi_{hh} = E(X_{h+1}U)/EU^2$.

Step 3. With $V = X_{h+1} - P_F X_{h+1}$, $E(X_{h+1}U) = E(VU)$. As γ is even, $(X_2,\dots,X_{h+1})$ and $(X_h,\dots,X_1)$ have the same covariance matrix, so $EV^2 = EU^2$ and $\phi_{hh} = E(VU)/\sqrt{EV^2 EU^2} = \operatorname{Corr}(V,U)$. ■

Example 2.5.2 (PACF of an AR(1))

For a stationary AR(1), $\rho(h) = \phi^h$ and $\phi_{11} = \rho(1) = \phi$. At $h = 2$:

$$\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix}\begin{pmatrix}\phi_{21}\\ \phi_{22}\end{pmatrix} = \begin{pmatrix}\rho(1)\\ \rho(2)\end{pmatrix} \implies \begin{pmatrix}\phi_{21}\\ \phi_{22}\end{pmatrix} = \frac{1}{1-\phi^2}\begin{pmatrix} 1 & -\phi \\ -\phi & 1\end{pmatrix}\begin{pmatrix}\phi\\ \phi^2\end{pmatrix} = \begin{pmatrix}\phi\\ 0\end{pmatrix},$$

so $\phi_{22} = 0$; likewise $\phi_{hh} = 0$ for all $h \geq 2$.

Proposition 2.5.B (PACF of a causal AR(p); supplement)

If $\{X_t\}$ is a causal AR(p) with $\phi_p \neq 0$ and $\sigma^2 > 0$, then $\phi_{pp} = \phi_p$ and $\phi_{hh} = 0$ for all $h > p$.

Proof. [Supplement]

Step 1. Γ_h is nonsingular: for $a \neq 0$ let i be the largest index with $a_i \neq 0$. By causality $X_i = \epsilon_i + \sum_{k\geq 1} a_k' \epsilon_{i-k}$ and $X_1,\dots,X_{i-1}$ depend on ϵ_u, $u < i$, so $Y = \sum_{j\leq i} a_j X_j = a_i\epsilon_i + W$ with $W \perp \epsilon_i$ and $a^T\Gamma_h a = \operatorname{Var}(Y) \geq a_i^2\sigma^2 > 0$.

Step 2. For $h \geq p$, $\hat{X} = \phi_1 X_h + \cdots + \phi_p X_{h+1-p}$ has error $\epsilon_{h+1} \perp X_1,\dots,X_h$ (causality), so it is the projection, with coefficients $(\phi_1,\dots,\phi_p,0,\dots,0)$.

Step 3. By Proposition 2.5.A, ϕ_{hh} is the coefficient on X_1: ϕ_p if $h = p$, 0 if $h > p$. ■

So the AR(p) PACF cuts off after lag p and the MA(q) ACF after lag q, the other functions decaying geometrically (Figure 1); this guides the choice of p, q (Chapters 3–4).

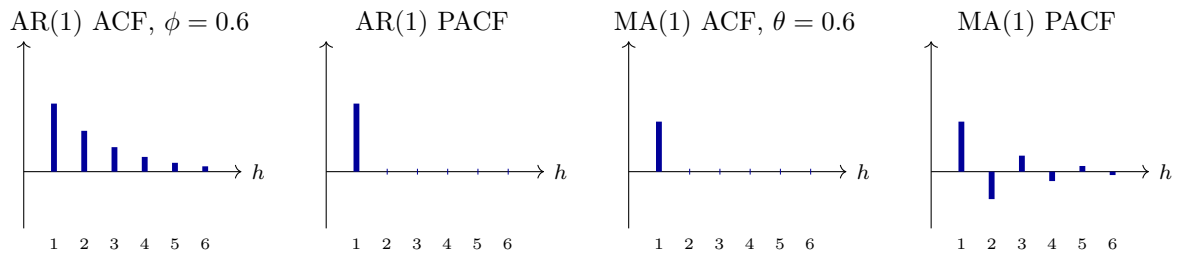


Figure 1: ACF and PACF of an AR(1) and an MA(1): the AR(1) PACF and MA(1) ACF cut off after lag 1.

2.6 Selected exercises

Motivation. Exercises on sums of MA processes, models sharing an ACF, and invertibility of ARMA(1,1).

Exercise 2.8 (Sum of independent MA(1) and MA(2))

Let $X_t = \epsilon_t + a\epsilon_{t-1}$ with $\{\epsilon_t\} \sim IID(0, \sigma_1^2)$ and $Y_t = \eta_t + b_1\eta_{t-1} + b_2\eta_{t-2}$ with $\{\eta_t\} \sim IID(0, \sigma_2^2)$, the two noise sequences being independent, and $\sigma_1^2 + \sigma_2^2 > 0$. Show that $Z_t = X_t + Y_t$ can be regarded as an MA(2) model: there are a white noise $\{u_t\} \sim WN(0, v)$ with $v > 0$ and constants c_1, c_2 such that

$$Z_t = u_t + c_1 u_{t-1} + c_2 u_{t-2} \qquad \text{for all } t.$$

Solution. Work in L^2 with inner product $E(UV)$; $\perp$ means uncorrelated. Let $H_t = \overline{\text{span}}\{Z_s : s \le t\}$ and P_t the projection onto H_t.

Step 1. By (2.15) and Exercise 1.7, Z is stationary, mean 0, $\gamma_Z = \gamma_X + \gamma_Y$, so $\gamma_Z(h) = 0$ for $|h| \ge 3$ and $\gamma_Z(0) = (1 + a^2)\sigma_1^2 + (1 + b_1^2 + b_2^2)\sigma_2^2 > 0$. Hence $Z_t \perp H_{t-3}$.

Step 2. (Innovations.) $u_t = Z_t - P_{t-1}Z_t \in H_t$, $u_t \perp H_{t-1}$; for $s < t$, $u_s \in H_{t-1}$, so $E(u_s u_t) = 0$.

Step 3. (Shift invariance.) Let α_m be the minimum-norm solution of $\Gamma_m \alpha_m = (\gamma_Z(1), \dots, \gamma_Z(m))^T$. As in Step 1 of Proposition 2.5.A, $\sum_i \alpha_{m,i} Z_{t-i}$ is the projection onto $\text{span}(Z_{t-1}, \dots, Z_{t-m})$, with t-free coefficients; these converge in L^2 to $P_{t-1}Z_t$. By continuity, $E(u_t u_s)$ and $E(u_t Z_s)$ depend only on $t - s$; so $v = Eu_t^2$ is constant and $\{u_t\} \sim WN(0, v)$.

Step 4. ($v > 0$.) If $v = 0$, $Z_s \in H_{s-1}$ for all s, so $H_t = H_{t-3}$; then $Z_t \perp H_t \ni Z_t$, i.e. $EZ_t^2 = 0$, a contradiction.

Step 5. $Z_s = u_s + P_{s-1}Z_s$ gives $H_s = H_{s-1} \oplus \text{span}(u_s)$; for $s = t-1, t-2$,

$$H_{t-1} = H_{t-3} \oplus \text{span}(u_{t-2}) \oplus \text{span}(u_{t-1}).$$

Step 6. Projecting Z_t,

$$Z_t = u_t + P_{t-1}Z_t = u_t + \frac{E(Z_t u_{t-1})}{v}\, u_{t-1} + \frac{E(Z_t u_{t-2})}{v}\, u_{t-2} + P_{H_{t-3}} Z_t,$$

the last term 0 by Step 1 and $c_j = E(Z_t u_{t-j})/v$ free of t by Step 3. Since $u_{t-2} - Z_{t-2} \in H_{t-3} \perp Z_t$, $c_2 = \gamma_Z(2)/v = b_2\sigma_2^2/v$.

Here $\{u_t\}$ is white noise, not necessarily i.i.d. ■

Exercise 2.9 (Two filters with the same ACF)

Let $\{Y_t\}$ be stationary with mean 0 and ACVF γ_Y, not a.s. zero, and let

$$X_t = Y_t - 0.4Y_{t-1}, \qquad W_t = Y_t - 2.5Y_{t-1}.$$

(a) Express the ACVFs of $\{X_t\}$ and $\{W_t\}$ through γ_Y.
(b) Show that $\{X_t\}$ and $\{W_t\}$ have the same ACF.

Solution.

(a) For $V_t = Y_t - cY_{t-1}$,

$$\begin{aligned}\mathrm{Cov}(V_t, V_{t+h}) &= \gamma_Y(h) - c\,\gamma_Y(h-1) - c\,\gamma_Y(h+1) + c^2\gamma_Y(h)\\ &= (1+c^2)\gamma_Y(h) - c\{\gamma_Y(h-1) + \gamma_Y(h+1)\},\end{aligned}$$

so with $c = 0.4$ and $c = 2.5$,

$$\begin{aligned}\gamma_X(h) &= 1.16\,\gamma_Y(h) - 0.4\{\gamma_Y(h-1) + \gamma_Y(h+1)\},\\ \gamma_W(h) &= 7.25\,\gamma_Y(h) - 2.5\{\gamma_Y(h-1) + \gamma_Y(h+1)\}.\end{aligned}$$

(b)

Step 1. $7.25 = 6.25 \times 1.16$ and $2.5 = 6.25 \times 0.4$, so $\gamma_W = 6.25\,\gamma_X$ (generally $1 + c^{-2} = c^{-2}(1+c^2)$, $c^{-1} = c^{-2}c$).

Step 2. $\gamma_X(0) > 0$: otherwise $Y_t = 0.4Y_{t-1} = 0.4^n Y_{t-n}$ a.s., so $\gamma_Y(0) = 0.16^n\gamma_Y(0)$ for all n and $Y_t = 0$ a.s.

Step 3. Hence $\rho_W(h) = 6.25\gamma_X(h)/(6.25\gamma_X(0)) = \rho_X(h)$; this generalizes Example 2.4.2. ■

Exercise 2.10 (Invertibility of ARMA(1,1))

Let $\{X_t\}$ be the stationary solution of $X_t - \phi X_{t-1} = \epsilon_t + \theta\epsilon_{t-1}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $\sigma^2 > 0$, $|\phi| < 1$. Find the condition for $\{X_t\}$ to be invertible, i.e. for $\epsilon_t = \sum_{j\geq 0}\pi_j X_{t-j}$ a.s. for some $\sum_j |\pi_j| < \infty$.

Solution. The condition is $|\theta| < 1$.

(a) Sufficiency.

Step 1. If $|\theta| < 1$, the root $-1/\theta$ of $\theta(z)$ lies outside the unit disc; apply Proposition 2.4.B.

Step 2. Explicitly, for $|z| < 1/|\theta|$,

$$\frac{\phi(z)}{\theta(z)} = (1-\phi z)\sum_{k\geq 0}(-\theta)^k z^k = 1 + \sum_{j\geq 1}\{(-\theta)^j - \phi(-\theta)^{j-1}\}z^j,$$

so $\pi_0 = 1$, $\pi_j = -(\phi+\theta)(-\theta)^{j-1}$, absolutely summable.

(b) Necessity. Suppose $\epsilon_t = \pi(B)X_t$ with $\sum|\pi_j| < \infty$ and $|\theta| \geq 1$.

Step 1. By Proposition 2.4.A and Lemma 2.2.A, $X_t = a(B)\epsilon_t$ with $a = \theta/\phi$ on $|z| \leq 1$, and $\epsilon_t = c(B)\epsilon_t$ with $c = \pi a$, $\sum|c_k| < \infty$.

Step 2. Covariance with ϵ_{t-k} gives $\sigma^2\mathbb{1}\{k=0\} = c_k\sigma^2$, so $c(z) \equiv 1$ and

$$\pi(z)\,\theta(z) = \phi(z), \qquad |z| \leq 1.$$

Step 3. At $z_0 = -1/\theta$ ($|z_0| \leq 1$), $\phi(z_0) = 1 + \phi/\theta = 0$, so $|\phi| = |\theta| \geq 1$, a contradiction.

If $\phi = -\theta$ (common factor), $X_t = \epsilon_t$, consistent with the criterion. ■

Chapter 3. Inference for Stationary ARMA Models

3.1 Estimation of the mean

Motivation. The CLT for $\bar{X}$ survives dependence, but with an asymptotic variance that depends on the model and can differ greatly from $\mathrm{Var}(X_t)$.

The subscript n is dropped ($\bar{X} = \bar{X}_n$) and limits are as $n \to \infty$. Consider the AR(1) model

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad |\phi| < 1.$$

By (2.5),

$$X_t = \mu + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}. \tag{3.1}$$

Theorem 3.1.1 (CLT for the sample mean of an AR(1))

If $\{X_t\}$ is the stationary series (3.1) and $\hat{\mu} = \bar{X} = n^{-1}\sum_{t=1}^{n} X_t$, then

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{(1-\phi)^2}\Big).$$

Proof.

Step 1. (Decomposition.) Put $b_j = \sum_{k \geq j+1} \phi^k = \phi^{j+1}/(1-\phi)$ and $y_t = \sum_{j \geq 0} b_j \epsilon_{t-j}$. Since $\sum |b_j| < \infty$, $\{y_t\}$ is stationary (Lemma 2.2.A). As $b_{j-1} - b_j = \phi^j$ and $b_0 = \phi/(1-\phi)$,

$$\begin{aligned} X_t - \mu &= \epsilon_t + \sum_{j \geq 1} (b_{j-1} - b_j)\epsilon_{t-j} = \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - b_0\epsilon_t + \sum_{j \geq 0} b_j \epsilon_{t-1-j} - \sum_{j \geq 1} b_j \epsilon_{t-j} \\ &= \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - y_t + y_{t-1}. \end{aligned}$$

Step 2. (Telescoping.) Summing over t,

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} (X_t - \mu) = \underbrace{\frac{1}{1-\phi} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \epsilon_t}_{\text{(I)}} \underbrace{- \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0}_{\text{(II)}}.$$

Step 3. By the i.i.d. CLT, (I) $\xrightarrow{d} N(0, \sigma^2/(1-\phi)^2)$, since $\sum_j \phi^j = 1/(1-\phi)$.

Step 4. $E(\text{II})^2 \leq \frac{2}{n}(Ey_n^2 + Ey_0^2) = \frac{4}{n} Ey_0^2 \to 0$, so (II) $\xrightarrow{p} 0$ by Chebyshev's inequality. Slutsky's theorem gives the claim. ∎

Remark 3.1.2 (General linear processes, ARMA)

For $X_t = \mu + \sum_{j \geq 0} a_j \epsilon_{t-j}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $|a_j| \leq c\rho^j$ ($0 < \rho < 1$; every causal ARMA by Proposition 2.4.A), put $b_j = \sum_{k \geq j+1} a_k$, $|b_j| \leq c\rho^{j+1}/(1-\rho)$, $y_t = \sum_j b_j \epsilon_{t-j}$. As above $X_t - \mu = (\sum_j a_j)\epsilon_t - y_t + y_{t-1}$, so

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}}\Big(\sum_{j=0}^{\infty} a_j\Big) \sum_{t=1}^{n} \epsilon_t - \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0 \xrightarrow{d} N\Big(0, \Big(\sum_{j=0}^{\infty} a_j\Big)^2 \sigma^2\Big).$$

To test $H_0 : \mu = 0$ at level α, reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2}\,|\sum_j a_j|\,\sigma$ (i.i.d.: $z_{\alpha/2}\sigma$), where $P(|Z| \leq z_{\alpha/2}) = 1-\alpha$. For AR(1), $\sum_j a_j = 1/(1-\phi)$ is huge for $\phi \approx 1$; the $100(1-\alpha)\%$ interval is

$$\bar{X} - \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)} \leq \mu \leq \bar{X} + \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)}.$$

3.2 Sample autocovariance, autocorrelation and partial autocorrelation

Motivation. Bartlett's theorem tells how far a sample autocorrelation can stray from its true value by chance.

Definition 3.2.1 (Sample autocovariance function)

For $|h| < n$, the *sample autocovariance function* is

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(X_t - \bar{X})(X_{t+|h|} - \bar{X}).$$

For stationary ARMA series $\hat{\gamma}(h) \to \gamma(h)$ a.s. (ergodic theorem). It is reliable only for small lags (rule of thumb $|h| \leq \sqrt{n}$); the divisor n makes $[\hat{\gamma}(i-j)]$ nonnegative definite.

Definition 3.2.2 (Sample autocorrelation function)

The *sample autocorrelation function* (SACF), the estimator of $\rho(h)$, is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Like ρ, it satisfies $|\hat{\rho}(h)| \leq 1$.

Definition 3.2.3 (Sample partial autocorrelation)

For fixed h, the last component $\hat{\phi}_{hh}$ of the solution of

$$\begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \cdots & \hat{\rho}(h-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \cdots & \hat{\rho}(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}(h-1) & \hat{\rho}(h-2) & \cdots & \hat{\rho}(0) \end{pmatrix} \begin{pmatrix} \hat{\phi}_{h1} \\ \hat{\phi}_{h2} \\ \vdots \\ \hat{\phi}_{hh} \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \\ \vdots \\ \hat{\rho}(h) \end{pmatrix}$$

is the *sample partial autocorrelation coefficient*; as a function of h it is the *sample partial autocorrelation function* (SPACF).

Theorem 3.2.4 (Bartlett)

Let $\{X_t\}$ be stationary with

$$X_t = \mu + \sum_{j=0}^{\infty} a_j \epsilon_{t-j}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad \sum_{j=0}^{\infty} |a_j| < \infty, \quad E\epsilon_t^4 < \infty.$$

Then for each fixed $h \geq 1$,

$$\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W), \qquad \hat{\boldsymbol{\rho}}(h) = (\hat{\rho}(1), \ldots, \hat{\rho}(h))^T, \quad \boldsymbol{\rho}(h) = (\rho(1), \ldots, \rho(h))^T,$$

where $W = (w_{ij})_{i,j=1}^h$ is given by *Bartlett's formula*

$$\begin{aligned} w_{ij} &= \sum_{k=-\infty}^{\infty} \Big\{ \rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j) + 2\rho(i)\rho(j)\rho^2(k) \\ &\qquad - 2\rho(i)\rho(k)\rho(k+j) - 2\rho(j)\rho(k)\rho(k+i) \Big\} \\ &= \sum_{k=1}^{\infty} \{\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)\}\{\rho(k+j) + \rho(k-j) - 2\rho(j)\rho(k)\}. \end{aligned}$$

Proof. [Supplement] Two standard limit theorems are quoted. Write $\eta = E\epsilon_t^4/\sigma^4$ and, without loss of generality, $\mu = 0$. Let $\gamma^*(h) = n^{-1}\sum_{t=1}^n X_t X_{t+h}$.

Step 1. (Fourth moments.) Since $E\epsilon_i\epsilon_j\epsilon_k\epsilon_l = \sigma^4(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + (\eta - 3)\sigma^4\delta_{ijkl}$, expanding the four series ($a_j = 0$ for $j < 0$) gives

$$\begin{aligned} E[X_t X_{t+p} X_s X_{s+q}] &= \gamma(p)\gamma(q) + \gamma(s-t)\gamma(s-t+q-p) + \gamma(s-t+q)\gamma(s-t-p) \\ &\quad + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+s-t} a_{i+s-t+q}. \end{aligned}$$

Step 2. (Covariances.) Summing over $s, t = 1, \ldots, n$ and grouping by $k = s - t$,

$$\begin{aligned} n\,\mathrm{Cov}(\gamma^*(p), \gamma^*(q)) &= \sum_{|k|<n} \Big(1 - \frac{|k|}{n}\Big) \Big[\gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p) \\ &\qquad + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q}\Big]. \end{aligned}$$

All terms are absolutely summable in k (as $\sum|\gamma(k)| \leq \sigma^2(\sum|a_j|)^2$), so by dominated convergence the limit is

$$V_{pq} = (\eta - 3)\gamma(p)\gamma(q) + \sum_{k=-\infty}^{\infty} [\gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p)],$$

using $\sum_k \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q} = \sigma^{-4}\gamma(p)\gamma(q)$.

Step 3. (Joint CLT for γ^*.) For the truncated process $X_t^{(m)} = \sum_{j=0}^m a_j \epsilon_{t-j}$, the vectors $(X_t^{(m)} X_{t+l}^{(m)})_{l=0}^h$ form a strictly stationary $(m+h)$-dependent sequence, so the CLT for m-dependent sequences (Hoeffding–Robbins) together with the Cramér–Wold device gives $\sqrt{n}(\boldsymbol{\gamma}_m^* - \boldsymbol{\gamma}_m) \xrightarrow{d} N(0, V^{(m)})$, where $V^{(m)} \to V$ as $m \to \infty$ by Step 2 applied to $a^{(m)}$. Applying the computation of Step 2 to the difference $X_t - X_t^{(m)}$ gives $\limsup_n n\,\mathrm{Var}(\gamma^*(l) - \gamma_m^*(l)) \leq C\sum_{j>m}|a_j| \to 0$ as $m \to \infty$. By the standard approximation lemma (Billingsley, Thm. 3.2), $\sqrt{n}(\boldsymbol{\gamma}^* - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$, with $\boldsymbol{\gamma}^* = (\gamma^*(0), \ldots, \gamma^*(h))^T$.

Step 4. (From γ^* to $\hat{\gamma}$.) Expanding the product in Definition 3.2.1,

$$\hat{\gamma}(l) - \gamma^*(l) = -\frac{1}{n}\sum_{t=n-l+1}^{n} X_t X_{t+l} - \bar{X} \cdot \frac{1}{n}\sum_{t=1}^{n-l}(X_t + X_{t+l}) + \Big(1 - \frac{l}{n}\Big)\bar{X}^2.$$

The first term has L^1 norm $\leq l\gamma(0)/n$. The other two are bounded by $C\bar{X}^2 + O_p(n^{-1})|\bar{X}|$, and $\bar{X} = O_p(n^{-1/2})$ by Remark 3.1.2. Hence $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*) \xrightarrow{p} 0$ and $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$.

Step 5. (Delta method.) $\hat{\rho}(i) = g_i(\hat{\boldsymbol{\gamma}})$ with $g_i(x) = x_i/x_0$, whose gradient at $\boldsymbol{\gamma}$ has entries $-\rho(i)/\gamma(0)$ (in x_0) and $1/\gamma(0)$ (in x_i). Hence $\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W)$ with

$$w_{ij} = \frac{V_{ij} - \rho(i)V_{0j} - \rho(j)V_{i0} + \rho(i)\rho(j)V_{00}}{\gamma(0)^2}.$$

The $(\eta - 3)$ terms contribute $(\eta - 3)[\rho(i)\rho(j) - \rho(i)\rho(j) - \rho(j)\rho(i) + \rho(i)\rho(j)] = 0$, so W does not depend on η. With $T_{pq} = V_{pq}/\gamma(0)^2$ (without the η term) and the shifts $k \mapsto k + i$, one gets $T_{ij} = \sum_k [\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)]$, $T_{0j} = 2\sum_k \rho(k)\rho(k+j)$, $T_{i0} = 2\sum_k \rho(k)\rho(k+i)$ and $T_{00} = 2\sum_k \rho^2(k)$, which yields the first form of w_{ij}.

Step 6. (Second form.) Expand the product in the second form, sum over $k \geq 1$, and use $\rho(-k) = \rho(k)$ to rewrite each one-sided sum as half of a two-sided sum (the $k = 0$ terms cancel); this reproduces the first form. ■

Corollary 3.2.A (White-noise and MA(q) bands; supplement)

(1) If $\{X_t\} \sim IID(0, \sigma^2)$ with $EX_t^4 < \infty$, then $\sqrt{n}\,\hat{\boldsymbol{\rho}}(h) \xrightarrow{d} N(0, I_h)$.
(2) If $\{X_t\}$ is MA(q), then for $i > q$, $\sqrt{n}\,\hat{\rho}(i) \xrightarrow{d} N(0,\, 1 + 2\sum_{k=1}^{q} \rho^2(k))$.

Proof. [Supplement]

Step 1. For i.i.d. data $\rho(k) = 0$ for $k \neq 0$. In the second form of Bartlett's formula, the factor $\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)$ with $k \geq 1$, $i \geq 1$ is nonzero only for $k = i$, where it equals 1. Thus $w_{ij} = \mathbb{1}\{i = j\}$.

Step 2. If $i > q$ then $\rho(i) = 0$ and $\rho(k+i) = 0$ for $k \geq 1$, so $w_{ii} = \sum_{k\geq 1} \rho^2(k-i) = \sum_{m=1-i}^{\infty} \rho^2(m) = \rho^2(0) + 2\sum_{k=1}^{q} \rho^2(k)$, since $\rho(m) = 0$ for $|m| > q$. ■

For white noise about 95% of the $\hat{\rho}(h)$ lie within $\pm 1.96/\sqrt{n}$; for an MA(q) the bands at lags $> q$ widen by $(1 + 2\sum_{k\leq q} \rho^2(k))^{1/2}$.

3.3 The method of moments

Motivation. Equating population and sample autocorrelations gives, for AR models, the linear Yule–Walker system.

The moment equations are nonlinear for MA/ARMA, so consider the causal AR(p) model

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

By causality,

$$E[(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p})X_{t-j}] = E\epsilon_t X_{t-j} = 0, \qquad j = 1, \ldots, p,$$

which gives the *Yule–Walker equations* $\gamma(j) = \phi_1\gamma(j-1) + \cdots + \phi_p\gamma(j-p)$, $j = 1, \ldots, p$; in matrix form

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & \rho(0) & \cdots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix}. \tag{3.2}$$

Replacing $\rho(j)$ by $\hat{\rho}(j)$ gives the *Yule–Walker (Y-W) estimator* $\hat{\boldsymbol{\phi}}_{YW}$; by Definition 3.2.3, the last component of the order-h Y-W estimator is $\hat{\phi}_{hh}$.

Theorem 3.3.1 (Asymptotic normality of the Y-W estimator)

Let $\hat{\boldsymbol{\phi}}_{YW} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T$ be the Y-W estimator of $\boldsymbol{\phi}$ in a causal AR(p) model. Then

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2),$$

where Γ is the $p \times p$ matrix with $\Gamma_{ij} = \gamma(i-j)$.

Lemma 3.3.A (Basic limits for causal AR(p); supplement)

Let $\{X_t\}$ be a causal AR(p) with $\{\epsilon_t\} \sim IID(0, \sigma^2)$, and $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$. Then
(a) $\frac{1}{n}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s., and $\hat{\gamma}(h) \to \gamma(h)$ a.s. for each h;
(b) $\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2\Gamma)$;
(c) Γ is nonsingular.

Proof. [Supplement]

Step 1. (a) $X_t = \sum_j a_j \epsilon_{t-j}$ is a fixed function of the shifted i.i.d. sequence, so $\{X_t\}$ and $X_{t-i}X_{t-j}$ are strictly stationary and ergodic; Birkhoff gives $n^{-1}\sum X_{t-i}X_{t-j} \to \gamma(i-j)$, $\bar{X} \to 0$ and $\hat{\gamma}(h) \to \gamma(h)$ a.s.

Step 2. (b) Let $\mathcal{F}_t = \sigma(\epsilon_s : s \le t)$, $c \in \mathbb{R}^p$, $\xi_t = c^T\mathbf{X}_{t-1}\epsilon_t$: stationary ergodic, $E(\xi_t \mid \mathcal{F}_{t-1}) = c^T\mathbf{X}_{t-1}E\epsilon_t = 0$, $E\xi_t^2 = \sigma^2 c^T\Gamma c$. The martingale-difference CLT (Billingsley–Ibragimov) gives $n^{-1/2}\sum \xi_t \xrightarrow{d} N(0, \sigma^2 c^T\Gamma c)$; apply Cramér–Wold.

Step 3. (c) Step 1 of Proposition 2.5.B: $c^T\Gamma c \ge c_i^2\sigma^2 > 0$ for $c \ne 0$. ■

Proof of Theorem 3.3.1. [Supplement] Let $\hat{\Gamma} = [\hat{\gamma}(i-j)]_{i,j=1}^{p}$ and $\hat{\boldsymbol{\gamma}}_p = (\hat{\gamma}(1), \ldots, \hat{\gamma}(p))^T$, so $\hat{\Gamma}\hat{\boldsymbol{\phi}}_{YW} = \hat{\boldsymbol{\gamma}}_p$ (multiply (3.2) with hats by $\hat{\gamma}(0)$).

Step 1. Then $\hat{\Gamma}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}$, whose j-th entry is $\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)$.

Step 2. Replacing $\hat{\gamma}(l)$ by $n^{-1}\sum_t X_tX_{t+l}$ costs $O(p)$ boundary terms $O_p(1/n)$ and terms $\bar{X} \cdot n^{-1}\sum X_t = O_p(1/n)$ (Remark 3.1.2), negligible after $\times\sqrt{n}$:

$$\sqrt{n}\Big(\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)\Big) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\Big(X_t - \sum_k \phi_k X_{t-k}\Big) + o_p(1) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\epsilon_t + o_p(1).$$

Step 3. By the Lemma, $\sqrt{n}(\hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \sigma^2\Gamma)$ and $\hat{\Gamma} \to \Gamma$ (nonsingular). By Slutsky,

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\Gamma}^{-1}\sqrt{n}(\hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2\Gamma\Gamma^{-1}) = N(0, \Gamma^{-1}\sigma^2).$$

■

Example 3.3.3 (Y-W estimators for AR(1) and AR(2))

For AR(1), $\rho(1) = \phi$ gives $\hat{\phi} = \hat{\rho}(1)$. For AR(2) the Y-W equations are

$$\rho(1) = \phi_1 + \rho(1)\phi_2, \qquad \rho(2) = \rho(1)\phi_1 + \phi_2,$$

so

$$\hat{\phi}_1 = \frac{\hat{\rho}(1)(1-\hat{\rho}(2))}{1-\hat{\rho}^2(1)}, \qquad \hat{\phi}_2 = \frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)}.$$

Corollary 3.3.B (Sample PACF of an AR(p); supplement)

For a causal AR(p) and each $h > p$, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. An AR(p) is also an AR(h) with $\boldsymbol{\phi}^{(h)} = (\phi_1, \ldots, \phi_p, 0, \ldots, 0)^T$, and $\hat{\phi}_{hh}$ is the last component of the order-h Y-W estimator. By Theorem 3.3.1, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0, \sigma^2(\Gamma_h^{-1})_{hh})$.

Step 2. $1/(\Gamma_h^{-1})_{hh} = \mathrm{Var}(X_{t-h} - P(X_{t-h} \mid X_{t-h+1}, \ldots, X_{t-1}))$, which by time reversal equals the one-step error of X_t from $X_{t-1}, \ldots, X_{t-h+1}$, i.e. σ^2 as $h - 1 \geq p$ (Proposition 2.5.B, Step 2). So $\sigma^2(\Gamma_h^{-1})_{hh} = 1$. ■

So $\pm 1.96/\sqrt{n}$ SPACF bands beyond lag p choose the AR order.

3.4 Least squares estimators

Motivation. Regressing X_t on its lags gives the same limit law as Yule–Walker despite dependent regressors.

For AR(1) $X_t = \phi X_{t-1} + \epsilon_t$, the *least squares estimator* minimizes $S(\phi) = \sum_{t=2}^n (X_t - \phi X_{t-1})^2$:

$$\hat{\phi} = \frac{\sum_{t=2}^n X_t X_{t-1}}{\sum_{t=2}^n X_{t-1}^2}, \qquad \sqrt{n}(\hat{\phi} - \phi) = \frac{n^{-1/2}\sum_{t=2}^n X_{t-1}\epsilon_t}{n^{-1}\sum_{t=2}^n X_{t-1}^2}.$$

Here $EX_{t-1}\epsilon_t = 0$, $\mathrm{Var}(\sum_{t=2}^n X_{t-1}\epsilon_t) = (n-1)\sigma^2 EX_1^2$ (cross terms vanish as $\epsilon_t \perp X_{s-1}\epsilon_s X_{t-1}$, $s < t$), and

$$\frac{1}{\sqrt{n}}\sum_{t=2}^n X_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2 EX_1^2) \tag{3.3}$$

(Lemma 3.3.A(b)), while the denominator tends to $\gamma(0) = EX_1^2$.

Theorem 3.4.1 (LS estimator in AR(1))

If $\hat{\phi}$ is the least squares estimator of ϕ in a stationary AR(1) model, then

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N\big(0, (EX_1^2)^{-1}\sigma^2\big) = N(0, 1 - \phi^2).$$

Proof. [Supplement] Numerator $\xrightarrow{d}$ $N(0, \sigma^2 EX_1^2)$ by (3.3), denominator $\to$ $EX_1^2 > 0$ a.s. (Lemma 3.3.A(a)); Slutsky gives $N(0, \sigma^2/EX_1^2)$, and $EX_1^2 = \sigma^2/(1-\phi^2)$ (Example 2.2.3). (Case $p = 1$ of Theorem 3.4.2.) ■

For AR(p) with $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$, the LS estimator is

$$\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1}X_t,$$

so that, using $X_t = \mathbf{X}_{t-1}^T\boldsymbol{\phi} + \epsilon_t$,

$$\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t.$$

Here $E\mathbf{X}_{t-1}\epsilon_t = 0$ and $\mathrm{Cov}(\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t) = (n-p)\Gamma\sigma^2$ with $\Gamma_{ij} = \gamma(i-j)$.

Theorem 3.4.2 (LS estimator in AR(p))

If $\hat{\boldsymbol{\phi}}$ is the least squares estimator of $\boldsymbol{\phi}$ in a stationary (causal) AR(p) model, then

$$\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2).$$

Proof. [Supplement]

Step 1. By Lemma 3.3.A(b), $n^{-1/2}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \Gamma\sigma^2)$.

Step 2. By Lemma 3.3.A(a),(c), $n^{-1}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s. with Γ nonsingular, so the inverse exists for large n and converges to Γ^{-1}.

Step 3. Slutsky: $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} \Gamma^{-1}N(0, \Gamma\sigma^2) = N(0, \Gamma^{-1}\sigma^2)$. ■

So Y-W and LS have the same limit law (Theorems 3.3.1, 3.4.2). With $\hat{\Gamma}_{ij} = \hat{\gamma}(i-j)$ and consistent $\hat{\sigma}^2$,

$$\frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \xrightarrow{d} \chi^2(p), \tag{3.4}$$

since $Z_n = \sqrt{n}\,\Gamma^{1/2}(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})/\sigma \xrightarrow{d} N(0, I_p)$ and the statistic is $Z_n^T Z_n + o_p(1)$. A $100(1-\alpha)\%$ confidence region is

$$S_\alpha = \Big\{\boldsymbol{\phi} \in \mathbb{R}^p : \frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \le \chi^2(p, 1-\alpha)\Big\},$$

where $P(\chi^2(p) \ge \chi^2(p, 1-\alpha)) = \alpha$.

3.5 Maximum likelihood estimators

Motivation. The Gaussian AR likelihood factorizes into one-step conditionals; dropping the first factor gives the conditional MLE, which is least squares.

Assume $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$ (the usual working assumption even for non-normal data).

3.5.1 Exact likelihood of AR(1)

Let $X_t = \phi X_{t-1} + \epsilon_t$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, $|\phi| < 1$. Then $EX_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2/(1-\phi^2)$, so

$$f_{X_1}(x_1; \phi, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2/(1-\phi^2)}} \exp\Big\{\frac{-x_1^2}{2\sigma^2/(1-\phi^2)}\Big\}.$$

As $\epsilon_2 \perp X_1$,

$$P(X_2 \le x \mid X_1 = x_1) = P(\phi X_1 + \epsilon_2 \le x \mid X_1 = x_1) = P(\epsilon_2 \le x - \phi x_1),$$

so $X_2 \mid X_1 = x_1 \sim N(\phi x_1, \sigma^2)$; by the Markov structure $f_{X_t|X_{t-1},\ldots,X_1} = f_{X_t|X_{t-1}}$ is the $N(\phi x_{t-1}, \sigma^2)$ density. Hence

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n; \phi, \sigma^2) = f_{X_1}(x_1; \phi, \sigma^2)\prod_{t=2}^{n} f_{X_t|X_{t-1}}(x_t \mid x_{t-1}; \phi, \sigma^2), \tag{3.5}$$

and the log-likelihood is

$$\begin{aligned} L(\phi, \sigma^2) = &-\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\frac{\sigma^2}{1-\phi^2} - \frac{x_1^2}{2\sigma^2/(1-\phi^2)} \\ &- \frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2. \end{aligned} \tag{3.6}$$

The score equations are nonlinear in ϕ, so (3.6) is maximized numerically. For AR(p) the first factor is the $N(0, \Gamma_p)$ density of $(X_1, \ldots, X_p)$; ARMA MLEs are also numerical.

3.5.2 Conditional likelihood

The *conditional maximum likelihood estimator* drops f_{X_1} from (3.5):

$$L_c(\phi, \sigma^2) = -\frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2.$$

Its maximizers are

$$\hat{\phi} = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=2}^{n}(X_t - \hat{\phi}X_{t-1})^2.$$

Proposition 3.5.A (Conditional MLE = LS for AR(p); supplement)

For a Gaussian causal AR(p), the conditional MLE of $\boldsymbol{\phi}$ given $X_1, \dots, X_p$ equals the LS estimator $\hat{\boldsymbol{\phi}}$, the conditional MLE of σ^2 is $\hat{\sigma}^2 = (n-p)^{-1}\sum_{t=p+1}^{n}(X_t - \mathbf{X}_{t-1}^T\hat{\boldsymbol{\phi}})^2 \to \sigma^2$ a.s., and $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2)$. The exact MLE has the same limit distribution.

Proof. [Supplement]

Step 1. $L_c = -\frac{n-p}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}S(\boldsymbol{\phi})$, $S(\boldsymbol{\phi}) = \sum_{t=p+1}^{n}(X_t - \mathbf{X}_{t-1}^T\boldsymbol{\phi})^2$, is maximized in $\boldsymbol{\phi}$ by the LS estimator for every σ^2; $\partial L_c/\partial\sigma^2 = -\frac{n-p}{2\sigma^2} + \frac{S(\hat{\boldsymbol{\phi}})}{2\sigma^4} = 0$ gives $\hat{\sigma}^2 = S(\hat{\boldsymbol{\phi}})/(n-p)$.

Step 2. Writing $X_t - \mathbf{X}_{t-1}^T\hat{\boldsymbol{\phi}} = \epsilon_t - \mathbf{X}_{t-1}^T(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})$ and using the normal equations, $S(\hat{\boldsymbol{\phi}})/(n-p) = n^{-1}\sum\epsilon_t^2 - (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T(n^{-1}\sum\mathbf{X}_{t-1}\mathbf{X}_{t-1}^T)(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) + o(1) \to \sigma^2$ (SLLN, Lemma 3.3.A, Theorem 3.4.2); the law of $\hat{\boldsymbol{\phi}}$ is Theorem 3.4.2.

Step 3. (Exact MLE, sketch.) The extra log-density of $(X_1, \dots, X_p)$ has $O_p(1)$ derivative while the Hessian of L_c is $\approx -n\Gamma/\sigma^2$; a one-step expansion shows the maximizers differ by $O_p(1/n)$, so the limits coincide. ■

3.6 Prediction

Motivation. The mean-square optimal forecast is a conditional expectation, computed recursively for ARMA models.

Given $X_1, \dots, X_n$, the *predictor* of X_{n+h} is

$$f(X_1, \dots, X_n) := E(X_{n+h} \mid X_1, \dots, X_n).$$

Proposition 3.6.A (Optimality of the conditional mean; supplement)

If $EX_{n+h}^2 < \infty$, then $E\big(X_{n+h} - g(X_1, \dots, X_n)\big)^2$ over all measurable g with $Eg^2 < \infty$ is minimized by $g = f$.

Proof. [Supplement] With $Y = X_{n+h}$, $\mathbf{X} = (X_1, \dots, X_n)$,

$$E(Y - g(\mathbf{X}))^2 = E(Y - f(\mathbf{X}))^2 + E(f(\mathbf{X}) - g(\mathbf{X}))^2 + 2E\big[(Y - f(\mathbf{X}))(f(\mathbf{X}) - g(\mathbf{X}))\big],$$

and the cross term is $2E\big[(f-g)(\mathbf{X})\,E(Y - f(\mathbf{X}) \mid \mathbf{X})\big] = 0$. So $E(Y-g)^2 \geq E(Y-f)^2$, with equality iff $g(\mathbf{X}) = f(\mathbf{X})$ a.s. ■

For Gaussian $\{X_t\}$, $f = \sum_{i=1}^{n} a_i X_i$ (Section 1.2) with a_i minimizing $E(X_{n+h} - \sum_i a_i X_i)^2$. In general f is nonlinear, but linear predictors are used in practice (Durbin–Levinson algorithm; for ARMA, functions of the parameters).

From the infinite past, $\hat{X}_t(h) = E(X_{t+h} \mid X_t, X_{t-1}, \dots)$.

Example 3.6.1 (Forecasting an AR(1) with mean)

Let $(1-\phi B)(X_t-\mu)=\epsilon_t$, $\{\epsilon_t\}\sim IID(0,\sigma^2)$, $|\phi|<1$, so $X_t=\mu+\phi(X_{t-1}-\mu)+\epsilon_t$. As $\epsilon_{t+1}\perp X_t, X_{t-1},\dots,$

$$\hat{X}_t(1)=E(\mu+\phi(X_t-\mu)+\epsilon_{t+1}\mid X_t,X_{t-1},\dots)=\mu+\phi(X_t-\mu).$$

Similarly $\hat{X}_t(h)=\mu+\phi(\hat{X}_t(h-1)-\mu)$, hence

$$\hat{X}_t(h)=\mu+\phi^h(X_t-\mu)\longrightarrow\mu\quad(h\to\infty).$$

In practice μ,ϕ are replaced by $\hat{\mu},\hat{\phi}$.

For a stationary invertible ARMA(1,1), $(1-\phi B)(X_t-\mu)=(1+\theta B)\epsilon_t$, $X_{t+1}=\mu(1-\phi)+\phi X_t+\epsilon_{t+1}+\theta\epsilon_t$, and by invertibility $E(\epsilon_t\mid X_t,X_{t-1},\dots)=\epsilon_t$:

$$\hat{X}_t(1)=\mu(1-\phi)+\phi X_t+\theta\epsilon_t,\qquad \hat{X}_t(h)=\mu(1-\phi)+\phi\hat{X}_t(h-1),\quad h\geq 2.$$

In practice ϵ_t is replaced by the residual $\hat{\epsilon}_t$ and μ,ϕ,θ by estimates.

3.7 Selected exercises

Motivation. Exercises on the mean interval, hand computation of sample ACF/PACF, and $|\hat{\rho}(h)|\leq 1$.

Exercise 3.1 (Inference on the mean of an AR(1))

In an AR(1) model $X_t-\mu=\phi(X_{t-1}-\mu)+\epsilon_t$ with $\phi=0.3$ and $\sigma^2=1$, a sample of size $n=100$ gives $\bar{X}=0.1$.
(a) Find an approximate 95% confidence interval for μ.
(b) Test $H_0:\mu=0$ against $H_1:\mu\neq 0$ at level 5%.

Solution.

(a) By Theorem 3.1.1, $\sqrt{n}(\bar{X}-\mu)(1-\phi)/\sigma\xrightarrow{d}N(0,1)$, so $P\big(|\bar{X}-\mu|\leq z_{0.025}\sigma/(\sqrt{n}(1-\phi))\big)\to 0.95$. Here

$$\frac{z_{0.025}\,\sigma}{\sqrt{n}(1-\phi)}=\frac{1.96\times 1}{10\times 0.7}=0.28,$$

so the interval is $0.1\pm 0.28=[-0.18,\,0.38]$.

(b) Reject when $|\sqrt{n}\bar{X}|\geq z_{0.025}\sigma/(1-\phi)=1.96/0.7=2.8$; here $|\sqrt{n}\bar{X}|=1<2.8$, so H_0 is not rejected.

An i.i.d. treatment with $\gamma(0)=1/0.91$ gives half-width $1.96\times(0.91)^{-1/2}/10\approx 0.205$, too narrow. ■

Exercise 3.2 (Sample ACVF, ACF and PACF by hand)

For the data

$$(x_1,\dots,x_{10})=(16,\ 9,\ 10,\ 3,\ 5,\ 12,\ 14,\ 7,\ 9,\ 11),$$

compute $\hat{\gamma}(h)$ and $\hat{\rho}(h)$ for $h=0,1,2,3$, and $\hat{\phi}_{hh}$ for $h=1,2,3$.

Solution.

Step 1. $\bar{x} = 96/10 = 9.6$, and the deviations $d_t = x_t - \bar{x}$ are

$$6.4,\ -0.6,\ 0.4,\ -6.6,\ -4.6,\ 2.4,\ 4.4,\ -2.6,\ -0.6,\ 1.4.$$

Step 2. By Definition 3.2.1, $\hat{\gamma}(h) = \frac{1}{10}\sum_{t=1}^{10-h} d_t d_{t+h}$. The lagged cross-product sums are 140.40, 12.44, −43.92, −50.88 for $h = 0, 1, 2, 3$, so

$$\hat{\gamma}(0) = 14.04, \quad \hat{\gamma}(1) = 1.244, \quad \hat{\gamma}(2) = -4.392, \quad \hat{\gamma}(3) = -5.088,$$

and, dividing by $\hat{\gamma}(0)$,

$$\hat{\rho}(0) = 1, \quad \hat{\rho}(1) = 0.0886, \quad \hat{\rho}(2) = -0.3128, \quad \hat{\rho}(3) = -0.3624.$$

Step 3. By Definition 3.2.3, $\hat{\phi}_{11} = \hat{\rho}(1) = 0.0886$, and solving the 2×2 system as in Example 2.5.2,

$$\hat{\phi}_{22} = \frac{\hat{\rho}(2) - \hat{\rho}(1)^2}{1 - \hat{\rho}(1)^2} = \frac{-0.3128 - 0.0079}{1 - 0.0079} = -0.3232.$$

Step 4. For $h = 3$, Cramer's rule gives $\hat{\phi}_{33} = \det \hat{R}_3^* / \det \hat{R}_3$ with $\hat{R}_3 = [\hat{\rho}(i-j)]_{i,j=1}^3$ and $\hat{R}_3^*$ its last column replaced by $(\hat{\rho}(1), \hat{\rho}(2), \hat{\rho}(3))^T$. Numerically $\det \hat{R}_3 = 0.8815$ and $\det \hat{R}_3^* = -0.2947$, so $\hat{\phi}_{33} = -0.3344$.

All values lie within $\pm 1.96/\sqrt{10} = \pm 0.62$ (Corollary 3.2.A): no significant autocorrelation. ■

Exercise 3.4 (Sample autocorrelations are bounded by one)

Let $x_1, \ldots, x_n$ be not all equal.
(a) Show that $|\hat{\rho}(h)| \le 1$ for all $|h| < n$.
(b) Show that the matrix $\hat{\Gamma}_k = [\hat{\gamma}(i-j)]_{i,j=1}^k$ is nonnegative definite for every $k \ge 1$.

Solution. Let $y_t = x_t - \bar{x}$ for $1 \le t \le n$ and $y_t = 0$ for all other integers t. Since $\hat{\gamma}(h) = \hat{\gamma}(-h)$ and the extra terms vanish,

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t \in \mathbb{Z}} y_t y_{t+h}, \qquad h \in \mathbb{Z}.$$

(a)

Step 1. $\hat{\gamma}(0) = n^{-1}\sum_t y_t^2 > 0$ because the x_t are not all equal, so $\hat{\rho}(h)$ is defined.

Step 2. By the Cauchy–Schwarz inequality for finite sums,

$$\Big|\sum_t y_t y_{t+h}\Big| \le \Big(\sum_t y_t^2\Big)^{1/2}\Big(\sum_t y_{t+h}^2\Big)^{1/2} = \sum_t y_t^2,$$

since shifting the index does not change $\sum_t y_t^2$. Dividing by $n\hat{\gamma}(0)$ gives $|\hat{\rho}(h)| \le 1$.

(b) For $a \in \mathbb{R}^k$, substituting $s = t - j$ in the inner sum (all sums are finite),

$$\sum_{i,j=1}^k a_i a_j \hat{\gamma}(i-j) = \frac{1}{n}\sum_{i,j} a_i a_j \sum_s y_{s+j} y_{s+i} = \frac{1}{n}\sum_{s \in \mathbb{Z}}\Big(\sum_{i=1}^k a_i y_{s+i}\Big)^2 \ge 0.$$

(b) also implies (a) via the 2×2 minor $\hat{\gamma}(0)^2 - \hat{\gamma}(h)^2 \ge 0$; zero-padding is why the divisor is n. ■

Chapter 4. Nonstationary Time Series Models and Model Selection

4.1 Autoregressive integrated moving average models

Motivation. Series that wander without a fixed level are differenced until stationary and then fitted by ARMA: the ARIMA class.

The *random walk*

$$X_t = X_{t-1} + \epsilon_t, \qquad \text{equivalently} \qquad X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t,$$

reads $(1-B)X_t = \epsilon_t$, i.e. $\nabla X_t = \epsilon_t$ with $\nabla = 1 - B$.

Example (the random walk is not stationary)

With $X_0 = 0$ and $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$, $EX_t = 0$ but $\text{Var}(X_t) = \text{Cov}(X_t, X_{t+h}) = t\sigma^2$ $(h \geq 0)$ depends on t, while $\nabla X_t = \epsilon_t$ is white noise; $\phi(z) = 1 - z$ has its root on the unit circle.

If $Z_t = (1-B)^d X_t$ is stationary ARMA, $\phi(B)Z_t = \theta(B)\epsilon_t$, then $\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t$.

Definition 4.1.1 (ARIMA(p,d,q) model)

A time series $\{X_t\}$ follows an *autoregressive integrated moving average model* of orders p, d, q, written $\{X_t\} \sim \text{ARIMA}(p,d,q)$, if

$$\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2),$$

where $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. In practice $d \in \{0, 1, 2\}$.

ϕ, θ are assumed causal and invertible (no zeros in $|z| \leq 1$); IID errors are needed for the inference below.

Remark 4.1.2

The random walk is ARIMA$(0,1,0)$. If the differenced series has a nonzero mean μ, the model is written

$$\phi(B)((1-B)^d X_t - \mu) = \theta(B)\epsilon_t, \qquad \mu = E(1-B)^d X_t.$$

Polynomial trends. For $X_t = t + \epsilon_t$, $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$,

$$(1-B)X_t = t + \epsilon_t - (t-1) - \epsilon_{t-1} = 1 + \epsilon_t - \epsilon_{t-1},$$

so $\{X_t\}$ is ARIMA$(0,1,1)$ with $\mu = 1$ and non-invertible $\theta(B) = 1 - B$. For $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$, $(1-B)$ lowers the degree by one ($a\{t^k - (t-1)^k\} = ak\, t^{k-1} + \cdots$), so $(1-B)^r X_t = a_r r! + (1-B)^r \epsilon_t$ is stationary. The converse fails; a known polynomial trend is better fitted directly.

4.2 Unit root tests

Motivation. Testing for a root at $z = 1$ needs Dickey–Fuller critical values, since under a unit root LS converges at rate n to a Brownian-motion functional.

4.2.1 The AR(1) case

Let $X_t = \phi X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$, and test

$$H_0 : \phi = 1 \qquad \text{vs} \qquad H_1 : |\phi| < 1.$$

With $\hat{\phi}_{LS} = \sum_{t=1}^{n} X_{t-1}X_t / \sum_{t=1}^{n} X_{t-1}^2$, let

$$T_n = n(\hat{\phi}_{LS} - 1). \tag{4.1}$$

Under H_0, $\hat{\phi}_{LS} - 1 = \sum X_{t-1}\epsilon_t / \sum X_{t-1}^2$, so

$$T_n = \frac{n^{-1}\sum_{t=1}^{n} X_{t-1}\epsilon_t}{n^{-2}\sum_{t=1}^{n} X_{t-1}^2}. \tag{4.2}$$

Define the partial-sum process

$$W_n(s) = \frac{1}{\sqrt{n}\,\sigma} \sum_{j=1}^{[ns]} \epsilon_j, \qquad 0 \le s \le 1; \tag{4.3}$$

under H_0, $X_t = \sigma\sqrt{n}\, W_n(t/n)$.

Definition 4.2.1 (Brownian motion)

A stochastic process $\{W(s) : s \ge 0\}$ with continuous sample paths is a (standard) *Brownian motion* if
(1) $W(0) = 0$;
(2) for all $t > s \ge 0$, $W(t) - W(s) \sim N(0, t-s)$;
(3) for all $0 \le t_1 < \cdots < t_k$, the increments $W(t_1)$, $W(t_2) - W(t_1), \ldots, W(t_k) - W(t_{k-1})$ are independent.

$W_n \xrightarrow{d} W$ means weak convergence in $D[0,1]$ (Skorokhod topology): $Ef(W_n) \to Ef(W)$ for bounded continuous f; then $h(W_n) \xrightarrow{d} h(W)$ for h continuous at a.e. path of W.

Theorem 4.2.2 (Functional central limit theorem)

Let $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$ and define W_n by (4.3). Then, as $n \to \infty$, $W_n \xrightarrow{d} W$, a standard Brownian motion on $[0,1]$.

Proof. [Supplement] Let $S_k = \epsilon_1 + \cdots + \epsilon_k$, $W_n(s) = S_{[ns]}/(\sigma\sqrt{n})$. We verify finite-dimensional convergence and tightness (Prokhorov).

Step 1 (finite-dimensional laws). For $0 = s_0 < \cdots < s_k \le 1$, $m_j = [ns_j] - [ns_{j-1}]$, $m_j/n \to s_j - s_{j-1}$,

$$W_n(s_j) - W_n(s_{j-1}) = \sqrt{\frac{m_j}{n}} \cdot \frac{1}{\sigma\sqrt{m_j}} \sum_{i=[ns_{j-1}]+1}^{[ns_j]} \epsilon_i \xrightarrow{d} N(0, s_j - s_{j-1})$$

(CLT, Slutsky). The increments use disjoint blocks, hence are independent and converge jointly to independent $N(0, s_j - s_{j-1})$; summing gives $(W_n(s_1), \ldots, W_n(s_k)) \xrightarrow{d} (W(s_1), \ldots, W(s_k))$.

Step 2 (a maximal inequality). By Billingsley (Thm. 8.4) it suffices that for every $\varepsilon > 0$ there are $\lambda > 1$, n_0 with

$$P\Big(\max_{k \le n} |S_k| \ge \lambda\sigma\sqrt{n}\Big) \le \frac{\varepsilon}{\lambda^2} \qquad (n \ge n_0).$$

Etemadi: $P(\max_{k\le n}|S_k|\ge 3a)\le 3\max_{k\le n}P(|S_k|\ge a)$; take $a=\mu\sigma\sqrt{n}$, $\mu=\lambda/3$. By the CLT, $P(|S_k|\ge\mu\sigma\sqrt{k})\le 2P(|N|\ge\mu)\le 6/\mu^4$ for $k\ge k_\mu$ ($N\sim N(0,1)$, $EN^4=3$). Hence

(i) for $k_\mu\le k\le n$: $P(|S_k|\ge\mu\sigma\sqrt{n})\le P(|S_k|\ge\mu\sigma\sqrt{k})\le 6/\mu^4$;

(ii) for $k<k_\mu$: by Chebyshev, $P(|S_k|\ge\mu\sigma\sqrt{n})\le k/(\mu^2 n)\le k_\mu/(\mu^2 n)\le 1/\mu^4$ once $n\ge\mu^2 k_\mu$.

So $P(\max_{k\le n}|S_k|\ge\lambda\sigma\sqrt{n})\le 3\cdot 7/\mu^4=1701/\lambda^4\le\varepsilon/\lambda^2$ for $\lambda^2\ge 1701/\varepsilon$.

Step 3 (conclusion). Every subsequence has a weakly convergent further subsequence whose limit has the finite-dimensional laws of W; these determine the law, so $W_n\xrightarrow{d}W$. ■

Proposition (Limit of the unit-root statistics)

Under $H_0:\phi=1$, with T_n as in (4.1),

$$T_n\xrightarrow{d}T:=\frac{\frac12(W(1)^2-1)}{\int_0^1 W(s)^2\,ds}=\frac{\int_0^1 W(s)\,dW(s)}{\int_0^1 W(s)^2\,ds}.$$

Under $H_1:|\phi|<1$, $P(|T_n|\ge c)\to 1$ for every constant c; that is, the test rejecting for large $|T_n|$ is consistent.

Proof. [Supplement]

Step 1 (numerator). $X_t^2-X_{t-1}^2=2X_{t-1}\epsilon_t+\epsilon_t^2$; summing with $X_0=0$,

$$\frac1n\sum_{t=1}^n X_{t-1}\epsilon_t=\frac12\Big(\frac{X_n^2}{n}-\frac1n\sum_{t=1}^n\epsilon_t^2\Big)=\frac{\sigma^2}{2}\Big(W_n(1)^2-\frac{1}{n\sigma^2}\sum_{t=1}^n\epsilon_t^2\Big).$$

Step 2 (denominator). $W_n(s)=X_{t-1}/(\sigma\sqrt{n})$ on $[(t-1)/n,t/n)$, so

$$\frac{1}{n^2}\sum_{t=1}^n X_{t-1}^2=\sigma^2\sum_{t=1}^n\int_{(t-1)/n}^{t/n}W_n(s)^2\,ds=\sigma^2\int_0^1 W_n(s)^2\,ds.$$

Step 3 (joint limit). $w\mapsto(w(1),\int_0^1 w^2)$ is continuous at continuous w, so $(W_n(1),\int_0^1 W_n^2)\xrightarrow{d}(W(1),\int_0^1 W^2)$; with $n^{-1}\sum\epsilon_t^2\xrightarrow{p}\sigma^2$ and Slutsky, all three converge jointly.

Step 4 (ratio). As $\int_0^1 W^2>0$ a.s., $(a,b,c)\mapsto\frac12(a^2-c)/b$ is continuous at the limit, giving $T_n\xrightarrow{d}\frac12(W(1)^2-1)/\int_0^1 W^2$; Itô's formula $W(1)^2=2\int_0^1 W\,dW+1$ gives the second form.

Step 5 (consistency). If $|\phi|<1$, $T_n=n(\hat\phi_{LS}-\phi)+n(\phi-1)$ with $n(\hat\phi_{LS}-\phi)=O_p(\sqrt{n})$ (Chapter 3), so $T_n/n\xrightarrow{p}\phi-1<0$ and $P(|T_n|\ge c)\to 1$. ■

Thus the rate is n (super-consistency) and T is left-skewed, not normal. Reject when $|T_n|>c_\alpha$, $P(|T|>c_\alpha)=\alpha$, with quantiles tabulated by simulation.

Dickey–Fuller test. The studentized version is

$$DF_n=\Big(\frac{\sum_{t=1}^n X_{t-1}^2}{\hat\sigma^2}\Big)^{1/2}(\hat\phi_{LS}-1),\qquad \hat\sigma^2=\frac1n\sum_{t=1}^n(X_t-\hat\phi_{LS}X_{t-1})^2.$$

Under H_0,

$$DF_n=\frac{n^{-1}\sum_{t=1}^n X_{t-1}\epsilon_t}{\hat\sigma\,(n^{-2}\sum_{t=1}^n X_{t-1}^2)^{1/2}}\xrightarrow{d}DF:=\frac{\frac12(W(1)^2-1)}{(\int_0^1 W(s)^2ds)^{1/2}},$$

by Steps 1–3 and $\hat\sigma^2\xrightarrow{p}\sigma^2$: with $X_t-\hat\phi_{LS}X_{t-1}=\epsilon_t-(\hat\phi_{LS}-1)X_{t-1}$,

$$\hat\sigma^2=\frac1n\sum\epsilon_t^2-2(\hat\phi_{LS}-1)\cdot\frac1n\sum X_{t-1}\epsilon_t+(\hat\phi_{LS}-1)^2\cdot\frac1n\sum X_{t-1}^2,$$

and $\hat{\phi}_{LS} - 1 = O_p(n^{-1})$, $n^{-1}\sum X_{t-1}\epsilon_t = O_p(1)$, $n^{-1}\sum X_{t-1}^2 = O_p(n)$ make the last two terms $O_p(n^{-1})$. Reject when $|DF_n| \geq DF_\alpha$, $P(|DF| \geq DF_\alpha) = \alpha$.

4.2.2 The AR(p) case: augmented Dickey–Fuller test

For $\phi(B)X_t = \epsilon_t$, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$, test $\phi(1) = 0$.

Lemma (Error-correction form)

For any $\phi_1, \ldots, \phi_p$, with $\Delta X_t = X_t - X_{t-1}$,

$$X_t - \sum_{i=1}^{p} \phi_i X_{t-i} = X_t - \beta X_{t-1} - \sum_{j=1}^{p-1} \psi_j \Delta X_{t-j}, \qquad \beta = \sum_{i=1}^{p} \phi_i = 1 - \phi(1), \quad \psi_j = -\sum_{i=j+1}^{p} \phi_i.$$

In particular $\phi(1) = 0$ iff $\beta = 1$, and then $\phi(B) = \psi(B)(1 - B)$ with $\psi(B) = 1 - \psi_1 B - \cdots - \psi_{p-1} B^{p-1}$.

Proof. [Supplement] Using $X_{t-i} = X_{t-1} - \sum_{j=1}^{i-1} \Delta X_{t-j}$ and exchanging sums,

$$\sum_{i=1}^{p} \phi_i X_{t-i} = \Big(\sum_{i=1}^{p} \phi_i\Big) X_{t-1} - \sum_{i=1}^{p} \phi_i \sum_{j=1}^{i-1} \Delta X_{t-j} = \beta X_{t-1} - \sum_{j=1}^{p-1} \Big(\sum_{i=j+1}^{p} \phi_i\Big) \Delta X_{t-j}.$$

If $\beta = 1$ the right side is $\Delta X_t - \sum_j \psi_j \Delta X_{t-j} = \psi(B)(1 - B)X_t$. ■

So the AR(p) model is the regression

$$X_t = \beta X_{t-1} + \psi_1 \Delta X_{t-1} + \cdots + \psi_{p-1} \Delta X_{t-p+1} + \eta_t, \qquad \{\eta_t\} \sim \mathrm{IID}(0, \tau^2), \tag{4.4}$$

with $\beta = 1$, $\eta_t = \epsilon_t$ under the unit root. The *augmented Dickey–Fuller* (ADF) test rejects for large $|(\hat{\beta} - 1)/\,\mathrm{sd}(\hat{\beta})|$; its limit law is the same DF.

4.3 Model selection and diagnostics

Motivation. Orders are identified by sample ACF/PACF or information criteria, and the fit is checked by overfitting and residuals.

4.3.1 The Box–Jenkins method

Three stages:

(1) *identification* of p, q, (2) *estimation* (Chapter 3), (3) *diagnostic checking* (Section 4.3.3), returning to (1) on failure (Figure 2).

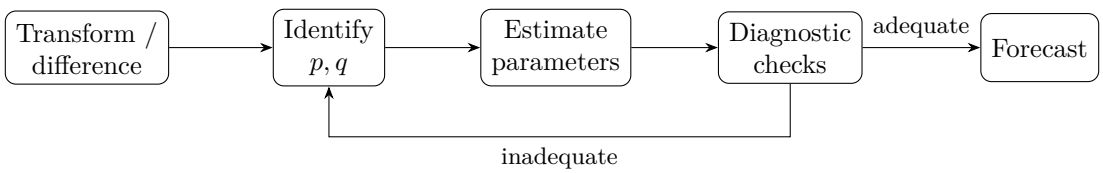


Figure 2: The Box–Jenkins modelling cycle.

Making the series stationary. Non-constant spread: Box–Cox $(y^\lambda - 1)/\lambda$ $(\lambda \neq 0)$, $\sqrt{y}$, $1/y$ or $\log y$ (standard for economic data). Trend: difference. A slowly decaying SACF indicates nonstationarity; a unit-root test (Section 4.2) decides. Usually $d \in \{0, 1, 2\}$.

Identification by ACF and PACF. Compare SACF and SPACF with Table 1: the SACF guides q, the SPACF p. By parsimony ($p + q + 2$ parameters) usually $p, q \leq 3$.

Model	ACF	PACF
AR(p)	decays exponentially or as a damped sine	cuts off after lag p
MA(q)	cuts off after lag q	decays exponentially or as a damped sine
ARMA(p, q)	decays after lag $q - p$	decays after lag $p - q$

Table 1: ACF and PACF of ARMA models (Table 4.3.1).

AR(p): for $k \geq p + 1$, $\hat{\phi}_{kk} \approx N(0, 1/n)$, so $H_0 : \phi_{kk} = 0$ is rejected when $|\hat{\phi}_{kk}| > z_{\alpha/2}/\sqrt{n}$.

MA(q): reject $H_0 : \rho(k) = 0$ when $|\hat{\rho}(k)| > z_{\alpha/2}\,\text{S.E.}(\hat{\rho}(k))$, where by Bartlett

$$\text{S.E.}(\hat{\rho}(k)) \approx \Big(\frac{1 + 2\sum_{j=1}^{q} \hat{\rho}(j)^2}{n}\Big)^{1/2}, \qquad k \geq q + 1;$$

if no $k \geq q+1$ rejects, MA(q) is tentatively chosen. Mixed ARMA orders are chosen by the criteria below.

4.3.2 Model selection criteria

Consider the nested models

$$M_1 = \{N(\mu, 1) : -\infty < \mu < \infty\}, \qquad M_2 = \{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

Choosing M_2 when M_1 holds wastes a parameter; the reverse omits an essential one, which is worse. With maximized likelihoods $MLE_1 = f_1(X_1, \ldots, X_n; \hat{\mu})$, $MLE_2 = f_2(X_1, \ldots, X_n; \hat{\mu}, \hat{\sigma}^2)$, select the smaller of

$$AIC(M_1) = -2\log(MLE_1) + 2 \times 1, \qquad AIC(M_2) = -2\log(MLE_2) + 2 \times 2.$$

In general

$$AIC(\text{model}) = -2\log(MLE) + 2\,(\text{number of parameters in the model}), \tag{4.5}$$

$$BIC(\text{model}) = -2\log(MLE) + \log(n)\,(\text{number of parameters in the model}).$$

The likelihood always grows with the number of parameters; the penalty enforces parsimony. AIC estimates the expected Kullback–Leibler discrepancy and applies directly to ARMA models.

[Supplement] AIC versus BIC

For a true model M and $M' \supset M$ with one extra parameter, $\Lambda_n = 2\log(MLE') - 2\log(MLE) \xrightarrow{d} \chi^2_1$. AIC picks M' iff $\Lambda_n > 2$ (limit probability $P(\chi^2_1 > 2) \approx 0.157$); BIC iff $\Lambda_n > \log n$ (probability $\to 0$). If M is false and M' true, $\Lambda_n/n \to$ a positive KL divergence, so both reject M. Hence BIC is consistent, AIC overfits but is efficient for prediction.

4.3.3 Model diagnostics

An adequate model is used for forecasting; otherwise return to identification.

(1) Overfitting. For a tentative ARIMA(p,d,q), fit ARIMA$(p+1,d,q)$ and ARIMA$(p,d,q+1)$ and check:

(a) are the added parameters judged to be zero?

(b) are the estimates of the original parameters essentially unchanged?

E.g. for MA(1), fit ARMA(1,1) and MA(2): ϕ_1, θ_2 should be insignificant (asymptotically normal with mean 0 under the tentative model) and the three $\hat{\theta}_1$ should agree.

(2) Residual analysis. For the stationary ARMA(p,q) model

$$\phi(B)(X_t - \mu) = \theta(B)\epsilon_t, \tag{4.6}$$

the residuals

$$e_t = \hat{\theta}(B)^{-1}\hat{\phi}(B)(X_t - \hat{\mu}), \qquad t = 1, \ldots, n,$$

(with $X_t = 0$ for $t \le 0$) should behave like IID errors. Check:

(a) zero mean ($\bar{e}$ vs its standard error), (b) constant variance (residual plot), (c) independence, via

$$\rho_e(k) = \frac{\sum_{t=1}^{n-k}(e_t - \bar{e})(e_{t+k} - \bar{e})}{\sum_{t=1}^{n}(e_t - \bar{e})^2}.$$

Under H_0: "(4.6) is true", $\mathrm{Var}(\rho_e(k)) \approx 1/n$; reject when $|\rho_e(k)| \ge z_{\alpha/2}\,\mathrm{S.E.}(\rho_e(k))$. An ARMA fit to the residuals should give ARMA(0,0).

Lag-by-lag tests inflate the overall type I error; the simultaneous *portmanteau* (*Ljung–Box*) statistic

$$Q = n(n+2)\sum_{h=1}^{K}\frac{\rho_e^2(h)}{n-h}$$

is approximately $\chi^2_{K-p-q-1}$ under H_0; reject when Q exceeds its upper $100\alpha\%$ point.

The *Durbin–Watson* statistic tests lag-one correlation against AR(1) errors:

$$DW = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \approx 2(1 - \rho_e(1)),$$

as the numerator is $\sum_{t\ge2} e_t^2 + \sum_{t\ge2} e_{t-1}^2 - 2\sum_{t\ge2} e_t e_{t-1} \approx 2\sum e_t^2 - 2\sum e_t e_{t-1}$. So $0 \le DW \le 4$; $DW \approx 2$: none, < 2: positive, > 2: negative correlation (tables: Durbin and Watson, 1971).

4.4 Selected exercises

Motivation. Exercises on differencing trends, the random walk, Brownian motion and over-differencing.

Exercise 4.1 (Differencing a polynomial trend)

Let $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ with $r \ge 1$, $a_r \ne 0$, $\{\epsilon_t\} \sim WN(0, \sigma^2)$. Show that
(a) $(1-B)^r t^r = r!$;
(b) $(1-B)^r \epsilon_t = \sum_{j=0}^{r} \binom{r}{j}(-1)^j \epsilon_{t-j}$;
(c) $(1-B)^r X_t$ is stationary, and find its mean and variance.

Solution.

(a)

Step 1. For degree $k \geq 1$, by the binomial theorem

$$(1-B)t^k = t^k - (t-1)^k = -\sum_{i=0}^{k-1} \binom{k}{i}(-1)^{k-i}t^i = k\,t^{k-1} + (\text{terms of degree} \leq k-2),$$

so $(1-B)$ maps degree k, leading coefficient c, to degree $k-1$, leading coefficient kc; $(1-B)c = 0$.

Step 2. By induction, $(1-B)^j t^r$ has degree $r-j$ and leading coefficient $r(r-1)\cdots(r-j+1)$; $j = r$ gives $r!$.

(b) 1 and B commute, so $(1-B)^r = \sum_{j=0}^{r}\binom{r}{j}(-1)^j B^j$; apply to ϵ_t.

(c)

Step 1. For $k < r$, $(1-B)^r t^k = (1-B)^{r-k}k! = 0$. By (a),(b),

$$Z_t := (1-B)^r X_t = a_r\, r! + \sum_{j=0}^{r} c_j \epsilon_{t-j}, \qquad c_j = \binom{r}{j}(-1)^j.$$

Step 2. As in (2.15), $\gamma_Z(h) = \sigma^2 \sum_{j=0}^{r-|h|} c_j c_{j+|h|}$ for $|h| \leq r$, 0 otherwise: $\{Z_t\}$ is stationary with $EZ_t = a_r\, r!$.

Step 3. By Vandermonde,

$$\mathrm{Var}(Z_t) = \sigma^2 \sum_{j=0}^{r}\binom{r}{j}^2 = \binom{2r}{r}\sigma^2.$$

■

Exercise 4.2 (Markov property of the random walk)

Let $X_t = X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$ with values in a countable set. Show that, whenever the conditioning events have positive probability,

$$P(X_n = i_n \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}) = P(X_n = i_n \mid X_{n-1} = i_{n-1}).$$

Solution.

Step 1. $X_0, \ldots, X_{n-1}$ are functions of $\epsilon_1, \ldots, \epsilon_{n-1}$, hence independent of ϵ_n; so are $A = \{X_0 = i_0, \ldots, X_{n-1} = i_{n-1}\}$ and $A' = \{X_{n-1} = i_{n-1}\}$.

Step 2. On A, $X_n = i_{n-1} + \epsilon_n$, so

$$P(X_n = i_n, A) = P(\epsilon_n = i_n - i_{n-1}, A) = P(\epsilon_n = i_n - i_{n-1})P(A),$$

i.e. $P(X_n = i_n \mid A) = P(\epsilon_1 = i_n - i_{n-1})$.

Step 3. Likewise $P(X_n = i_n \mid A') = P(\epsilon_1 = i_n - i_{n-1})$.

For general ϵ_t with d.f. F, the freezing lemma gives $P(X_n \leq x \mid X_0, \ldots, X_{n-1}) = F(x - X_{n-1}) = P(X_n \leq x \mid X_{n-1})$. ■

Exercise 4.3 (Quadratic trend plus a local linear trend)

Let

$$X_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \mu_t + \epsilon_t, \qquad \mu_t = \mu_{t-1} + \beta_t + \eta_t, \qquad \beta_t = \beta_{t-1} + \zeta_t,$$

where $\{\epsilon_t\} \sim WN(0, \sigma_\epsilon^2)$, $\{\eta_t\} \sim WN(0, \sigma_\eta^2)$, $\{\zeta_t\} \sim WN(0, \sigma_\zeta^2)$ are mutually independent. Difference $\{X_t\}$ to obtain a stationary series and find its mean and variance.

Solution.

Step 1. $(1-B)\beta_t = \zeta_t$ and $(1-B)\mu_t = \beta_t + \eta_t$, so

$$(1-B)^2\mu_t = (1-B)\beta_t + (1-B)\eta_t = \zeta_t + \eta_t - \eta_{t-1}.$$

Step 2. By Exercise 4.1, $(1-B)^2(\alpha_0 + \alpha_1 t + \alpha_2 t^2) = 2\alpha_2$, hence

$$D_t := (1-B)^2 X_t = 2\alpha_2 + \zeta_t + (\eta_t - \eta_{t-1}) + (\epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}).$$

Step 3. The three uncorrelated finite MAs have ACVFs (2.15) (coefficients (1), $(1,-1)$, $(1,-2,1)$); by Exercise 1.7, $\{D_t\}$ is stationary with

$$\gamma_D(0) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2, \quad \gamma_D(\pm 1) = -\sigma_\eta^2 - 4\sigma_\epsilon^2, \quad \gamma_D(\pm 2) = \sigma_\epsilon^2,$$

and $\gamma_D(h) = 0$ for $|h| \geq 3$.

Step 4. Hence $ED_t = 2\alpha_2$ and $\mathrm{Var}(D_t) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2$.

One difference is not enough: $(1-B)X_t = \alpha_1 - \alpha_2 + 2\alpha_2 t + \beta_t + \eta_t + \epsilon_t - \epsilon_{t-1}$ has a linear mean and contains the random walk β_t. ■

Exercise 4.8 (Properties of Brownian motion)

Let $\{W(s) : s \geq 0\}$ be a standard Brownian motion (Definition 4.2.1). Show that
(a) $EW(t)W(s) = t \wedge s$;
(b) $Z(s) = W(\eta s)/\sqrt{\eta}$ is a Brownian motion for every $\eta > 0$;
(c) $G(s) = sW(1/s)$ for $s > 0$, $G(0) = 0$, is a Brownian motion.

Solution.

(a) For $s \leq t$, $W(s) \perp W(t) - W(s)$, both mean 0, $EW(s)^2 = s$: $EW(t)W(s) = EW(s)^2 + E[W(s)(W(t) - W(s))] = s$.

Characterization. A process with continuous paths, $G(0) = 0$, and jointly normal mean-zero finite-dimensional laws with $\mathrm{Cov}(G(s), G(t)) = s \wedge t$ is a Brownian motion; conversely W has these properties.

Step 1. For $t > s$, $G(t) - G(s) \sim N(0,\, t - 2(s \wedge t) + s) = N(0, t-s)$.

Step 2. For $t_0 < \cdots < t_k$ the increments Δ_i are jointly normal and, for $i < j$ $(t_i \leq t_{j-1})$,

$$\mathrm{Cov}(\Delta_i, \Delta_j) = t_i \wedge t_j - t_i \wedge t_{j-1} - t_{i-1} \wedge t_j + t_{i-1} \wedge t_{j-1} = t_i - t_i - t_{i-1} + t_{i-1} = 0,$$

hence independent (Exercise 1.6); $t_0 = 0$ gives (3).

Step 3. Conversely, $(W(t_1), \ldots, W(t_k))$ is a linear image of independent normal increments, with covariance $s \wedge t$ by (a).

(b) Z is continuous, $Z(0) = 0$, jointly normal, and $\mathrm{Cov}(Z(s), Z(t)) = (\eta s \wedge \eta t)/\eta = s \wedge t$.

(c)

Step 1. $\mathrm{Cov}(G(s), G(t)) = st\,(1/s \wedge 1/t) = s \wedge t$; laws are jointly normal; paths are continuous on $(0, \infty)$.

Step 2. Continuity at 0 means $W(u)/u \to 0$ a.s.; the SLLN gives $W(n)/n \to 0$.

Step 3. $M_k = \sup_{u \in [k, k+1]} |W(u) - W(k)|$ has the law of $M_0 = \sup_{u \leq 1} |W(u)|$ (same Gaussian covariance), and Doob's L^2 inequality gives $EM_0^2 \leq 4EW(1)^2 = 4$, so $EM_0 \leq 2$.

Step 4. $\sum_{k \geq 1} P(M_k > \varepsilon k) = \sum_{k \geq 1} P(M_0/\varepsilon > k) \leq E(M_0)/\varepsilon < \infty$; Borel–Cantelli gives $M_k/k \to 0$ a.s.

Step 5. For $u \in [k, k+1]$, $|W(u)|/u \le (|W(k)| + M_k)/k \to 0$, so $G(s) \to 0$ as $s \downarrow 0$; the characterization applies. ■

Exercise 4.9 (Over-differencing an MA(2))

Let $X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, be stationary, and suppose one fits $(1-B)X_t = Y_t$. Find the ACVF of $\{Y_t\}$.

Solution.

Step 1. $Y_t = \epsilon_t + (\theta_1 - 1)\epsilon_{t-1} + (\theta_2 - \theta_1)\epsilon_{t-2} - \theta_2\epsilon_{t-3}$, an MA(3) with $c_0 = 1$, $c_1 = \theta_1 - 1$, $c_2 = \theta_2 - \theta_1$, $c_3 = -\theta_2$.

Step 2. By Exercise 2.9(a) with $c = 1$, $\gamma_Y(h) = 2\gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1)$, with $\gamma_X(0) = (1 + \theta_1^2 + \theta_2^2)\sigma^2$, $\gamma_X(\pm 1) = (\theta_1 + \theta_1\theta_2)\sigma^2$, $\gamma_X(\pm 2) = \theta_2\sigma^2$:

$$\begin{aligned} \gamma_Y(0) &= 2\sigma^2(1 + \theta_1^2 + \theta_2^2 - \theta_1 - \theta_1\theta_2), \\ \gamma_Y(\pm 1) &= \sigma^2(2\theta_1 + 2\theta_1\theta_2 - \theta_2 - 1 - \theta_1^2 - \theta_2^2), \\ \gamma_Y(\pm 2) &= \sigma^2(2\theta_2 - \theta_1 - \theta_1\theta_2), \qquad \gamma_Y(\pm 3) = -\theta_2\sigma^2, \end{aligned}$$

and $\gamma_Y(h) = 0$ for $|h| \ge 4$ (agreeing with $\sigma^2 \sum_j c_j c_{j+|h|}$).

Step 3. The MA polynomial $(1 - z)\theta(z)$ vanishes at $z = 1$: over-differencing destroys invertibility. For $\theta_1 = \theta_2 = 0$, $\rho_Y(1) = -\frac{1}{2}$; a lag-one SACF near -0.5 warns of unnecessary differencing. ■

Chapter 5. Seasonal Time Series Models

Many series repeat a pattern with known period s ($s = 12$ monthly, $s = 4$ quarterly): *seasonality*. This chapter adapts ARIMA models to seasonal data.

5.1 Time series with a seasonal component

Motivation. A periodic component is not removed by ordinary differencing, but differencing at lag s removes it.

Suppose

$$X_t = S_t + Z_t, \tag{5.1}$$

with *seasonal component* $S_t = S_{t+s}$, $S_1, \dots, S_s$ independent, $ES_t = 0$, $\mathrm{Var}(S_t) = \sigma_t^2 < \infty$, and $\{Z_t\}$ stationary, mean 0, ACVF γ_Z, independent of $\{S_t\}$. Then

$$\begin{aligned} EX_{t+h}X_t &= E(S_{t+h} + Z_{t+h})(S_t + Z_t) = ES_{t+h}S_t + \gamma_Z(h) \\ &= \begin{cases} \sigma_t^2 + \gamma_Z(h), & h = 0, \pm s, \pm 2s, \dots, \\ \gamma_Z(h), & \text{otherwise.} \end{cases} \end{aligned}$$

This depends on t, and even with equal σ_t^2 it does not decay along $s, 2s, \dots$, so no causal ARMA fits $\{X_t\}$.

The *seasonal difference operator*

$$\nabla_s = 1 - B^s$$

gives (*seasonal differencing*)

$$Y_t = (1 - B^s)X_t = X_t - X_{t-s} = S_t + Z_t - (S_{t-s} + Z_{t-s}) = Z_t - Z_{t-s},$$

which is stationary. Fitting $\phi(B)Y_t = \theta(B)\epsilon_t$, $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$,

$$\phi(B)\nabla_s X_t = \theta(B)\epsilon_t. \tag{5.2}$$

Definition 5.1.1 (Seasonal difference model)

The model (5.2) is called the *seasonal difference model*.

5.2 Time series with trend and seasonality

Motivation. One seasonal difference removes seasonality and lowers the trend degree by one; ordinary differences remove the rest.

Let

$$X_t = T_t + S_t + Z_t,$$

with mutually independent trend, seasonal and stationary parts,

$$T_t = \beta_0 + \beta_1 t + \cdots + \beta_r t^r \quad (r \geq 1,\ \beta_r \neq 0), \qquad S_t = S_{t+s} \text{ for all } t.$$

Then $\nabla_s X_t = \nabla_s T_t + \nabla_s Z_t$ since $\nabla_s S_t = 0$.

Lemma (Removing a polynomial trend)

With T_t and Z_t as above,
(1) $(1-B)^{r-1}\nabla_s T_t = \mu$ for the constant $\mu = s\,r!\,\beta_r$;
(2) $(1-B)^{r-1}\nabla_s Z_t$ is stationary with mean 0.
Consequently

$$(1-B)^{r-1}\nabla_s X_t = (1-B)^{r-1}\nabla_s T_t + (1-B)^{r-1}\nabla_s Z_t = \mu + (1-B)^{r-1}\nabla_s Z_t. \tag{5.3}$$

Proof. [Supplement]

Step 1. For $P(t) = at^k + \cdots$, $k \geq 1$, and $m \geq 1$: $t^k - (t-m)^k = km\,t^{k-1} + \cdots$, so $P(t) - P(t-m)$ has degree $k-1$ and leading coefficient akm (0 if $k = 0$).

Step 2. With $m = s$ once and $m = 1$ another $r-1$ times, the leading coefficient becomes $\beta_r rs \cdot (r-1)! = s\,r!\,\beta_r$ at degree 0: (1).

Step 3. $(1-B)^{r-1}\nabla_s = \sum_{j=0}^{m} c_j B^j$, $m = r-1+s$, so $U_t = \sum_j c_j Z_{t-j}$ has $EU_t = 0$ and $\mathrm{Cov}(U_{t+h}, U_t) = \sum_{j,k} c_j c_k \gamma_Z(h-j+k)$, free of t: (2). Linearity gives (5.3). ■

Fitting ARMA to $(1-B)^{r-1}\nabla_s X_t - \mu$:

$$\phi(B)\{(1-B)^{r-1}\nabla_s X_t - \mu\} = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2). \tag{5.4}$$

Definition 5.2.1 (Seasonal difference model with trend)

The model (5.4) is called the *seasonal difference model with trend*.

For $r = 1$ one seasonal difference suffices; in practice ∇_s is often followed by one ∇.

5.3 Seasonal ARIMA models

Motivation. Stochastic seasonality, where an observation depends on the one a season earlier, is modelled by ARMA recursions in B^s.

The model

$$X_t = \phi X_{t-s} + \epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.5}$$

has, for $|\phi| < 1$, the stationary solution

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-si}. \tag{5.6}$$

It has no component S_t, yet $\gamma_X(h) = 0$ unless $s \mid h$: a seasonal ARMA (SARMA) effect. Allowing seasonal differencing gives SARIMA.

Definition 5.3.1 (SARIMA$(P, D, Q)_s$ model)

A time series $\{X_t\}$ follows a *seasonal ARIMA model* SARIMA$(P, D, Q)_s$ if

$$\Phi(B^s)\{(1-B^s)^D X_t - \mu\} = \Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.7}$$

where

$$\Phi(z) = 1 - \Phi_1 z - \cdots - \Phi_P z^P, \qquad \Theta(z) = 1 + \Theta_1 z + \cdots + \Theta_Q z^Q$$

satisfy the causality and invertibility conditions. Usually $\mu = 0$ is assumed.

It suits data correlated only at the same seasonal position; $D \neq 0$ handles a seasonal trend. Models (5.4) and (5.7) differ; the data decide.

Example 5.3.2 (ACF of seasonal MA and AR models)

(1) For MA$(2)_s$, $X_t = \epsilon_t + \theta_1\epsilon_{t-s} + \theta_2\epsilon_{t-2s}$, the MA(2) ACVF sits on lags $0, s, 2s$:

$$\gamma_X(h) = \begin{cases} (1+\theta_1^2+\theta_2^2)\sigma^2, & h = 0, \\ (\theta_1 + \theta_1\theta_2)\sigma^2, & h = s, \\ \theta_2\sigma^2, & h = 2s, \\ 0, & \text{otherwise } (h \geq 0). \end{cases}$$

(2) For AR$(1)_s$, $X_t - \phi X_{t-s} = \epsilon_t$, $|\phi| < 1$, (5.6) gives $\gamma_X(sj) = \sigma^2 \sum_{i\geq 0} \phi^i\phi^{i+j} = \phi^j\sigma^2/(1-\phi^2)$, $\gamma_X(h) = 0$ if $s \nmid h$:

$$\rho(h) = \begin{cases} \phi^{h/s}, & h = 0, s, 2s, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

5.4 Multiplicative seasonal models

Motivation. Multiplying a seasonal operator in B^s by an ordinary one in B captures both within- and across-season dependence with few parameters.

Definition 5.4.1 (Multiplicative seasonal ARIMA model)

A time series $\{X_t\}$ follows a *multiplicative seasonal* ARIMA$(p,d,q)(P,D,Q)_s$ model if

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2),$$

with ϕ, θ as in Definition 4.1.1 and Φ, Θ as in Definition 5.3.1.

It contains Definitions 5.1.1, 5.2.1, 5.3.1; its ACF shows a short-lag pattern (ϕ, θ) repeated around $s, 2s, \ldots$ (Φ, Θ).

Example 5.4.2 (The ARIMA$(0,1,1)(0,1,1)_s$ model)

The model $\nabla_s(1-B)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ reads

$$X_t - X_{t-1} - X_{t-s} + X_{t-s-1} = \epsilon_t + \theta\epsilon_{t-1} + \Theta\epsilon_{t-s} + \theta\Theta\epsilon_{t-s-1}.$$

$W_t = \nabla_s(1-B)X_t$ is the stationary ARIMA$(0,0,1)(0,0,1)_s$. For $s \geq 3$,

$$\gamma_W(0) = (1+\theta^2)(1+\Theta^2)\sigma^2, \qquad \gamma_W(1) = \theta(1+\Theta^2)\sigma^2,$$
$$\gamma_W(s) = \Theta(1+\theta^2)\sigma^2, \qquad \gamma_W(s-1) = \gamma_W(s+1) = \theta\Theta\sigma^2,$$

and $\gamma_W(h) = 0$ for $2 \leq h \leq s-2$ and for $h \geq s+2$ (Figure 3).

Example 5.4.3 (The ARIMA$(0,0,1)(1,0,1)_s$ model)

Let $(1-\Phi B^s)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$, $|\Phi| < 1$, $s \geq 4$, and $X_t = (1+\theta B)Y_t$ with

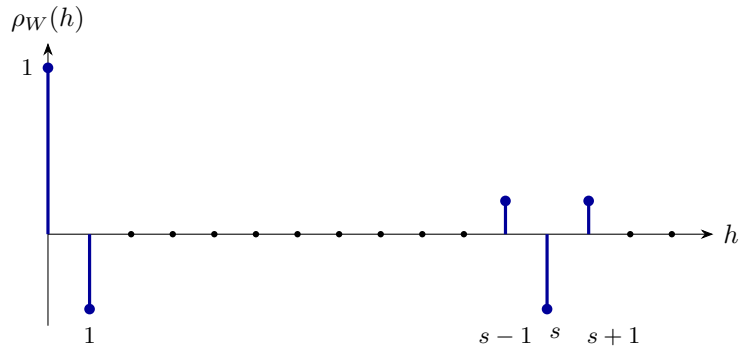


Figure 3: ACF of $W_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ ($\theta, \Theta < 0$, $s = 12$): nonzero only at lags $0, 1, s-1, s, s+1$.

SARMA$(1,1)_s$ $Y_t = \Phi Y_{t-s} + \epsilon_t + \Theta\epsilon_{t-s}$. Using $\mathrm{Cov}(Y_t, \epsilon_t) = \sigma^2$, $\mathrm{Cov}(Y_t, \epsilon_{t-s}) = (\Phi+\Theta)\sigma^2$,

$$\gamma_Y(0) = \Phi^2\gamma_Y(0) + (1+\Theta^2)\sigma^2 + 2\Phi\Theta\sigma^2 \implies \gamma_Y(0) = \Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_Y(s) = \Phi\gamma_Y(0) + \Theta\sigma^2 = \Big\{\Theta + \Phi + \frac{\Phi(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$

$\gamma_Y(ks) = \Phi^{k-1}\gamma_Y(s)$, $k \geq 1$, $\gamma_Y(h) = 0$ if $s \nmid h$. With $\gamma_X(h) = (1+\theta^2)\gamma_Y(h) + \theta\{\gamma_Y(h-1) + \gamma_Y(h+1)\}$,

$$\gamma_X(0) = (1+\theta^2)\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2, \qquad \gamma_X(1) = \theta\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_X(s) = (1+\theta^2)\gamma_Y(s), \qquad \gamma_X(s-1) = \gamma_X(s+1) = \theta\gamma_Y(s),$$

$\gamma_X(2) = \cdots = \gamma_X(s-2) = 0$, $\gamma_X(h) = \Phi\gamma_X(h-s)$ for $h \geq s+2$: the pattern around s recurs around $2s, 3s, \ldots$, damped by Φ.

5.5 Selected exercises

Motivation. Exercises on ψ-weights and autocovariances of seasonal models, and seasonal versus unrestricted MA.

Exercise 5.2 (ψ-weights of a seasonal model)

Let $(1-\Phi B^s)X_t = (1+\theta B)\epsilon_t$ with $|\Phi| < 1$ and $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$. Find the coefficients a_j in the representation $X_t = \sum_{j=0}^{\infty} a_j\epsilon_{t-j}$ of the stationary solution.

Solution.

Step 1. $1 - \Phi z^s$ vanishes only on $|z| = |\Phi|^{-1/s} > 1$, so by Proposition 2.4.A the unique stationary solution has $a(z) = (1+\theta z)/(1-\Phi z^s)$.

Step 2. For $|z| < |\Phi|^{-1/s}$,

$$a(z) = \sum_{i=0}^{\infty} \Phi^i z^{si} + \theta\sum_{i=0}^{\infty} \Phi^i z^{si+1}.$$

Step 3: $s \geq 2$. The exponents si, $si'+1$ never coincide, so

$$a_{si} = \Phi^i, \qquad a_{si+1} = \theta\Phi^i \quad (i = 0, 1, 2, \ldots), \qquad a_j = 0 \text{ if } j \bmod s \notin \{0, 1\},$$

i.e. $X_t = \sum_{i\geq 0} \Phi^i(\epsilon_{t-si} + \theta\epsilon_{t-si-1})$.

Step 4: $s = 1$. ARMA(1,1): $a_0 = 1$, $a_j = \Phi^{j-1}(\Phi + \theta)$ for $j \geq 1$.

Step 5: check. $a_j - \Phi a_{j-s} = \mathbb{1}\{j = 0\} + \theta\,\mathbb{1}\{j = 1\}$, the coefficient form of $(1 - \Phi z^s)a(z) = 1 + \theta z$. ■

Exercise 5.3 (MA$(1)_{12}$ versus MA(12))

Can the MA$(1)_{12}$ model $X_t = \epsilon_t + \Theta\epsilon_{t-12}$ be regarded as an MA(12) model? Which of the two formulations is preferable, and why?

Solution.

Step 1: it is an MA(12) model. with $\theta_1 = \cdots = \theta_{11} = 0$, $\theta_{12} = \Theta$, and

$$\gamma(0) = (1 + \Theta^2)\sigma^2, \qquad \gamma(12) = \Theta\sigma^2, \qquad \gamma(h) = 0 \text{ for } h \geq 1,\ h \neq 12,$$

since equal-time noise products occur only for $h \in \{0, \pm 12\}$.

Step 2: the same invertibility condition. $1 + \Theta z^{12} = 0$ iff $|z| = |\Theta|^{-1/12}$, outside the unit circle iff $|\Theta| < 1$, the MA$(1)_{12}$ condition.

Step 3: which to prefer. The unrestricted MA(12) estimates eleven zero coefficients, adding $O(n^{-1/2})$ error to fitted ACVFs and forecasts without improving fit (LR statistic $\approx \chi^2(11)$; AIC charges 22, BIC $11 \log n$). By parsimony MA$(1)_{12}$ is preferable and states the structure directly. Only genuine SACF correlation at lags $1, \ldots, 11$ argues otherwise, and then a multiplicative model (Example 5.4.2) is tried first. ■

Exercise 5.4 (ACF of ARIMA$(1,0,1)(0,0,1)_s$)

The book asks for the autocovariance functions of four seasonal models; parts (1), (2) and (4) are worked out in Examples 5.4.3 (the process Y_t there), 5.4.2 and 5.4.3. Part (3): let $s \geq 2$, $|\phi| < 1$ and

$$(1 - \phi B)X_t = (1 + \theta B)(1 + \Theta B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2).$$

Find the autocovariance function of the stationary solution.

Solution.

Step 1: reduction to ARMA(1,1). Let $(1-\phi B)Y_t = (1+\theta B)\epsilon_t$ (causal) and $X_t = Y_t + \Theta Y_{t-s}$. Then

$$(1 - \phi B)X_t = (1 - \phi B)Y_t + \Theta(1 - \phi B)Y_{t-s} = (1 + \theta B)\epsilon_t + \Theta(1 + \theta B)\epsilon_{t-s} = (1 + \theta B)(1 + \Theta B^s)\epsilon_t,$$

and X is stationary, the unique stationary solution by Proposition 2.4.A.

Step 2: ACVF of Y. By Example 2.4.3, with $c = \gamma_Y(1)$,

$$\gamma_Y(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad c = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}, \qquad \gamma_Y(h) = c\,\phi^{|h|-1}\ (h \neq 0).$$

Step 3: general formula. By bilinearity,

$$\begin{aligned}\gamma_X(h) &= \text{Cov}(Y_{t+h} + \Theta Y_{t+h-s},\ Y_t + \Theta Y_{t-s}) \\ &= (1 + \Theta^2)\gamma_Y(h) + \Theta\{\gamma_Y(h - s) + \gamma_Y(h + s)\}.\end{aligned}$$

Step 4: explicit values. For $h \geq 0$:

$$\gamma_X(0) = (1 + \Theta^2)\gamma_Y(0) + 2\Theta c\,\phi^{s-1},$$

$$\gamma_X(h) = c\{(1+\Theta^2)\phi^{h-1} + \Theta(\phi^{s-h-1} + \phi^{s+h-1})\}, \quad 1 \le h \le s-1,$$
$$\gamma_X(s) = (1+\Theta^2)c\,\phi^{s-1} + \Theta\{\gamma_Y(0) + c\,\phi^{2s-1}\},$$
$$\gamma_X(h) = c\{(1+\Theta^2)\phi^{h-1} + \Theta(\phi^{h-s-1} + \phi^{h+s-1})\}, \quad h \ge s+1.$$

So $\gamma_X(h) = \phi\,\gamma_X(h-1)$ for $h \ge s+2$, with a seasonal spike $\Theta\gamma_Y(0)$ at $h = s$; $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$. ■

Chapter 6. Conditional Heteroscedasticity Models

6.1 Introduction

Motivation. Financial returns are nearly uncorrelated, yet their size is predictable, so the conditional variance (*volatility*) becomes the object of inference.

Volatility clustering means large moves come in bursts, i.e. the conditional variance depends on the past. For AR(1) $y_t = \phi y_{t-1} + \epsilon_t$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$,

$$E(y_t \mid y_{t-1}) = \phi y_{t-1}, \qquad \mathrm{Var}(y_t \mid y_{t-1}) = \sigma^2,$$

constant. With $y_t = 0.1 y_{t-1} + \epsilon_t$, $\epsilon_t \mid \epsilon_{t-1} \sim N(0,\, 1 + 0.5\epsilon_{t-1}^2)$, a large shock raises the next variance (Figure 4).

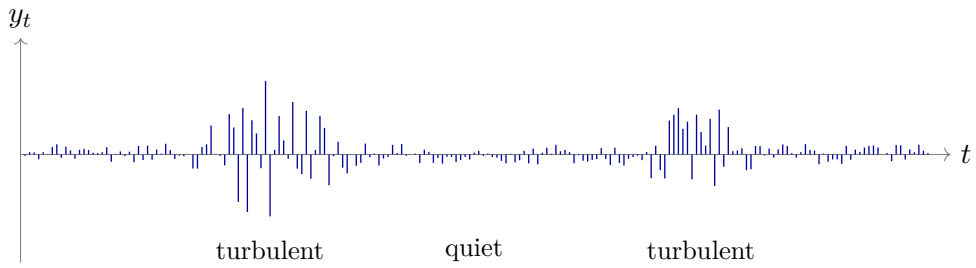


Figure 4: Volatility clustering: large moves come in bursts.

6.2 ARCH models

Motivation. ARCH makes the conditional variance a function of past squares, giving uncorrelated heavy-tailed data whose squares are AR.

With $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ (the book's I_{t-1}), we model

$$y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \qquad \sigma_t^2 = \sigma_t^2(y_{t-1}, y_{t-2}, \dots), \tag{6.1}$$

often with finitely many lags $y_{t-1}, \dots, y_{t-q}$.

6.2.1 ARCH(1)

Let $\{\xi_t\} \sim IID\, N(0,1)$ and

$$y_t = \sigma_t \xi_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2, \qquad \alpha_0 > 0,\ 0 \le \alpha_1 < 1, \tag{6.2}$$

with ξ_t independent of σ_t. Then

$$P(y_t \le x \mid y_{t-1} = y) = P(\xi_t \le x/\sqrt{\alpha_0 + \alpha_1 y^2}) = \Phi(x/\sqrt{\alpha_0 + \alpha_1 y^2}), \tag{6.3}$$

so $y_t \mid y_{t-1}, y_{t-2}, \cdots \sim N(0, \sigma_t^2)$: the *ARCH(1) model*, an instance of (6.1). Often only $\xi_t \sim IID(0,1)$ is assumed.

Iterating $y_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2$,

$$y_t^2 = (\alpha_0 + \alpha_1(\alpha_0 + \alpha_1 y_{t-2}^2)\xi_{t-1}^2)\xi_t^2 = \cdots = \alpha_0 \sum_{j=0}^{n} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2 + \alpha_1^{n+1} \xi_t^2 \cdots \xi_{t-n}^2 y_{t-n-1}^2, \tag{6.4}$$

which suggests

$$y_t^2 = \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2, \qquad \sigma_t^2 = \alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2, \tag{6.5}$$

i.e. the candidate solution

$$y_t = \xi_t \Big(\alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2 \Big)^{1/2}, \tag{6.6}$$

a nonlinear function of $\xi_t, \xi_{t-1}, \ldots$.

Proposition (Properties of ARCH(1), [Supplement])

Let $\alpha_0 > 0$, $0 \le \alpha_1 < 1$ and $\{\xi_t\} \sim IID(0,1)$.
(1) The series in (6.6) converges a.s.; $\{y_t\}$ is a strictly stationary solution of (6.2), σ_t^2 is a function of $\xi_{t-1}, \xi_{t-2}, \ldots$, and (σ_t^2, y_t) is independent of $\xi_{t+1}, \xi_{t+2}, \ldots$.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \mathrm{Var}(y_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and

$$Ey_t^2 = E\sigma_t^2 = \frac{\alpha_0}{1-\alpha_1}; \tag{6.7}$$

$\{y_t\}$ is a stationary white noise.
(3) If $\xi_t \sim N(0,1)$, then $Ey_t^4 < \infty$ if and only if $3\alpha_1^2 < 1$, and then the kurtosis is

$$\frac{Ey_t^4}{(Ey_t^2)^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3 \quad (\alpha_1 > 0). \tag{6.8}$$

(4) If $Ey_t^4 < \infty$, then $\nu_t = y_t^2 - \sigma_t^2$ is a white noise and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t, \tag{6.9}$$

so $\{y_t^2\}$ is an AR(1) process.

Proof.

Step 1. Let $h_t = \alpha_0 + \alpha_0\alpha_1 \sum_{j\ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2 \ge 0$. As $E(\xi_{t-1}^2 \cdots \xi_{t-j-1}^2) = 1$, $Eh_t = \alpha_0/(1-\alpha_1) < \infty$, so the series converges a.s., and

$$\alpha_0 + \alpha_1 \xi_{t-1}^2 h_{t-1} = \alpha_0 + \alpha_0\alpha_1 \xi_{t-1}^2 + \alpha_0 \sum_{j\ge 0} \alpha_1^{j+2} \xi_{t-1}^2 \xi_{t-2}^2 \cdots \xi_{t-j-2}^2 = h_t,$$

so $\sigma_t^2 = h_t$, $y_t = \sigma_t \xi_t$ satisfy $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$. Being a fixed function of the shifted i.i.d. sequence, $\{y_t\}$ is strictly stationary; independence is immediate.

Step 2. $\sigma_t = (\alpha_0 + \alpha_1 y_{t-1}^2)^{1/2}$ is $\mathcal{F}_{t-1}$-measurable and $\xi_t \perp \mathcal{F}_{t-1} \subset \sigma(\xi_{t-1}, \ldots)$, so $E(y_t \mid \mathcal{F}_{t-1}) = \sigma_t E\xi_t = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and $Ey_t^2 = Eh_t = \alpha_0/(1-\alpha_1)$. For $h \ge 1$, $E(y_t y_{t+h}) = E(y_t\, E(y_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$.

Step 3. $y_t^2 = \alpha_0 \sum_{j\ge 0} \alpha_1^j P_j$, $P_j = \xi_t^2 \cdots \xi_{t-j}^2$, $\|P_j\|_2 = 3^{(j+1)/2}$. If $\sqrt{3}\,\alpha_1 < 1$, Minkowski gives $\|y_t^2\|_2 \le \alpha_0 \sum_j \alpha_1^j 3^{(j+1)/2} < \infty$. Conversely, if $Ey_t^4 < \infty$, $m_4 := E\sigma_t^4$ is finite and constant, and with $Ey_{t-1}^4 = 3m_4$,

$$m_4 = E(\alpha_0 + \alpha_1 y_{t-1}^2)^2 = \alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1-\alpha_1} + 3\alpha_1^2 m_4, \quad \text{so} \quad (1-3\alpha_1^2)m_4 = \alpha_0^2 \frac{1+\alpha_1}{1-\alpha_1} > 0,$$

forcing $3\alpha_1^2 < 1$. Then $Ey_t^4 = 3m_4$ and

$$\frac{Ey_t^4}{(Ey_t^2)^2} = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)} \cdot \frac{(1-\alpha_1)^2}{\alpha_0^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3 \quad (\alpha_1 > 0).$$

Step 4. $\nu_t = \sigma_t^2(\xi_t^2 - 1)$ has $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E\nu_t^2 = E\sigma_t^4\, E(\xi_t^2-1)^2 < \infty$; a stationary square-integrable martingale difference is white noise ($E\nu_t\nu_{t+h} = E(\nu_t E(\nu_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$). Finally $y_t^2 = \sigma_t^2 + \nu_t = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t$. ■

Parts (1)–(2) need no normality. By (3) the marginal law is *leptokurtic*; e.g. 10,000 draws from $\sigma_t^2 = 0.2 + 0.8y_{t-1}^2$ give a histogram far more peaked and heavy-tailed than the normal.

Definition 6.2.1 (ARCH(q) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2, \tag{6.10}$$

with $\alpha_0 > 0$ and $\alpha_i \geq 0$ $(1 \leq i \leq q)$, then $\{y_t\}$ follows an *ARCH(q) model*. If moreover $1 - \alpha_1 z - \cdots - \alpha_q z^q \neq 0$ for all $|z| \leq 1$ (equivalently $\sum_{i=1}^q \alpha_i < 1$), then $\{y_t\}$ is strictly stationary and

$$\mathrm{Var}(y_t) = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i}. \tag{6.11}$$

Proof of the stationarity statement. [Supplement] Let $S = \sum_{i=1}^q \alpha_i$, $A(z) = \sum_i \alpha_i z^i$.

Step 1: the root condition is $S < 1$. If $S < 1$, $|A(z)| \leq S < 1$ on $|z| \leq 1$. If $S \geq 1$, $A(0) = 0$, $A(1) \geq 1$ give $A(z) = 1$ for some $z \in (0,1]$.

Step 2: construction. Let $h_t^{(0)} = 0$, $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^q \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)}$. By induction $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \dots) \geq 0$ is nondecreasing in m (nonnegative coefficients). As $h_{t-i}^{(m)} \perp \xi_{t-i}$, $Eh_t^{(m+1)} = \alpha_0 + S\, Eh_t^{(m)}$, so $Eh_t^{(m)} \leq \alpha_0/(1-S)$.

Step 3: limit. $\sigma_t^2 = \lim_m h_t^{(m)}$ has $E\sigma_t^2 = \alpha_0/(1-S)$ (monotone convergence) and satisfies $\sigma_t^2 = \alpha_0 + \sum_i \alpha_i \xi_{t-i}^2 \sigma_{t-i}^2$; with $y_t = \sigma_t \xi_t$ this is (6.10), strictly stationary as in Proposition , and $\mathrm{Var}(y_t) = E\sigma_t^2 E\xi_t^2$ gives (6.11).

Step 4: uniqueness. For another stationary solution $\tilde\sigma_t^2$ (function of past ξ, $E\tilde\sigma_t^2 < \infty$), $D_t = \sigma_t^2 - \tilde\sigma_t^2 = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i}$ gives $E|D_0| \leq S\, E|D_0|$, so $D_t = 0$ a.s. ■

As for ARCH(1), $\nu_t = y_t^2 - \sigma_t^2$ is a martingale difference and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2 + \nu_t,$$

so choosing q is choosing an AR order for $\{y_t^2\}$ (Chapter 4).

6.3 GARCH models

Motivation. Adding lagged conditional variances, as ARMA adds lagged noise to AR, gives a parsimonious alternative to high-order ARCH.

Definition 6.3.1 (GARCH(1,1) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_0 > 0,\ \alpha_1, \beta_1 \geq 0, \tag{6.12}$$

then $\{y_t\}$ follows a *GARCH(1,1) model*. If $\alpha_1 + \beta_1 < 1$, $\{y_t\}$ is (second-order) stationary; and if

$$E\log(\beta_1 + \alpha_1 \xi_t^2) < 0,$$

then $\{(y_t, \sigma_t)\}$ is strictly stationary (Nelson, 1990; Bougerol and Picard, 1992).

Proof of the stationarity statements. [Supplement] With $a(x) = \beta_1 + \alpha_1 x^2$, $\sigma_t^2 = \alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2$, so

$$\sigma_t^2 = \alpha_0\Big[1 + \sum_{k=1}^{K}\prod_{i=1}^{k} a(\xi_{t-i})\Big] + \prod_{i=1}^{K+1} a(\xi_{t-i})\,\sigma_{t-K-1}^2. \tag{6.13}$$

Step 1: strict stationarity. Let $\gamma = E\log a(\xi_t) < 0$ ($\le \log Ea < \infty$; if $\gamma = -\infty$ use $\max(\log a, -M)$). By the SLLN $\left(\prod_{i=1}^{k} a(\xi_{t-i})\right)^{1/k} \to e^{\gamma} < 1$, so by the root test

$$\sigma_t^2 := \alpha_0\Big[1 + \sum_{k=1}^{\infty}\prod_{i=1}^{k} a(\xi_{t-i})\Big]$$

converges a.s.; it satisfies $\alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2 = \sigma_t^2$, and (y_t, σ_t), a fixed function of the shifted i.i.d. sequence, is strictly stationary. Uniqueness: for another such $\tilde\sigma_t^2$, (6.13) gives $\tilde\sigma_t^2 - \sigma_t^2 = \prod_{i=1}^{K} a(\xi_{t-i})\,(\tilde\sigma_{t-K}^2 - \sigma_{t-K}^2)$; the product $\to 0$ a.s. and the second factor is tight, so the (K-free) left side is 0.

Step 2: second-order stationarity. If $\alpha_1 + \beta_1 < 1$, Jensen gives $E\log a(\xi_t) \le \log(\alpha_1 + \beta_1) < 0$, so Step 1 applies, and $E\prod_{i=1}^{k} a(\xi_{t-i}) = (\alpha_1 + \beta_1)^k$ gives

$$Ey_t^2 = E\sigma_t^2 = \alpha_0 \sum_{k\ge 0}(\alpha_1 + \beta_1)^k = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} < \infty;$$

with $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $\{y_t\}$ is stationary white noise. ∎

If $\alpha_1 + \beta_1 = 1$ and ξ_t^2 is nondegenerate, strict Jensen gives $E\log(\beta_1 + \alpha_1\xi_t^2) < 0$: strictly stationary with $Ey_t^2 = \infty$.

Fits to daily returns (e.g. $r_t = 100\log(P_{t+1}/P_t)$) typically give $\alpha_1 + \beta_1$ just below 1. $\alpha_1 + \beta_1 = 1$ is the *IGARCH model*; for $\alpha_1 + \beta_1 \approx 1$ the variance is *persistent*:

Proposition (Variance forecasts, [Supplement])

In the GARCH(1,1) model let $\phi = \alpha_1 + \beta_1$. For $k \ge 0$,

$$E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 \sum_{j=0}^{k-1}\phi^j + \phi^k\sigma_t^2, \tag{6.14}$$

which has the form $c + \phi^k\sigma_t^2$. If $\phi \approx 1$, the effect of today's variance on the forecast barely decays as k grows.

Proof. For $s \ge t$, as $\sigma_s^2 \in \mathcal{F}_{s-1}$ and $E(y_s^2 \mid \mathcal{F}_{s-1}) = \sigma_s^2$,

$$E(\sigma_{s+1}^2 \mid \mathcal{F}_{s-1}) = \alpha_0 + \alpha_1 E(y_s^2 \mid \mathcal{F}_{s-1}) + \beta_1\sigma_s^2 = \alpha_0 + \phi\sigma_s^2.$$

Induction on k with the tower property: $E(\sigma_{t+k+1}^2 \mid \mathcal{F}_{t-1}) = \alpha_0 + \phi E(\sigma_{t+k}^2 \mid \mathcal{F}_{t-1}) = \alpha_0\sum_{j=0}^{k}\phi^j + \phi^{k+1}\sigma_t^2$. ∎

6.3.1 Asymmetric models

Returns react more strongly to bad news.

Definition (Threshold GARCH(1,1))

Let $x^+ = \max\{0, x\}$ and $x^- = \max\{-x, 0\}$. If $y_t = \sigma_t \xi_t$ with $\{\xi_t\} \sim IID(0,1)$ and

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(y_{t-1}^+)^2 + \alpha_{12}(y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0,$$

then $\{y_t\}$ follows a *threshold* (or *asymmetric*) *GARCH(1,1) model*. Equivalently, $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \gamma_1 (y_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2$ with $\alpha_1 = \alpha_{11}$, $\gamma_1 = \alpha_{12} - \alpha_{11}$.

$\alpha_{12} > \alpha_{11}$ (typical for stocks) means negative returns raise the next variance more.

6.3.2 Gaussian quasi-maximum likelihood

For $\xi_t \sim N(0,1)$, initial values y_0, σ_0^2 (commonly y_1^2) and $\theta = (\alpha_0, \alpha_1, \beta_1)^T$, compute

$$\sigma_t^{2*}(\theta) = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^{2*}(\theta), \qquad t = 1, \dots, n.$$

With $L(\theta) = \prod_{t=1}^n f(y_t \mid y_{t-1}, \dots)$,

$$f(y_t \mid y_{t-1}, \dots) = (2\pi\sigma_t^{2*})^{-1/2} \exp\{-y_t^2/(2\sigma_t^{2*})\},$$

the *Gaussian MLE* $\hat\theta$ maximizes

$$l(\theta) = \log L(\theta) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{t=1}^n \Big\{\log \sigma_t^{2*}(\theta) + \frac{y_t^2}{\sigma_t^{2*}(\theta)}\Big\}. \tag{6.15}$$

For non-normal ξ_t it is the *quasi-MLE* (QMLE), still consistent and asymptotically normal.

Proposition (Asymptotic normality of the QMLE, [Supplement])

Let $\{y_t\}$ be a strictly stationary GARCH(1,1) series with true parameter θ_0 in the interior of a compact parameter set, $\kappa_\xi = E\xi_t^4 < \infty$, and suppose $\hat\theta \to \theta_0$ in probability. Let $D_t = \sigma_t^{-2}\, \partial\sigma_t^2/\partial\theta$ at θ_0 and

$$J = E(D_t D_t^T) = -2\,E\Big[\frac{\partial^2 \log f(y_t \mid y_{t-1}, \dots)}{\partial\theta\, \partial\theta^T}\Big],$$

assumed positive definite. Then

$$\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, \Gamma), \qquad \Gamma = (\kappa_\xi - 1)J^{-1}$$

(Francq and Zakoian, 2004). For normal ξ_t, $\kappa_\xi = 3$ and $\Gamma = 2J^{-1}$.

Proof. [Supplement] Taken from Francq and Zakoian: initial values are negligible ($\sigma_t^{2*} - \sigma_t^2 = O(\beta_1^t)$), and D_t and second derivatives of $\log\sigma_t^2$ have uniformly bounded moments near θ_0. Let $l_t(\theta) = -\frac{1}{2}\{\log\sigma_t^2 + y_t^2/\sigma_t^2\}$.

Step 1: score. $\partial l_t/\partial\theta = \frac{1}{2}(y_t^2/\sigma_t^2 - 1)\sigma_t^{-2}\partial\sigma_t^2/\partial\theta$, at θ_0 equal to $s_t = \frac{1}{2}(\xi_t^2 - 1)D_t$, a stationary ergodic martingale difference ($D_t \in \mathcal{F}_{t-1}$) with $E(s_t s_t^T) = \frac{\kappa_\xi - 1}{4}J$; the martingale CLT gives $n^{-1/2}\sum_{t=1}^n s_t \xrightarrow{d} N(0, \frac{\kappa_\xi - 1}{4}J)$.

Step 2: Hessian.

$$\frac{\partial^2 l_t}{\partial\theta\,\partial\theta^T} = \frac{1}{2}\Big(\frac{y_t^2}{\sigma_t^2} - 1\Big)\frac{\partial}{\partial\theta^T}\Big(\sigma_t^{-2}\frac{\partial\sigma_t^2}{\partial\theta}\Big) - \frac{1}{2}\,\frac{y_t^2}{\sigma_t^2}\,\sigma_t^{-4}\frac{\partial\sigma_t^2}{\partial\theta}\frac{\partial\sigma_t^2}{\partial\theta^T}.$$

At θ_0, $E(\xi_t^2 \mid \mathcal{F}_{t-1}) = 1$, so its mean is $-\frac{1}{2}J$; by the ergodic theorem and the moment bound, $\frac{1}{n}\sum_t \partial^2 l_t(\bar\theta_n)/\partial\theta\partial\theta^T \xrightarrow{p} -\frac{1}{2}J$ for $\bar\theta_n \to \theta_0$.

Step 3: expansion. For an interior maximizer, $0 = n^{-1/2}\sum_t s_t + (\frac{1}{n}\sum_t \partial^2 l_t(\bar{\theta}_n))\sqrt{n}(\hat{\theta}-\theta_0)$, so $\sqrt{n}(\hat{\theta}-\theta_0) = 2J^{-1}n^{-1/2}\sum_t s_t + o_p(1)$, with covariance $2J^{-1}\cdot\frac{\kappa_\xi-1}{4}J\cdot 2J^{-1} = (\kappa_\xi-1)J^{-1}$. ■

Example 6.3.2 (Gaussian MLE for ARCH(1))

Given y_0, the conditional log-likelihood of ARCH(1) is

$$l(\alpha_0,\alpha_1) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{t=1}^{n}\Big\{\log(\alpha_0+\alpha_1 y_{t-1}^2) + \frac{y_t^2}{\alpha_0+\alpha_1 y_{t-1}^2}\Big\},$$

maximized numerically (no closed form).

6.3.3 GARCH(p,q)

Definition 6.3.3 (GARCH(p,q) model)

If $\{y_t\}$ satisfies

$$y_t = \xi_t\sigma_t, \quad \{\xi_t\}\sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q}\alpha_i y_{t-i}^2 + \sum_{j=1}^{p}\beta_j\sigma_{t-j}^2, \quad \alpha_0>0,\ \alpha_i,\beta_j\geq 0, \tag{6.16}$$

then $\{y_t\}$ follows a *GARCH(p,q) model.*

Proposition (Stationarity and ARMA form of GARCH(p,q), [Supplement])

Let $S = \sum_{i=1}^{q}\alpha_i + \sum_{j=1}^{p}\beta_j$.
(1) If $S<1$, (6.16) has a unique strictly stationary solution with σ_t^2 a function of $\xi_{t-1},\xi_{t-2},\ldots$ and $E\sigma_t^2<\infty$; it satisfies $Ey_t^2 = E\sigma_t^2 = \alpha_0/(1-S)$, and $\{y_t\}$ and $\{\sigma_t^2\}$ are stationary (Bollerslev, 1986). Conversely, a solution with $E\sigma_t^2$ finite and constant in t forces $S<1$.
(2) With $\nu_t = y_t^2-\sigma_t^2$, a martingale difference (white noise if $Ey_t^4<\infty$),

$$\Big(1-\sum_{i=1}^{q}\alpha_i B^i - \sum_{j=1}^{p}\beta_j B^j\Big)y_t^2 = \alpha_0 + \nu_t - \sum_{j=1}^{p}\beta_j\nu_{t-j}, \tag{6.17}$$

so $\{y_t^2\}$ follows an ARMA$(\max\{p,q\},p)$ model.

Proof.

Step 1: construction. $h_t^{(0)}=0$, $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^{q}\alpha_i\xi_{t-i}^2 h_{t-i}^{(m)} + \sum_{j=1}^{p}\beta_j h_{t-j}^{(m)}$. As for Definition 6.2.1, $h_t^{(m)}\geq 0$ is a nondecreasing function of past ξ's with $Eh^{(m+1)} = \alpha_0 + S\,Eh^{(m)}\uparrow\alpha_0/(1-S)$; the limit σ_t^2 solves (6.16), is strictly stationary, and $E(y_t\mid\mathcal{F}_{t-1})=0$ makes $\{y_t\}$ white noise.

Step 2: uniqueness. $D_t = \sigma_t^2-\tilde{\sigma}_t^2 = \sum_i\alpha_i\xi_{t-i}^2 D_{t-i} + \sum_j\beta_j D_{t-j}$ gives $E|D_0|\leq S\,E|D_0|$, so $D=0$.

Step 3: necessity. Expectations in (6.16) give $m=\alpha_0+Sm$ with $\alpha_0,m>0$, so $S<1$.

Step 4: ARMA form. $E(\nu_t\mid\mathcal{F}_{t-1})=0$ as in Proposition (4). Substituting $\sigma_{t-k}^2 = y_{t-k}^2-\nu_{t-k}$ $(k=0,\ldots,p)$ in (6.16),

$$y_t^2-\nu_t = \alpha_0 + \sum_{i=1}^{q}\alpha_i y_{t-i}^2 + \sum_{j=1}^{p}\beta_j(y_{t-j}^2-\nu_{t-j}),$$

which is (6.17), with AR degree $\max\{p,q\}$ and MA degree p. ■

So p, q are chosen by AIC/BIC (Chapter 4); GARCH(1,1) usually suffices.

Remark 6.3.4 (threshold AR(1)–GARCH(1,1))

A useful combination of conditional mean and asymmetric variance:

$$y_t = \phi y_{t-1} + \epsilon_t, \qquad \epsilon_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1),$$

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(\epsilon_{t-1}^+)^2 + \alpha_{12}(\epsilon_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0.$$

Heavy-tailed ξ_t (standardized t: *t-GARCH*) make the Gaussian QMLE less efficient, so testing normality matters (Section 6.4). A log-scale variant:

Definition 6.3.5 (EGARCH(1,1) model)

If $\{y_t\}$ satisfies $y_t = \sigma_t \xi_t$, $\{\xi_t\} \sim IID(0,1)$, and

$$\log \sigma_t^2 = \alpha_0 + \theta \frac{y_{t-1}}{\sigma_{t-1}} + \gamma\Big\{\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big| - E\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big|\Big\} + \beta \log \sigma_{t-1}^2,$$

where $\alpha_0, \theta, \gamma \in \mathbb{R}$ and $|\beta| < 1$, then $\{y_t\}$ follows an *EGARCH(1,1) model* (exponential GARCH).

No sign restrictions are needed; $\theta < 0$ means bad news raises volatility more. Stochastic volatility models are not treated.

6.4 Testing normality

Motivation. Non-normal GARCH errors cost the Gaussian MLE efficiency; Jarque–Bera detects them via skewness and kurtosis.

For i.i.d. $\xi_1, \ldots, \xi_n$ and H_0: $\xi_1, \ldots, \xi_n$ *are normal*, the *Jarque–Bera statistic* is

$$T_{JB} = n\Big(\frac{S_n^2}{6} + \frac{(K_n - 3)^2}{24}\Big), \qquad S_n = \frac{m_3}{m_2^{3/2}}, \quad K_n = \frac{m_4}{m_2^2}, \quad m_k = \frac{1}{n}\sum_{t=1}^{n}(\xi_t - \bar{\xi})^k, \tag{6.18}$$

with sample skewness $S_n \approx 0$ and kurtosis $K_n \approx 3$ under H_0.

Proposition (Null distribution of the Jarque–Bera statistic, [Supplement])

Under H_0, $T_{JB} \xrightarrow{d} \chi^2(2)$.

Proof. By invariance take $\xi_t \sim N(0,1)$; $\hat{\mu}_k = \frac{1}{n}\sum \xi_t^k$; $E\xi^2 = 1$, $E\xi^4 = 3$, $E\xi^6 = 15$, $E\xi^8 = 105$, odd moments 0.

Step 1: reduction. $\sqrt{n}\,\bar{\xi} = O_p(1)$, $\hat{\mu}_2 \to 1$, $\hat{\mu}_3 \to 0$ a.s., and

$$m_2 = \hat{\mu}_2 - \bar{\xi}^2, \quad m_3 = \hat{\mu}_3 - 3\bar{\xi}\hat{\mu}_2 + 2\bar{\xi}^3, \quad m_4 = \hat{\mu}_4 - 4\bar{\xi}\hat{\mu}_3 + 6\bar{\xi}^2\hat{\mu}_2 - 3\bar{\xi}^4,$$

so $\sqrt{n}\,m_3 = \sqrt{n}(\hat{\mu}_3 - 3\bar{\xi}) + o_p(1)$, $\sqrt{n}(m_4 - 3) = \sqrt{n}(\hat{\mu}_4 - 3) + o_p(1)$, $\sqrt{n}(m_2 - 1) = \sqrt{n}(\hat{\mu}_2 - 1) + o_p(1)$.

Step 2: delta method. $\sqrt{n}S_n = \sqrt{n}\,m_3 + o_p(1)$; $(x, y) \mapsto x/y^2$ has gradient $(1, -6)$ at $(3, 1)$, so $\sqrt{n}(K_n - 3) = \sqrt{n}(m_4 - 3) - 6\sqrt{n}(m_2 - 1) + o_p(1)$ and

$$\sqrt{n}\begin{pmatrix} S_n \\ K_n - 3 \end{pmatrix} = \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\begin{pmatrix} \xi_t^3 - 3\xi_t \\ \xi_t^4 - 6\xi_t^2 + 3 \end{pmatrix} + o_p(1).$$

Step 3: CLT. $\mathrm{Var}(\xi^3-3\xi)=15-18+9=6$, $\mathrm{Var}(\xi^4-6\xi^2)=105-180+108-9=24$, covariance an odd moment $=0$; so $\sqrt{n}S_n$, $\sqrt{n}(K_n-3)$ are asymptotically independent $N(0,6)$, $N(0,24)$ and $T_{JB}\xrightarrow{d}\chi^2(2)$. ■

For GARCH(1,1), residuals replace errors. Writing (6.12) as $(1-\beta_1 B)\sigma_t^2=\alpha_0+\alpha_1 y_{t-1}^2$,

$$\sigma_t^2=(1-\beta_1 B)^{-1}\alpha_0+\alpha_1(1-\beta_1 B)^{-1}y_{t-1}^2=\frac{\alpha_0}{1-\beta_1}+\alpha_1\sum_{j=1}^{\infty}\beta_1^{j-1}y_{t-j}^2,$$

and truncating with the Gaussian MLE,

$$\hat\xi_t=\frac{y_t}{\hat\sigma_t},\qquad \hat\sigma_t^2=\frac{\hat\alpha_0}{1-\hat\beta_1}+\hat\alpha_1\sum_{j=1}^{t-1}\hat\beta_1^{j-1}y_{t-j}^2,\qquad 1\le t\le n \tag{6.19}$$

(or the recursion (6.12) from $y_0=\sigma_0=y_1$; asymptotically equivalent). The residual statistic $\hat T_{JB}$ has the same $\chi^2(2)$ limit; reject if $\hat T_{JB}\ge c_\alpha$, $P(\chi^2(2)\ge c_\alpha)=\alpha$.

6.5 Change-point detection

Motivation. Ignoring a GARCH parameter shift typically yields a spurious IGARCH fit; the CUSUM of squared residuals detects it.

Test

H_0: $y_1,\dots,y_n$ follow a GARCH(1,1) model with no parameter change;
H_1: $y_1,\dots,y_n$ follow GARCH(1,1) but the parameters change at some time.

With residuals (6.19) from the Gaussian MLE under H_0, the *CUSUM statistic* is

$$\hat T_n=\max_{m\le k\le n}\frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{t=1}^{k}\hat\xi_t^2-\frac{k}{n}\sum_{t=1}^{n}\hat\xi_t^2\Big|,\qquad \hat\tau^2=\frac{1}{n}\sum_{t=1}^{n}\hat\xi_t^4-\Big(\frac{1}{n}\sum_{t=1}^{n}\hat\xi_t^2\Big)^2, \tag{6.20}$$

$\hat\tau^2$ estimating $\tau^2=\mathrm{Var}(\xi_1^2)$, m small and fixed (5–10). Under H_0,

$$\hat T_n\approx T_n=\max_{m\le k\le n}\frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{t=1}^{k}\xi_t^2-\frac{k}{n}\sum_{t=1}^{n}\xi_t^2\Big|, \tag{6.21}$$

small since both sums are $\approx k$. A change at k^* with $k^*/n\in(0,1)$ makes $\frac{1}{n}|\sum_{t\le k^*}\hat\xi_t^2-\frac{k^*}{n}\sum_{t\le n}\hat\xi_t^2|$ tend to a positive constant, so $\hat T_n\to\infty$.

Proposition (Limit of the CUSUM statistic, [Supplement])

Let ξ_t be i.i.d. with $E\xi_t^2=1$, $E\xi_t^4<\infty$, $\tau^2=\mathrm{Var}(\xi_t^2)>0$, and let $\hat\tau\to\tau$ in probability. Then

$$T_n\xrightarrow{d}T=\sup_{0\le u\le 1}|B^0(u)|,$$

where $B^0(u)=W(u)-uW(1)$ is a Brownian bridge and W a standard Brownian motion on $[0,1]$.

Proof.

Step 1. By Donsker, $W_n(u)=\frac{1}{\tau\sqrt{n}}\sum_{t\le\lfloor nu\rfloor}(\xi_t^2-1)\xrightarrow{d}W$ in $D[0,1]$.

Step 2. $\sum_{t\le k}\xi_t^2 - \frac{k}{n}\sum_{t\le n}\xi_t^2 = \sum_{t\le k}(\xi_t^2-1) - \frac{k}{n}\sum_{t\le n}(\xi_t^2-1)$, so with τ for $\hat\tau$ the statistic is $\max_{m\le k\le n}|W_n(k/n) - \frac{k}{n}W_n(1)|$, differing from $\sup_u|W_n(u)-uW_n(1)|$ by at most $|W_n(1)|/n + \max_{k<m}|W_n(k/n)| + m|W_n(1)|/n = o_p(1)$, as $\max_{k<m}|W_n(k/n)| \le \frac{1}{\tau\sqrt{n}}\sum_{t<m}|\xi_t^2-1| \to 0$.

Step 3. $x \mapsto \sup_u|x(u)-ux(1)|$ is continuous at continuous x, so $\sup_u|W_n(u)-uW_n(1)| \xrightarrow{d} T$; Slutsky with $\tau/\hat\tau \to 1$ gives $T_n \xrightarrow{d} T$. ■

T has the Kolmogorov law $P(T\le x) = 1 - 2\sum_{k\ge1}(-1)^{k-1}e^{-2k^2x^2}$, $P(T\ge 1.358)=0.05$: reject at 5% if $\hat T_n \ge 1.358$, and estimate the change point by

$$\hat k = \arg\max_{1\le k\le n}\hat T_n(k), \qquad \hat T_n(k) = \frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{j=1}^{k}\hat\xi_j^2 - \frac{k}{n}\sum_{j=1}^{n}\hat\xi_j^2\Big|. \tag{6.22}$$

Binary segmentation (retesting $y_1,\dots,y_{\hat k}$ and $y_{\hat k+1},\dots,y_n$) finds several changes. Power is highest for changes near $n/2$. An apparent IGARCH fit ($\hat\alpha_1+\hat\beta_1\approx 1$) should prompt a change-point test; many stock series show changes.

6.6 Value-at-Risk

Motivation. VaR is a low conditional quantile, in GARCH the volatility times an error quantile, estimable by quantile regression.

For X with d.f. F_X and $0<\tau<1$, the *τ-quantile* is

$$F_X^{-1}(\tau) = \min\{y : F_X(y)\ge\tau\}$$

(attained by right-continuity).

Proposition (Two properties of quantiles, [Supplement])

(1) For $a>0$ and $b\in\mathbb{R}$,

$$F_{aX+b}^{-1}(\tau) = aF_X^{-1}(\tau)+b. \tag{6.23}$$

(2) If $E|X|<\infty$ and $\rho_\tau(x) = x(\tau - I(x<0))$ (the *check function*), then

$$F_X^{-1}(\tau) = \min\Big\{m:\ m \text{ minimizes } E\rho_\tau(X-m)\Big\}. \tag{6.24}$$

Proof.

Step 1. $F_{aX+b}(ax+b) = F_X(x)$ and $x\mapsto ax+b$ is an increasing bijection, so

$$\min\{y : F_{aX+b}(y)\ge\tau\} = \min\{ax+b : F_X(x)\ge\tau\} = a\min\{x : F_X(x)\ge\tau\}+b.$$

Step 2. $\rho_\tau(x-m') - \rho_\tau(x-m) = \int_m^{m'}(I(x<u)-\tau)du$ for $m<m'$; by Fubini, with $g(m) = E\rho_\tau(X-m)$ (finite as $\rho_\tau(x)\le|x|$),

$$g(m')-g(m) = \int_m^{m'}(P(X<u)-\tau)du = \int_m^{m'}(F_X(u)-\tau)du$$

(integrands differ only at countably many atoms). With $q = F_X^{-1}(\tau)$: for $m<q$, $F_X<\tau$ on $[m,q)$, so $g(q)<g(m)$; for $m>q$, $F_X\ge\tau$ on $[q,m]$, so $g(m)\ge g(q)$. ■

So the sample quantile minimizes

$$\frac{1}{n}\sum_{t=1}^{n}\rho_\tau(X_t-m). \tag{6.25}$$

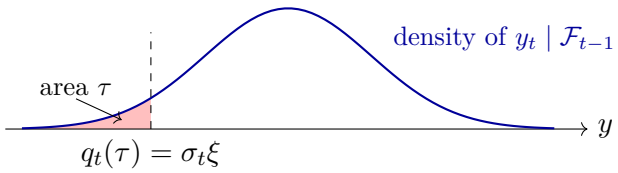


Figure 5: Conditional VaR: the τ-quantile of $y_t \mid \mathcal{F}_{t-1}$, scaling with σ_t.

Reparametrize GARCH(1,1) with the scale in the error variance ω:

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim iid(0,\omega),\ \omega > 0, \qquad \sigma_t^2 = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \alpha, \beta \geq 0,\ \alpha + \beta < 1.$$

As $\sigma_t \in \mathcal{F}_{t-1}$ and $\xi_t \perp \mathcal{F}_{t-1}$, (6.23) gives the *conditional quantile*

$$q_t(\tau) = \sigma_t F_\xi^{-1}(\tau) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2)^{1/2}\, \xi, \qquad \xi = F_\xi^{-1}(\tau),$$

a function $q_t(\xi, \alpha, \beta)$, and by the conditional (6.24)

$$q_t(\tau) = \min\{m :\ m \text{ minimizes } E(\rho_\tau(y_t - m) \mid \mathcal{F}_{t-1})\}.$$

As in (6.25), $(\hat{\xi}, \hat{\alpha}, \hat{\beta})$ minimizes

$$\frac{1}{n} \sum_{t=1}^{n} \rho_\tau(y_t - q_t^*(\xi, \alpha, \beta)), \qquad q_t^*(\xi, \alpha, \beta) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*})^{1/2} \xi,$$

with $\sigma_t^{2*} = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*}$ from arbitrary y_0, σ_0^{2*}. Then

$$\hat{q}_t(\tau) = q_t^*(\hat{\xi}, \hat{\alpha}, \hat{\beta}) =: \mathrm{VaR}_{Y_t \mid \mathcal{F}_{t-1}}(\tau)$$

is the *conditional Value-at-Risk* (Figure 5), with small τ such as 0.01 or 0.05.

6.7 Selected exercises

Motivation. Exercises proving the conditional law, uniqueness and strict stationarity used in the ARCH(1) construction.

Exercise 6.1 (Conditional law of ARCH(1))

Let X, Y be independent random elements (with values in $\mathbb{R}^k$ or $\mathbb{R}^\infty$) and f a measurable real function. It is known that

$$P(f(X,Y) \leq z \mid X = x) = P(f(x,Y) \leq z).$$

Prove this identity, and use it to prove (6.3).

Solution.

Step 1: the identity. Let $g(x) = P(f(x,Y) \leq z) = \int \mathbb{1}\{f(x,u) \leq z\}\, P_Y(du)$. As the law of (X,Y) is $P_X \otimes P_Y$, Tonelli gives g measurable and

$$P(f(X,Y) \leq z,\ X \in A) = \int_A \int \mathbb{1}\{f(x,u) \leq z\}\, P_Y(du)\, P_X(dx) = E[g(X)\mathbb{1}\{X \in A\}],$$

the defining property of $P(f(X,Y) \leq z \mid X = x)$.

Step 2: application. $y_{t-1} \perp \xi_t$ (Proposition (1)); with $X = y_{t-1}$, $Y = \xi_t$, $f(x,u) = u\,(\alpha_0+\alpha_1 x^2)^{1/2}$,

$$P(y_t \le z \mid y_{t-1} = y) = P(\xi_t(\alpha_0 + \alpha_1 y^2)^{1/2} \le z) = \Phi(z/\sqrt{\alpha_0 + \alpha_1 y^2}),$$

which is (6.3).

Step 3: the whole past. With $X = (y_{t-1}, y_{t-2}, \dots) \perp \xi_t$ and $f(x,u) = u(\alpha_0 + \alpha_1 x_1^2)^{1/2}$, $y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2)$. ■

Exercise 6.2 (Iterating the ARCH(1) recursion)

Consider the ARCH(1) model (6.2).
(a) Prove (6.4) for every $n \ge 0$.
(b) Let $\{y_t\}$ be a solution with $Ey_t^2 = c < \infty$ for all t, where each y_s is a function of $\xi_s, \xi_{s-1}, \dots$. Prove that y_t^2 and σ_t^2 are given by the series (6.5); consequently y_t equals (6.6).

Solution.

(a) Let $P_{t,j} = \xi_t^2 \cdots \xi_{t-j}^2$. For $n = 0$, $y_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2 = \alpha_0 P_{t,0} + \alpha_1 P_{t,0}\, y_{t-1}^2$. Induction: substituting $y_{t-n-1}^2 = (\alpha_0 + \alpha_1 y_{t-n-2}^2)\xi_{t-n-1}^2$,

$$\alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = \alpha_0 \alpha_1^{n+1} P_{t,n+1} + \alpha_1^{n+2} P_{t,n+1}\, y_{t-n-2}^2.$$

(b)

Step 1. With $S_n = \alpha_0 \sum_{j=0}^{n} \alpha_1^j P_{t,j}$ and $R_n = \alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = y_t^2 - S_n$, $R_n \ge 0$ is nonincreasing, so $R_n \downarrow R \ge 0$ a.s.

Step 2. $y_{t-n-1} \perp (\xi_t, \dots, \xi_{t-n})$, so

$$ER_n = \alpha_1^{n+1} E(\xi_t^2) \cdots E(\xi_{t-n}^2)\, Ey_{t-n-1}^2 = \alpha_1^{n+1} c \to 0;$$

hence $R = 0$ a.s. and $y_t^2 = \alpha_0 \sum_{j \ge 0} \alpha_1^j P_{t,j}$.

Step 3. At time $t-1$, with $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \alpha_0 \sum_{j \ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-1-j}^2,$$

and $y_t = \sigma_t \xi_t$ is (6.6): with Proposition (1), the *only* causal finite-variance solution. ■

Exercise 6.4 (Functions of an i.i.d. sequence)

Let $\{\xi_t\}$ be an i.i.d. sequence.
(a) For measurable $f_n : \mathbb{R}^n \to \mathbb{R}$, show that $Y_t^{(n)} = f_n(\xi_t, \dots, \xi_{t-n+1})$ is strictly stationary (in particular for $n = 2$).
(b) Let $f : \mathbb{R}^\infty \to \mathbb{R}$ be measurable and suppose $f_n(\xi_t, \dots, \xi_{t-n+1}) \to f(\xi_t, \xi_{t-1}, \dots)$ a.s. for every t. Show that $Y_t = f(\xi_t, \xi_{t-1}, \dots)$ is strictly stationary.

Solution.

(a) For times $t_1, \dots, t_m$, shift h, $L = \min_i t_i - n + 1$, $U = \max_i t_i$,

$$(Y_{t_1}^{(n)}, \dots, Y_{t_m}^{(n)}) = g(\boldsymbol{\xi}), \qquad (Y_{t_1+h}^{(n)}, \dots, Y_{t_m+h}^{(n)}) = g(\boldsymbol{\xi}^h),$$

with the same measurable g, $\boldsymbol{\xi} = (\xi_L, \dots, \xi_U)$, $\boldsymbol{\xi}^h = (\xi_{L+h}, \dots, \xi_{U+h})$. Both have law $\mu^{\otimes(U-L+1)}$, so the images agree in law (Definition 1.2.2); $n = 2$ is a special case.

(b) $\mathbf{V}_n = (Y_{t_1}^{(n)}, \dots, Y_{t_m}^{(n)}) \to \mathbf{V}$ and $\mathbf{V}_n^h \to \mathbf{V}^h$ a.s., hence in distribution; by (a) $\mathbf{V}_n \overset{d}{=} \mathbf{V}_n^h$, so by uniqueness of weak limits $\mathbf{V} \overset{d}{=} \mathbf{V}^h$. Taking f_n the truncation of (6.6) proves the strict stationarity in Proposition (1). ■

Chapter 7. Multivariate Time Series

Portfolios and their risk measures need joint models of several series. This chapter covers multivariate stationarity, VARMA (VAR) models, cointegration, multivariate GARCH and copula-based models.

7.1 Multivariate time series models

Motivation. Stationarity, autocovariance and ARMA carry over with matrices in place of scalars; new are cross-dependence and cointegration.

$\mathbf{Y}_t = (Y_{1t}, \dots, Y_{kt})^T$ is a *k-dimensional multivariate time series*; $\mathbf{x} \le \mathbf{y}$ means $x_i \le y_i$ for all i.

Definition 7.1.1 (Strict stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *strictly stationary* if for all integers $t_1, \dots, t_m$ and h ($m \ge 1$) and all $\mathbf{y}_1, \dots, \mathbf{y}_m \in \mathbb{R}^k$,

$$P(\mathbf{Y}_{t_1} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m} \le \mathbf{y}_m) = P(\mathbf{Y}_{t_1+h} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m+h} \le \mathbf{y}_m).$$

Definition 7.1.2 (Stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *stationary* if
(1) $E\mathbf{Y}_t = \boldsymbol{\mu}$ for all t;
(2) $EY_{it}^2 < \infty$ for all $1 \le i \le k$ and all t;
(3) $\operatorname{Cov}(Y_{it}, Y_{j,t+h}) = \operatorname{Cov}(Y_{i0}, Y_{jh})$ for all $1 \le i, j \le k$ and all t, h.

Definition 7.1.3 (Autocovariance function)

For a stationary $\{\mathbf{Y}_t\}$, the $k \times k$ matrix-valued function $\Gamma(\cdot) = (\Gamma_{ij}(\cdot))_{i,j=1}^k$ with

$$\Gamma_{ij}(h) = \operatorname{Cov}(Y_{it}, Y_{j,t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is the *autocovariance function*; that is, $\Gamma(h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ with $\boldsymbol{\mu} = E\mathbf{Y}_1$.

Theorem 7.1.4 (Properties of the autocovariance function)

The autocovariance function of a stationary series satisfies:
(1) $\Gamma(-h) = \Gamma(h)^T$;
(2) $|\Gamma_{ij}(h)| \le \sqrt{\Gamma_{ii}(0)}\sqrt{\Gamma_{jj}(0)}$;
(3) for all $n \ge 1$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^k$, $\sum_{i,j=1}^n \mathbf{a}_i^T \Gamma(i-j)\mathbf{a}_j \ge 0$.

Proof. [Supplement] (Book exercise.)

(1) With $s = t - h$, $\Gamma(-h) = E(\mathbf{Y}_{s+h} - \boldsymbol{\mu})(\mathbf{Y}_s - \boldsymbol{\mu})^T = \big[E(\mathbf{Y}_s - \boldsymbol{\mu})(\mathbf{Y}_{s+h} - \boldsymbol{\mu})^T\big]^T = \Gamma(h)^T$.

(2) Cauchy–Schwarz: $|\operatorname{Cov}(Y_{it}, Y_{j,t+h})| \le \sqrt{\operatorname{Var} Y_{it}}\sqrt{\operatorname{Var} Y_{j,t+h}} = \sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}$.

(3) As $\operatorname{Cov}(\mathbf{a}_i^T\mathbf{Y}_i, \mathbf{a}_j^T\mathbf{Y}_j) = \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j$,

$$0 \le \operatorname{Var}\Big(\sum_{i=1}^n \mathbf{a}_i^T\mathbf{Y}_i\Big) = \sum_{i,j=1}^n \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j = \sum_{i,j=1}^n \mathbf{a}_j^T\Gamma(i-j)\mathbf{a}_i,$$

transposing each scalar and using (1); relabel $i \leftrightarrow j$. ■

Definition 7.1.5 (Vector ARMA model)

If $\{\mathbf{Y}_t\}$ satisfies

$$\mathbf{Y}_t - \Phi_1 \mathbf{Y}_{t-1} - \cdots - \Phi_p \mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t + \Theta_1 \boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q \boldsymbol{\epsilon}_{t-q}, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \quad (7.1)$$

where $\Phi_1, \ldots, \Phi_p, \Theta_1, \ldots, \Theta_q, \Sigma$ are $k \times k$ matrices and Σ is positive definite, then $\{\mathbf{Y}_t\}$ follows a *vector ARMA(p, q) model* (VARMA(p, q)). With $q = 0$ it is a *VAR(p) model*.

With $\Phi(z) = I - \Phi_1 z - \cdots - \Phi_p z^p$, $\Theta(z) = I + \Theta_1 z + \cdots + \Theta_q z^q$, (7.1) is $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$; causality is now a condition on the zeros of $\det \Phi(z)$.

Theorem 7.1.6 (Causal VARMA)

If

$$\det(I - \Phi_1 z - \cdots - \Phi_p z^p) \neq 0 \qquad \text{for all complex } |z| \leq 1,$$

then the VARMA model (7.1) has a stationary solution $\{\mathbf{Y}_t\}$, it is unique, and it has the representation

$$\mathbf{Y}_t = \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}, \quad (7.2)$$

where $C_0 = I_k$ and the $k \times k$ matrices C_i are determined by

$$(I - \Phi_1 z - \cdots - \Phi_p z^p) \sum_{i=0}^{\infty} C_i z^i = I + \Theta_1 z + \cdots + \Theta_q z^q, \qquad |z| \leq 1. \quad (7.3)$$

With mean $\boldsymbol{\mu}$ the model reads $\mathbf{Y}_t - \boldsymbol{\mu} - \Phi_1(\mathbf{Y}_{t-1} - \boldsymbol{\mu}) - \cdots - \Phi_p(\mathbf{Y}_{t-p} - \boldsymbol{\mu}) = \boldsymbol{\epsilon}_t + \Theta_1 \boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q \boldsymbol{\epsilon}_{t-q}$.

Proof. [Supplement] (Omitted in the book.) Matrix products do not commute: $\Phi(z)$ stands on the left in (7.3).

Step 1: the coefficients. $\det \Phi(z) \neq 0$ on $|z| < 1 + \delta$ for some $\delta > 0$, so $\Phi(z)^{-1} = \operatorname{adj} \Phi(z) / \det \Phi(z)$ and $C(z) := \Phi(z)^{-1}\Theta(z) = \sum_{i \geq 0} C_i z^i$ are analytic there. Convergence at $z = 1 + \delta/2$ gives $\|C_i\| \leq K(1 + \delta/2)^{-i}$, so $\sum_i \|C_i\| < \infty$, $C_0 = I$, $\Phi(z)C(z) = \Theta(z)$; likewise $\Phi(z)^{-1} = \sum_i D_i z^i$, $\sum_i \|D_i\| < \infty$.

Step 2: existence. $E\|\boldsymbol{\epsilon}_t\| \leq (\operatorname{tr}\Sigma)^{1/2}$, so (7.2) converges absolutely a.s., and in L^2 since $\left(E\|\sum_{i=m}^{n} C_i \boldsymbol{\epsilon}_{t-i}\|^2\right)^{1/2} \leq (\operatorname{tr}\Sigma)^{1/2} \sum_{i=m}^{n} \|C_i\| \to 0$. Then $E\mathbf{Y}_t = \mathbf{0}$ and

$$E\mathbf{Y}_t \mathbf{Y}_{t+h}^T = \sum_{i,j} C_i E(\boldsymbol{\epsilon}_{t-i} \boldsymbol{\epsilon}_{t+h-j}^T) C_j^T = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T \qquad (h \geq 0),$$

free of t. With $\Phi(z) = \sum_{j=0}^{p} A_j z^j$ ($A_0 = I$, $A_j = -\Phi_j$),

$$\Phi(B)\mathbf{Y}_t = \sum_{j=0}^{p} A_j \sum_{i \geq 0} C_i \boldsymbol{\epsilon}_{t-j-i} = \sum_{m \geq 0} \Big(\sum_{j=0}^{\min(p,m)} A_j C_{m-j} \Big) \boldsymbol{\epsilon}_{t-m} = \sum_{m=0}^{q} \Theta_m \boldsymbol{\epsilon}_{t-m},$$

the inner sum being the z^m coefficient of $\Phi(z)C(z) = \Theta(z)$.

Step 3: uniqueness. For any stationary solution, applying $\sum_i D_i B^i$ (summable, composing like power series) to $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$ gives $\mathbf{Y}_t = \Phi(B)^{-1}\Theta(B)\boldsymbol{\epsilon}_t = \sum_i C_i \boldsymbol{\epsilon}_{t-i}$. ■

Example 7.1.7 (Coefficients of a bivariate ARMA(1,1))

Let $k=2$, $p=q=1$, $\Phi_1=\begin{pmatrix}0.5 & 0.5\\ 0 & 0.5\end{pmatrix}$ and $\Theta_1=\Phi_1^T$. Here $\det\Phi(z)=(1-0.5z)^2\neq 0$ for $|z|\le 1$. From (7.3),

$$\sum_{i=0}^{\infty}C_iz^i=\begin{pmatrix}1-0.5z & -0.5z\\ 0 & 1-0.5z\end{pmatrix}^{-1}\begin{pmatrix}1+0.5z & 0\\ 0.5z & 1+0.5z\end{pmatrix}$$
$$=(1-0.5z)^{-2}\begin{pmatrix}1 & 0.5z(1+0.5z)\\ 0.5z(1-0.5z) & 1-0.25z^2\end{pmatrix}.$$

With $(1-0.5z)^{-2}=\sum_{m\ge0}(m+1)2^{-m}z^m$,

$$C_0=I,\qquad C_i=2^{-i}\begin{pmatrix}i+1 & 2i-1\\ 1 & 2\end{pmatrix},\quad i=1,2,\ldots$$

e.g. the $(1,2)$ entry is $\frac{1}{2}\,i\,2^{-(i-1)}+\frac{1}{4}(i-1)2^{-(i-2)}=(2i-1)2^{-i}$.

7.1.1 Cointegration

Definition 7.1.8 (Integration order and cointegration)

If $(1-B)^d\mathbf{Y}_t=\left((1-B)^dY_{1t},\ldots,(1-B)^dY_{kt}\right)^T$ is stationary while $\mathbf{Y}_t,(1-B)\mathbf{Y}_t,\ldots,(1-B)^{d-1}\mathbf{Y}_t$ are nonstationary, we write $\mathbf{Y}_t\sim I(d)$. If $\mathbf{Y}_t\sim I(d)$ and, for some positive integer $\delta<d$ and some vector $\boldsymbol{\alpha}\in\mathbb{R}^k$, $\boldsymbol{\alpha}^T\mathbf{Y}_t\sim I(\delta)$, then $\mathbf{Y}_t$ is *cointegrated* with *cointegrating vector* $\boldsymbol{\alpha}$. (The most common case is $d=1$, $\delta=0$: a stationary combination of $I(1)$ series.)

Contrast *spurious regression*: regressing one of two *independent* random walks on the other typically gives a large R^2 and a significant slope. Cointegration is a genuine long-run equilibrium.

Example 7.1.10 (Consumption and income)

Let consumption C_t and disposable income I_t (real, per capita) be $I(1)$ with $C_t=a+bI_t+u_t$, $a,b\in(0,1)$, $Eu_t=0$, $u_t\sim I(0)$. Then $\boldsymbol{\alpha}^T\mathbf{Y}_t=C_t-bI_t=a+u_t\sim I(0)$ for $\mathbf{Y}_t=(C_t,I_t)^T$, $\boldsymbol{\alpha}=(1,-b)^T$: cointegrated.

Cointegrated series are written as a *vector error correction model*:

Definition 7.1.11 (VECM)

The VECM(p) with *cointegration rank* r ($\le k$) is

$$\Delta\mathbf{Y}_t=\boldsymbol{\delta}+\Pi\mathbf{Y}_{t-1}+\sum_{i=1}^{p-1}\Phi_i^*\Delta\mathbf{Y}_{t-i}+\boldsymbol{\epsilon}_t,\qquad\{\boldsymbol{\epsilon}_t\}\sim IID(\mathbf{0},\Sigma),\tag{7.4}$$

where $\Delta=1-B$ is the differencing operator, $\boldsymbol{\delta}$ is a $k\times1$ constant vector, $\Pi=\boldsymbol{\alpha}\boldsymbol{\beta}^T$ with $\boldsymbol{\alpha},\boldsymbol{\beta}$ $k\times r$ matrices, and the Φ_i^* are $k\times k$ matrices.

If $\mathbf{Y}_t\sim I(1)$, (7.4) forces $\boldsymbol{\alpha}\boldsymbol{\beta}^T\mathbf{Y}_{t-1}$, hence ($\boldsymbol{\alpha}$ of full column rank) $\boldsymbol{\beta}^T\mathbf{Y}_{t-1}$, to be stationary: the columns of $\boldsymbol{\beta}$ are r cointegrating vectors, and $\boldsymbol{\alpha}$ gives the speed of error correction.

Johansen (1988) tests $H_0 : r \le r_0$ against $H_1 : r > r_0$ by the *trace statistic*

$$\lambda_{\text{trace}}(r_0) = -n \sum_{i=r_0+1}^{k} \log(1 - \hat{\lambda}_i), \tag{7.5}$$

with $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_k$ the estimated eigenvalues associated with Π; under H_0 the eigenvalues beyond r_0 vanish, so large values reject. The null law is nonstandard and depends on deterministic terms, r_0 and the dimension.

7.2 Inference for stationary multivariate series

Motivation. Sample means and autocovariance matrices are consistent, and VAR(p) is fitted by equation-wise least squares.

Theorem 7.2.1 (Sample mean)

Let $\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}$ with $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma)$ and $k \times k$ matrices C_i satisfying $\sum_{i=0}^{\infty} \|C_i\| < \infty$, where $\|C\| = \sup\{\|C\mathbf{y}\| : \|\mathbf{y}\| \le 1\}$. Let $\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Y}_t$. Then:
(1) $P(\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu},\ n \to \infty) = 1$;
(2) if $\Gamma_{ii}(n) \to 0$ for all $1 \le i \le k$, then $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to 0$;
(3) if $\sum_{h=0}^{\infty} |\Gamma_{ii}(h)| < \infty$ for all $1 \le i \le k$, then

$$nE(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to \sum_{i=1}^{k} \sum_{h=-\infty}^{\infty} \Gamma_{ii}(h).$$

Proof. [Supplement] (Omitted in the book.) As in Step 2 of Theorem 7.1.6, the series converges a.s. and in L^2, and $\{\mathbf{Y}_t\}$ is stationary.

(1) $\mathbf{Y}_t = G(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_{t-1}, \dots)$ is strictly stationary ergodic with $E\|\mathbf{Y}_t\| < \infty$; Birkhoff gives $\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu}$ a.s.

(2) $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) = \sum_{i=1}^{k} \operatorname{Var}(\bar{Y}_i)$ and

$$\operatorname{Var}(\bar{Y}_i) = \frac{1}{n^2} \sum_{s,t=1}^{n} \Gamma_{ii}(t-s) = \frac{1}{n} \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big)\Gamma_{ii}(h) \le \frac{1}{n} \sum_{|h|<n} |\Gamma_{ii}(h)|,$$

a Cesàro mean tending to 0 when $\Gamma_{ii}(h) \to 0$.

(3) $n \operatorname{Var}(\bar{Y}_i) = \sum_{|h|<n}(1 - |h|/n)\Gamma_{ii}(h) \to \sum_h \Gamma_{ii}(h)$ by dominated convergence (bound $|\Gamma_{ii}(h)|$); sum over i. ∎

Theorem 7.2.2 (Sample autocovariance)

Let $\{\mathbf{Y}_t\}$ be as in Theorem 7.2.1. Then $\Gamma(h) = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T$ $(h \ge 0)$, and the sample autocovariance function

$$\hat{\Gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (\mathbf{Y}_t - \hat{\boldsymbol{\mu}})(\mathbf{Y}_{t+|h|} - \hat{\boldsymbol{\mu}})^T$$

(for $h < 0$ take the transpose) satisfies $P(\hat{\Gamma}(h) \to \Gamma(h),\ n \to \infty) = 1$; that is, $\hat{\Gamma}(\cdot)$ is a consistent estimator of $\Gamma(\cdot)$.

Proof. [Supplement] $\Gamma(h)$ is from Step 2 of Theorem 7.1.6. For $h \geq 0$, with $\mathbf{Y}_t - \hat{\boldsymbol{\mu}} = (\mathbf{Y}_t - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$,

$$\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T - \Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t - \boldsymbol{\mu})\Big)(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T$$
$$- (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})\Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_{t+h} - \boldsymbol{\mu})\Big)^T + \frac{n-h}{n}(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T.$$

$Z_t = (\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ is an integrable function of the shifted i.i.d. sequence, so $\frac{1}{n}\sum_{t\leq n-h} Z_t \to \Gamma(h)$ a.s.; the other terms vanish by Theorem 7.2.1(1). ■

7.2.1 Inference in VAR(p) models

For VAR(p) $\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t$, given $\mathbf{Y}_{-p+1}, \ldots, \mathbf{Y}_0$, let

$$\mathbf{y}_t = \begin{pmatrix} \mathbf{Y}_t \\ \vdots \\ \mathbf{Y}_{t-p+1} \end{pmatrix} (kp \times 1), \qquad \begin{aligned} Y &= (\mathbf{Y}_1, \ldots, \mathbf{Y}_n)\ (k \times n), & \Phi &= (\Phi_1, \ldots, \Phi_p)\ (k \times kp), \\ X &= (\mathbf{y}_0, \ldots, \mathbf{y}_{n-1})\ (kp \times n), & \epsilon &= (\boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)\ (k \times n), \end{aligned}$$

so that $Y = \Phi X + \epsilon$.

Proposition (Least squares for VAR(p), [Supplement])

If XX^T is invertible, then for every Φ

$$(Y-\Phi X)(Y-\Phi X)^T = (Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T + (\hat{\Phi}-\Phi)XX^T(\hat{\Phi}-\Phi)^T, \qquad \hat{\Phi} = YX^T(XX^T)^{-1}.$$

Hence $\hat{\Phi}$ minimizes $\sum_{t=1}^n \|\mathbf{Y}_t - \Phi\mathbf{y}_{t-1}\|^2 = \mathrm{tr}\{(Y - \Phi X)(Y - \Phi X)^T\}$ (and the determinant of the residual cross-product matrix), uniquely.

Proof. Expand $Y - \Phi X = (Y - \hat{\Phi}X) + (\hat{\Phi} - \Phi)X$; cross terms vanish as $(Y - \hat{\Phi}X)X^T = YX^T - YX^T(XX^T)^{-1}XX^T = 0$. The second term is positive semidefinite, zero only for $\Phi = \hat{\Phi}$; take traces or determinants. ■

So

$$\hat{\Phi} = YX^T(XX^T)^{-1}, \qquad \hat{\Sigma} = \frac{1}{n - kp}(Y - \hat{\Phi}X)(Y - \hat{\Phi}X)^T,$$

and $\hat{\Phi}$ equals OLS applied to each equation.

To choose p, fit VAR(i), $i = 1, \ldots, p_0$, with residuals $\hat{\boldsymbol{\epsilon}}_t^{(i)} = \mathbf{Y}_t - \hat{\Phi}_1^{(i)}\mathbf{Y}_{t-1} - \cdots - \hat{\Phi}_i^{(i)}\mathbf{Y}_{t-i}$,

$$\hat{\Sigma}_i = \frac{1}{n - 2i}\sum_{t=i+1}^{n} \hat{\boldsymbol{\epsilon}}_t^{(i)}\hat{\boldsymbol{\epsilon}}_t^{(i)T},$$

and minimize ($k^2 i$ AR parameters)

$$AIC(i) = \log|\hat{\Sigma}_i| + 2k^2 i/n \qquad \text{or} \qquad BIC(i) = \log|\hat{\Sigma}_i| + k^2 i \log(n)/n.$$

7.3 Multivariate GARCH models

Motivation. Portfolio volatility depends on a time-varying conditional covariance matrix.

For $\mathbf{Y}_t = (y_{1t}, \dots, y_{dt})^T$, $\sigma_{ijt} = \mathrm{Cov}(y_{it}, y_{jt} \mid \mathcal{F}_{t-1})$ and $\Sigma_t = (\sigma_{ijt})$, the simplest multivariate GARCH model is

$$\mathbf{Y}_t = \Sigma_t^{1/2} \boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, I_d), \qquad E(\mathbf{Y}_t \mathbf{Y}_t^T \mid \mathcal{F}_{t-1}) = \Sigma_t, \tag{7.6}$$

with Σ_t a function of the past. An unrestricted recursion for the $d(d+1)/2$ entries has $O(d^4)$ parameters, so restricted forms are used: BEKK (Baba–Engle–Kraft–Kroner), positive definite by a quadratic-form recursion, and CCC-GARCH, $\Sigma_t = D_t R D_t$ with $D_t = \mathrm{diag}(\sigma_{11t}^{1/2}, \dots, \sigma_{ddt}^{1/2})$ from univariate GARCH and constant correlation R (Bollerslev, 1990; Engle and Kroner, 1995).

7.4 SCOMDY models

Motivation. Univariate GARCH volatilities plus a copula for the standardized errors give a flexible semiparametric multivariate model.

A simple instance of (7.6): d GARCH(1,1) series with jointly normal errors,

$$y_{it} = \sigma_{it} \xi_{it}, \quad i = 1, \dots, d, \qquad \boldsymbol{\xi}_t = (\xi_{1t}, \dots, \xi_{dt})^T \text{ i.i.d. multivariate normal.} \tag{7.7}$$

For non-normal $\boldsymbol{\xi}_t$, the *semiparametric copula-based multivariate dynamic* (SCOMDY) model uses a copula.

Theorem (Sklar, continuous case, [Supplement])

Let $\mathbf{Y} = (Y_1, \dots, Y_d)^T$ have joint distribution function F and continuous marginals $F_1, \dots, F_d$. Then there is a function C on $[0,1]^d$, namely the distribution function of $(F_1(Y_1), \dots, F_d(Y_d))$, such that

$$F(\mathbf{y}) = C(F_1(y_1), \dots, F_d(y_d)), \qquad \mathbf{y} = (y_1, \dots, y_d)^T,$$

and C has uniform $(0,1)$ marginals (a *copula*).

Proof. For $u \in (0,1)$ let $x_u = \max\{x : F_i(x) = u\}$ (exists by continuity); $\{F_i(Y_i) \le u\} = \{Y_i \le x_u\}$ a.s., so $U_i = F_i(Y_i)$ is uniform. Also $\{Y_i \le y_i\} \subset \{F_i(Y_i) \le F_i(y_i)\}$ with null difference (F_i is flat there). Hence

$$F(\mathbf{y}) = P(Y_i \le y_i \ \forall i) = P(U_i \le F_i(y_i) \ \forall i) = C(F_1(y_1), \dots, F_d(y_d)).$$

■

The copula carries all the dependence and is modelled instead of F (Gaussian, Archimedean, Frank, Clayton). For $d = 2$ the *Gaussian copula* is

$$C(u, v; \rho) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)),$$

with Φ_ρ the standard bivariate normal d.f. with correlation ρ. An *Archimedean copula* is

$$C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$$

for a convex decreasing generator φ on $[0,1]$, $\varphi(1) = 0$: $\varphi(t) = \log(e^{-\theta} - 1) - \log(e^{-\theta t} - 1)$, $\theta \neq 0$ (*Frank*), $\varphi(t) = (t^{-\theta} - 1)/\theta$, $\theta > 0$ (*Clayton*); see Joe (1997).

Marginals are estimated empirically: $\hat{F}(\mathbf{y}) = C(\hat{F}_1(y_1), \dots, \hat{F}_d(y_d))$ with

$$\hat{F}_i(y) = \frac{1}{n} \sum_{j=1}^{n} I(Y_{ij} \le y), \qquad i = 1, \dots, d,$$

giving a conditional likelihood for (7.7) with copula errors. SCOMDY contains GARCH, VAR and Markov-switching models; Chen and Fan (2006) estimate the copula parameters by a Kullback–Leibler criterion, with asymptotics and model-selection tests.

7.5 Selected exercises

Motivation. Exercises on geometric decay of VARMA autocovariances and the ACVF of a bivariate ARMA(1,1).

Exercise 7.2 (Geometric decay of the autocovariances)

Let $\{\mathbf{Y}_t\}$ be the stationary solution of a VARMA model satisfying the condition of Theorem 7.1.6. Show that there are $\rho \in (0,1)$ and a constant K such that $|\Gamma_{ij}(h)| \le K\rho^{|h|}$ for all i, j and h.

Solution.

Step 1: decay of C_i. By Step 1 of Theorem 7.1.6, $\|C_i\| \le K_0\rho^i$ with $\rho = (1+\delta/2)^{-1} \in (0,1)$.

Step 2: bound on $\Gamma(h)$. By Theorem 7.2.2, for $h \ge 0$,

$$\|\Gamma(h)\| \le \sum_{i\ge 0}\|C_i\|\,\|\Sigma\|\,\|C_{i+h}\| \le \|\Sigma\|K_0^2\sum_{i\ge 0}\rho^i\rho^{i+h} = \frac{\|\Sigma\|K_0^2}{1-\rho^2}\,\rho^h.$$

Step 3: entries. $|\Gamma_{ij}(h)| = |\mathbf{e}_i^T\Gamma(h)\mathbf{e}_j| \le \|\Gamma(h)\|$, and $\Gamma(h) = \Gamma(-h)^T$ for $h<0$; so $K = \|\Sigma\|K_0^2/(1-\rho^2)$. ■

Exercise 7.3 (Autocovariances of a bivariate ARMA(1,1))

Find the autocovariance matrix function of the bivariate ARMA(1,1) model $\mathbf{Y}_t = \Phi\mathbf{Y}_{t-1} + \boldsymbol{\epsilon}_t + \Theta\boldsymbol{\epsilon}_{t-1}$ with

$$\Phi = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.6 \end{pmatrix}, \qquad \Theta = \begin{pmatrix} 0.5 & 0.6 \\ -0.7 & 0.8 \end{pmatrix}, \qquad \{\boldsymbol{\epsilon}_t\} \sim \text{IID } N\Big(\mathbf{0}, \begin{pmatrix} 1 & 0.71 \\ 0.71 & 2 \end{pmatrix}\Big).$$

Solution.

Step 1: causality. $\det(I - \Phi z) = (1-0.7z)(1-0.6z)$ has zeros $1/0.7$, $1/0.6$; by Theorem 7.1.6, $\mathbf{Y}_t = \sum_{i\ge 0} C_i\boldsymbol{\epsilon}_{t-i}$, so $E\mathbf{Y}_t = \mathbf{0}$, $E\mathbf{Y}_t\boldsymbol{\epsilon}_{t+j}^T = 0$ $(j \ge 1)$, $E\mathbf{Y}_t\boldsymbol{\epsilon}_t^T = \Sigma$.

Step 2: lag 0. With $\mathbf{U}_t = \boldsymbol{\epsilon}_t + \Theta\boldsymbol{\epsilon}_{t-1}$, $E\mathbf{Y}_{t-1}\mathbf{U}_t^T = \Sigma\Theta^T$, $E\mathbf{U}_t\mathbf{U}_t^T = \Sigma + \Theta\Sigma\Theta^T$, so

$$\Gamma(0) = \Phi\Gamma(0)\Phi^T + Q, \qquad Q = \Phi\Sigma\Theta^T + \Theta\Sigma\Phi^T + \Sigma + \Theta\Sigma\Theta^T.$$

Numerically $\Sigma\Theta^T = \begin{pmatrix} 0.926 & -0.132 \\ 1.555 & 1.103 \end{pmatrix}$, $\Phi\Sigma\Theta^T = \begin{pmatrix} 0.6482 & -0.0924 \\ 0.9330 & 0.6618 \end{pmatrix}$ ($\Theta\Sigma\Phi^T$ its transpose), $\Theta\Sigma\Theta^T = \begin{pmatrix} 1.3960 & 0.5958 \\ 0.5958 & 0.9748 \end{pmatrix}$, hence

$$Q = \begin{pmatrix} 3.6924 & 2.1464 \\ 2.1464 & 4.2984 \end{pmatrix}.$$

As Φ is diagonal, $\Gamma_{ij}(0) = Q_{ij}/(1-\phi_i\phi_j)$:

$$\Gamma(0) = \begin{pmatrix} 3.6924/0.51 & 2.1464/0.58 \\ 2.1464/0.58 & 4.2984/0.64 \end{pmatrix} = \begin{pmatrix} 7.2400 & 3.7007 \\ 3.7007 & 6.7163 \end{pmatrix}.$$

Step 3: lag 1 and beyond. For $h \ge 1$,

$$\Gamma(h) = E\mathbf{Y}_t\big(\Phi\mathbf{Y}_{t+h-1} + \boldsymbol{\epsilon}_{t+h} + \Theta\boldsymbol{\epsilon}_{t+h-1}\big)^T = \Gamma(h-1)\Phi^T + E(\mathbf{Y}_t\boldsymbol{\epsilon}_{t+h-1}^T)\Theta^T,$$

the last term being $\Sigma\Theta^T$ for $h = 1$ and 0 for $h \geq 2$:

$$\Gamma(1) = \Gamma(0)\Phi^T + \Sigma\Theta^T = \begin{pmatrix} 5.9940 & 2.0884 \\ 4.1455 & 5.1328 \end{pmatrix}, \qquad \Gamma(h) = \Gamma(1)(\Phi^T)^{h-1} \ (h \geq 1),$$

i.e. $\Gamma_{i1}(h) = 0.7^{h-1}\Gamma_{i1}(1)$, $\Gamma_{i2}(h) = 0.6^{h-1}\Gamma_{i2}(1)$, and $\Gamma(-h) = \Gamma(h)^T$. The contemporaneous correlation is $3.7007/\sqrt{7.24 \cdot 6.7163} \approx 0.531$; $\Gamma(1)$ is not symmetric (4.1455 vs 2.0884), mainly through $(\Sigma\Theta^T)_{21} = 1.555$ vs $(\Sigma\Theta^T)_{12} = -0.132$. ■

Chapter 8. Integer-Valued Time Series Models

Count time series take values in $\{0, 1, 2, \dots\}$ (trades per minute, monthly accidents, daily admissions). ARMA/GARCH fail since $\phi y_{t-1} + \epsilon_t$ is not integer-valued. Standard alternatives: INAR (McKenzie, 1985; Al-Osh and Alzaid, 1987) and INGARCH (Ferland et al., 2006), Poisson given the past; inference by Davis and Liu (2016), Lee and Lee (2019) (see Weiss, 2018). $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ (the book's I_t).

8.1 The INAR(1) model

Motivation. Binomial thinning (keep each of Y units with probability α) replaces multiplication by α and gives an integer-valued AR(1).

Definition 8.1.1 (Binomial thinning)

For a nonnegative integer-valued random variable Y and $\alpha \in (0, 1)$, the *thinning operator* $\circ$ is

$$\alpha \circ Y = \sum_{i=1}^{Y} w_i,$$

where the w_i are i.i.d. Bernoulli variables with success probability α, independent of Y (the *counting sequence*); $\alpha \circ Y = 0$ when $Y = 0$.

Proposition (Properties of thinning, [Supplement])

Let Y be nonnegative integer-valued with $EY^2 < \infty$, with probability generating function $G_Y(s) = Es^Y$, and let $\alpha, \beta \in (0, 1)$.
(1) $E(\alpha \circ Y \mid Y) = \alpha Y$ and $\operatorname{Var}(\alpha \circ Y \mid Y) = \alpha(1-\alpha)Y$; hence $E(\alpha \circ Y) = \alpha EY$ and $\operatorname{Var}(\alpha \circ Y) = \alpha(1-\alpha)EY + \alpha^2 \operatorname{Var} Y$.
(2) $G_{\alpha \circ Y}(s) = G_Y(1 - \alpha + \alpha s)$.
(3) With independent counting sequences, $\alpha \circ (\beta \circ Y) \overset{d}{=} (\alpha\beta) \circ Y$.
(4) If $Y \sim \text{Poisson}(\lambda)$ then $\alpha \circ Y \sim \text{Poisson}(\alpha\lambda)$.

Proof.

(1) Given $Y = y$, $\alpha \circ Y$ is Binomial(y, α), with mean αy and variance $\alpha(1-\alpha)y$. The unconditional formulas follow from $E(\cdot) = E(E(\cdot \mid Y))$ and $\operatorname{Var}(\cdot) = E \operatorname{Var}(\cdot \mid Y) + \operatorname{Var} E(\cdot \mid Y)$.

(2) $E(s^{\alpha \circ Y} \mid Y) = (Es^{w_1})^Y = (1 - \alpha + \alpha s)^Y$; take expectations.

(3) By (2) twice, $G_{\alpha \circ (\beta \circ Y)}(s) = G_{\beta \circ Y}(1-\alpha+\alpha s) = G_Y\big(1-\beta+\beta(1-\alpha+\alpha s)\big) = G_Y(1-\alpha\beta+\alpha\beta s)$, the generating function of $(\alpha\beta) \circ Y$.

(4) $G_Y(s) = e^{\lambda(s-1)}$, so $G_{\alpha \circ Y}(s) = e^{\lambda(1-\alpha+\alpha s-1)} = e^{\alpha\lambda(s-1)}$. ■

Definition 8.1.2 (INAR(1) model)

A series y_t, $t = 1, 2, \dots$, with values in $\{0, 1, 2, \dots\}$ follows the *INAR(1) model* if

$$y_t = \alpha \circ y_{t-1} + \epsilon_t, \tag{8.1}$$

where $\alpha \in (0, 1)$, the ϵ_t are i.i.d. nonnegative integer-valued random variables, and ϵ_t and the counting sequence of $\alpha \circ y_{t-1}$ are independent of each other and of $\mathcal{F}_{t-1}$. ($\alpha \circ y_{t-1}$: survivors of a branching process; ϵ_t: immigrants.)

$\mu = E\epsilon_t$, $\sigma_\epsilon^2 = \operatorname{Var}\epsilon_t$; usually ϵ_t is Poisson or geometric.

Proposition (Stationarity, moments and ACF of INAR(1), [Supplement])

Let $0 < \alpha < 1$ and $\sigma_\epsilon^2 < \infty$.
(1) (8.1) has a strictly stationary solution; every stationary solution has the marginal law of $\sum_{j=0}^{\infty} \alpha^j \circ \epsilon_{t-j}$ (independent thinnings), which is finite a.s.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = \alpha y_{t-1} + \mu$ and $\operatorname{Var}(y_t \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2$.
(3) $Ey_t = \dfrac{\mu}{1-\alpha}$ and $\operatorname{Var} y_t = \dfrac{\sigma_\epsilon^2 + \alpha\mu}{1-\alpha^2}$.
(4) $\gamma(h) = \operatorname{Cov}(y_t, y_{t+h}) = \alpha^{|h|}\gamma(0)$, so the ACF is $\rho(h) = \alpha^{|h|}$, as for an AR(1) process.

Proof.

Step 1: construction. Take i.i.d. ϵ_s ($s \in \mathbb{Z}$) and, independently, i.i.d. Bernoulli(α) indicators $B_{s,i,r}$ for $s < r$ and $i \geq 1$. Immigrant i of time s ($1 \leq i \leq \epsilon_s$) is *alive at time* $t \geq s$ if $B_{s,i,r} = 1$ for all $s < r \leq t$, and let y_t be the number of individuals alive at time t. Since $Ey_t = \sum_{j\geq 0} \alpha^j \mu < \infty$, y_t is finite a.s. Each individual alive at $t-1$ survives to t independently with probability α, via indicators $B_{\cdot,\cdot,t}$ independent of $\mathcal{F}_{t-1}$ and of ϵ_t; so the survivors form $\alpha \circ y_{t-1}$ and $y_t = \alpha \circ y_{t-1} + \epsilon_t$. The construction is shift-invariant, so $\{y_t\}$ is strictly stationary. The contributions of the cohorts $s = t - j$ are independent, and each is distributed as $\alpha^j \circ \epsilon_{t-j}$ (survival over j steps has probability α^j); hence $y_t \overset{d}{=} \sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$.

Step 2: uniqueness of the law. For any solution, iterating (8.1) and using Proposition (3) gives $y_t \overset{d}{=} \alpha^t \circ y_0 + \sum_{j=0}^{t-1} \alpha^j \circ \epsilon_{t-j}$ with all thinnings independent. As $P(\alpha^t \circ y_0 \geq 1) = E\left[1-(1-\alpha^t)^{y_0}\right] \to 0$ by dominated convergence, the law of y_t converges to that of $\sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$. If the solution is stationary its law does not change with t, so it equals this limit.

Step 3: conditional moments. Given $\mathcal{F}_{t-1}$, $\alpha \circ y_{t-1}$ is Binomial(y_{t-1}, α) and independent of ϵ_t, so Proposition (1) gives (2); in particular

$$E(y_t \mid y_{t-1}) = \alpha y_{t-1} + \mu. \tag{8.2}$$

Step 4: moments. The representation of Step 1 has finite variance, since $\operatorname{Var}(\alpha^j \circ \epsilon) = \alpha^j(1-\alpha^j)\mu + \alpha^{2j}\sigma_\epsilon^2$ is summable. Taking expectations in (8.2), $m = Ey_t$ solves $m = \alpha m + \mu$. By the variance decomposition and (2),

$$v = \operatorname{Var} y_t = E(\alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2) + \operatorname{Var}(\alpha y_{t-1} + \mu) = \alpha(1-\alpha)m + \sigma_\epsilon^2 + \alpha^2 v,$$

and $(1-\alpha)m = \mu$ gives $v(1-\alpha^2) = \alpha\mu + \sigma_\epsilon^2$.

Step 5: ACF. For $h \geq 1$, y_t is $\mathcal{F}_{t+h-1}$-measurable, so $\gamma(h) = \operatorname{Cov}(E(y_{t+h} \mid \mathcal{F}_{t+h-1}), y_t) = \operatorname{Cov}(\alpha y_{t+h-1} + \mu, y_t) = \alpha\gamma(h-1)$, whence $\gamma(h) = \alpha^h\gamma(0)$; and $\gamma(-h) = \gamma(h)$. ■

Example 8.1.3 (Poisson innovations)

Let $\epsilon_t \sim \text{Poisson}(\mu)$. By Proposition (4), $\alpha^j \circ \epsilon_{t-j} \sim \text{Poisson}(\alpha^j\mu)$, and a sum of independent Poisson variables is Poisson, so the stationary law is

$$y_t \sim \text{Poisson}\Big(\frac{\mu}{1-\alpha}\Big),$$

with mean = variance, as in Proposition (3) with $\sigma_\epsilon^2 = \mu$. Larger α: higher level, longer runs ($\rho(h) = \alpha^h$).

8.1.1 Conditional least squares

By (8.2), $(\hat\alpha, \hat\mu)$ minimizes $\sum_{t=1}^n (y_t - \alpha y_{t-1} - \mu)^2$ given $y_0, \dots, y_n$:

$$\begin{pmatrix} \hat\alpha \\ \hat\mu \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^n y_{t-1}^2 & \sum_{t=1}^n y_{t-1} \\ \sum_{t=1}^n y_{t-1} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{t=1}^n y_{t-1} y_t \\ \sum_{t=1}^n y_t \end{pmatrix}. \tag{8.3}$$

Proposition (Asymptotics of the least squares estimator, [Supplement])

Let $\{y_t\}$ be the stationary INAR(1) series with $\operatorname{Var} y_t > 0$ and $Ey_t^3 < \infty$. Put $\theta = (\alpha, \mu)^T$, $\mathbf{x}_t = (y_{t-1}, 1)^T$, $u_t = y_t - \alpha y_{t-1} - \mu$, $A = E(\mathbf{x}_t \mathbf{x}_t^T)$ and $V = E(u_t^2 \mathbf{x}_t \mathbf{x}_t^T)$. Then $\hat\theta \to \theta$ a.s. and

$$\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} N(0, A^{-1} V A^{-1}).$$

Proof.

Step 1. $\{y_t\}$ is a fixed function of an i.i.d. sequence (Step 1 above), hence stationary and ergodic. A is positive definite because $\det A = \operatorname{Var} y_{t-1} > 0$. Since $y_t = \theta^T \mathbf{x}_t + u_t$, (8.3) gives

$$\hat\theta - \theta = \Big(\frac{1}{n} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t^T \Big)^{-1} \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t u_t.$$

Step 2. By the ergodic theorem $\frac{1}{n}\sum \mathbf{x}_t \mathbf{x}_t^T \to A$ and $\frac{1}{n}\sum \mathbf{x}_t u_t \to E(\mathbf{x}_t u_t) = E(\mathbf{x}_t E(u_t \mid \mathcal{F}_{t-1})) = 0$ a.s., so $\hat\theta \to \theta$ a.s.

Step 3. $\{\mathbf{x}_t u_t\}$ is a stationary ergodic martingale difference with $E\|\mathbf{x}_t u_t\|^2 < \infty$ (because $E(u_t^2 \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha) y_{t-1} + \sigma_\epsilon^2$ and $Ey^3 < \infty$). The martingale CLT gives $n^{-1/2} \sum \mathbf{x}_t u_t \xrightarrow{d} N(0, V)$, and Slutsky's lemma completes the proof. ∎

8.2 The INGARCH(1,1) model

Motivation. INGARCH makes the count Poisson given the past with a GARCH-type intensity, which is also the conditional variance.

Let the conditional mean follow

$$\mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}, \qquad t = 1, \dots, n, \tag{8.4}$$

with $\omega > 0$, $\alpha, \beta > 0$, $\alpha + \beta < 1$, arbitrary initial $y_0 \in \{0, 1, \dots\}$, $\mu_0 > 0$, and

$$P(y_t = k \mid \mathcal{F}_{t-1}) = \frac{e^{-\mu_t} \mu_t^k}{k!}, \qquad k = 0, 1, \dots \tag{8.5}$$

Definition 8.2.1 (INGARCH(1,1) model)

The *INGARCH(1,1) model* is

$$y_t \mid \mathcal{F}_{t-1} \sim \text{Poisson}(\mu_t), \qquad \mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}. \tag{8.6}$$

The conditional variance is also μ_t.

$(1 - \beta B)\mu_t = \omega + \alpha y_{t-1}$, so

$$\mu_t = \frac{\omega}{1-\beta} + \alpha \sum_{j=1}^\infty \beta^{j-1} y_{t-j}, \tag{8.7}$$

a function of (ω, α, β) and past counts; cf. GARCH(1,1) with $\sigma_t^2 \leftrightarrow \mu_t$, $y_{t-1}^2 \leftrightarrow y_{t-1}$.

Proposition (Moments of INGARCH(1,1), [Supplement])

Let $\{y_t\}$ be a second-order stationary solution of (8.6) (Ferland et al., 2006, show that one exists when $\alpha+\beta<1$). Let $m=Ey_t$. Then
(1) $m=E\mu_t=\dfrac{\omega}{1-\alpha-\beta}$;
(2) $\operatorname{Var}\mu_t=\dfrac{\alpha^2 m}{1-(\alpha+\beta)^2}$ and $\operatorname{Var} y_t=m\dfrac{1-(\alpha+\beta)^2+\alpha^2}{1-(\alpha+\beta)^2}\geq m$, so the counts are overdispersed;
(3) for $h\geq 1$, $\operatorname{Cov}(y_t,y_{t-h})=(\alpha+\beta)^{h-1}(\alpha\operatorname{Var} y_t+\beta\operatorname{Var}\mu_t)$: the ACF decays geometrically at rate $\alpha+\beta$, like that of an ARMA(1,1).

Proof. Let $\nu_t=y_t-\mu_t$. Since μ_t is $\mathcal{F}_{t-1}$-measurable, $E(\nu_t\mid\mathcal{F}_{t-1})=0$ and $E(\nu_t^2\mid\mathcal{F}_{t-1})=\mu_t$; hence ν_t is uncorrelated with every square-integrable $\mathcal{F}_{t-1}$-measurable variable, and $E\nu_t^2=m$.

Step 1. Taking expectations in (8.4), $m=\omega+(\alpha+\beta)m$ (note $E\mu_t=Ey_t$).

Step 2. Substituting $y_{t-1}=\mu_{t-1}+\nu_{t-1}$, $\mu_t=\omega+(\alpha+\beta)\mu_{t-1}+\alpha\nu_{t-1}$, where μ_{t-1} and ν_{t-1} are uncorrelated. Hence $\operatorname{Var}\mu_t=(\alpha+\beta)^2\operatorname{Var}\mu_{t-1}+\alpha^2 m$, which by stationarity gives the formula for $\operatorname{Var}\mu_t$. Then $\operatorname{Var} y_t=\operatorname{Var}\mu_t+E\nu_t^2=\operatorname{Var}\mu_t+m$.

Step 3. For $h\geq 1$, y_{t-h} is $\mathcal{F}_{t-1}$-measurable, so $\operatorname{Cov}(y_t,y_{t-h})=\operatorname{Cov}(\mu_t,y_{t-h})$. For $h=1$: $\operatorname{Cov}(\omega+\alpha y_{t-1}+\beta\mu_{t-1},y_{t-1})=\alpha\operatorname{Var} y_{t-1}+\beta\operatorname{Cov}(\mu_{t-1},\mu_{t-1}+\nu_{t-1})=\alpha\operatorname{Var} y+\beta\operatorname{Var}\mu$. For $h\geq 2$: $\operatorname{Cov}(\mu_t,y_{t-h})=\operatorname{Cov}\big((\alpha+\beta)\mu_{t-1}+\alpha\nu_{t-1},y_{t-h}\big)=(\alpha+\beta)\operatorname{Cov}(\mu_{t-1},y_{t-h})=(\alpha+\beta)\operatorname{Cov}(y_{t-1},y_{t-h})$, using that ν_{t-1} is uncorrelated with y_{t-h}. Induction on h finishes the proof. ■

The conditional MLE of (ω,α,β) maximizes (log-likelihood up to a constant)

$$L(\omega,\alpha,\beta)=\prod_{t=1}^{n}\frac{e^{-\mu_t}\mu_t^{y_t}}{y_t!},\qquad l(\omega,\alpha,\beta)=\sum_{t=1}^{n}(y_t\log\mu_t-\mu_t),$$

with μ_t from (8.4); it is consistent and asymptotically normal (Lee and Lee, 2019). Negative binomial and one-parameter exponential families are also used as conditional laws.

8.3 Change-point tests

Motivation. A CUSUM test on the residuals, a martingale difference under the null, detects parameter changes.

Residuals (CUSUM of Section 6.5): for INAR(1),

$$\hat\epsilon_t=y_t-\hat\alpha y_{t-1}-\hat\mu,\tag{8.8}$$

and for INGARCH(1,1),

$$\hat\epsilon_t=y_t-\hat\mu_t,\qquad \hat\mu_t=\hat\omega+\hat\alpha y_{t-1}+\hat\beta\hat\mu_{t-1},\tag{8.9}$$

with arbitrary $y_0,\hat\mu_0$. The CUSUM statistic is

$$\hat T_n=\max_{1\leq k\leq n}\frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{t=1}^{k}\hat\epsilon_t-\frac{k}{n}\sum_{t=1}^{n}\hat\epsilon_t\Big|,\qquad \hat\tau^2=\frac{1}{n}\sum_{t=1}^{n}\hat\epsilon_t^2-\Big(\frac{1}{n}\sum_{t=1}^{n}\hat\epsilon_t\Big)^2.\tag{8.10}$$

Reject at 5% if $\hat T_n>1.358$; the maximizing k estimates the change point.

Proposition (Null limit of the residual CUSUM test, [Supplement])

Let $\{y_t\}$ be a stationary INAR(1) series satisfying the assumptions of Proposition , with no parameter change, and let $\hat{\epsilon}_t$ be the residuals (8.8) from the least squares estimator. Then $\hat{T}_n \xrightarrow{d} \sup_{0\le u\le 1}|B^0(u)|$, the Brownian-bridge limit of Proposition .

Proof.

Step 1: true innovations. The true residuals $u_t = y_t - \alpha y_{t-1} - \mu = (\alpha\circ y_{t-1} - \alpha y_{t-1}) + (\epsilon_t - \mu)$ form a stationary ergodic martingale difference sequence with variance $\tau^2 = Eu_t^2 = \alpha(1-\alpha)Ey_t + \sigma_\epsilon^2 > 0$. By the functional CLT for such martingale differences (Billingsley), $W_n(s) = \frac{1}{\tau\sqrt{n}}\sum_{t\le\lfloor ns\rfloor} u_t \xrightarrow{d} W$. Exactly as in the proof of Proposition (Steps 2–3), $\max_k \frac{1}{\tau\sqrt{n}}\left|\sum_{t\le k} u_t - \frac{k}{n}\sum_{t\le n} u_t\right| \xrightarrow{d} \sup_s |B^0(s)|$.

Step 2: estimation effect. $\hat{\epsilon}_t - u_t = (\alpha-\hat{\alpha})y_{t-1} + (\mu-\hat{\mu})$. Constants cancel in $\sum_{t\le k} c - \frac{k}{n}\sum_{t\le n} c = 0$, so

$$\frac{1}{\sqrt{n}}\Big|\sum_{t\le k}(\hat{\epsilon}_t - u_t) - \frac{k}{n}\sum_{t\le n}(\hat{\epsilon}_t - u_t)\Big| = \sqrt{n}|\hat{\alpha}-\alpha|\cdot\frac{1}{n}\Big|\sum_{t\le k}(y_{t-1}-m) - \frac{k}{n}\sum_{t\le n}(y_{t-1}-m)\Big|,$$

with $m = Ey_t$. Here $\sqrt{n}(\hat{\alpha}-\alpha) = O_p(1)$ by Proposition , and $\max_{k\le n}\frac{1}{n}|\sum_{t\le k}(y_{t-1}-m)| \to 0$ a.s. because $\frac{1}{k}\sum_{t\le k}(y_{t-1}-m)\to 0$ a.s. (ergodic theorem). So the maximum over k of this difference is $o_p(1)$.

Step 3: variance estimate. Similarly $\hat{\epsilon}_t = u_t + O(|\hat{\alpha}-\alpha|y_{t-1} + |\hat{\mu}-\mu|)$ and the ergodic theorem give $\hat{\tau}^2 \to \tau^2$ in probability. Slutsky's lemma combines Steps 1–3. ■

For INGARCH(1,1), $y_t - \mu_t$ are again martingale differences and the same limit holds for (8.9) (Lee and Lee, 2019), so 1.358 is used in both models.

8.4 Selected exercises

Motivation. For INGARCH(1,1): μ_t is a geometric sum of past counts, and y_t is ARMA(1,1) with martingale-difference noise.

Exercise 8.3 (The intensity as a weighted sum of past counts)

Let $\{y_t\}_{t\in\mathbb{Z}}$ be a second-order stationary solution of (8.6) with $\omega > 0$, $\alpha, \beta > 0$, $\alpha + \beta < 1$. Prove (8.7), the series converging a.s. and in mean.

Solution.

Step 1: finite iteration. For every $K \ge 1$,

$$\mu_t = \omega\sum_{j=0}^{K-1}\beta^j + \alpha\sum_{j=1}^{K}\beta^{j-1}y_{t-j} + \beta^K\mu_{t-K}.$$

For $K = 1$ this is (8.4). If it holds for K, inserting $\mu_{t-K} = \omega + \alpha y_{t-K-1} + \beta\mu_{t-K-1}$ into the last term gives $\beta^K\mu_{t-K} = \omega\beta^K + \alpha\beta^K y_{t-K-1} + \beta^{K+1}\mu_{t-K-1}$, which is the identity for $K+1$.

Step 2: the remainder. $\mu_t > 0$ and, by Proposition (1), $E\mu_t = m < \infty$ for all t. Hence $E|\beta^K\mu_{t-K}| = \beta^K m \to 0$ as $K\to\infty$, because $0 < \beta < 1$.

Step 3: the series. The terms $\beta^{j-1}y_{t-j}$ are nonnegative and $\sum_{j\ge1}\beta^{j-1}Ey_{t-j} = m/(1-\beta) < \infty$. By monotone convergence $S = \sum_{j\ge1}\beta^{j-1}y_{t-j}$ has finite mean, so it is finite a.s., the partial sums S_K converge to S a.s., and $E|S - S_K| = \sum_{j>K}\beta^{j-1}m \to 0$.

Step 4: conclusion. By Step 1, $D_K = \mu_t - \omega(1-\beta^K)/(1-\beta) - \alpha S_K = \beta^K \mu_{t-K}$. By Step 3, $D_K \to \mu_t - \omega/(1-\beta) - \alpha S$ a.s., while by Step 2 $D_K \to 0$ in mean, hence a.s. along a subsequence. The two limits agree, so $\mu_t = \omega/(1-\beta) + \alpha S$ a.s., which is (8.7).

Step 5: remark on initial values. If the recursion is started at $t = 1$ from arbitrary y_0, μ_0, Step 1 with $K = t$ gives

$$\mu_t = \omega \frac{1-\beta^t}{1-\beta} + \alpha \sum_{j=1}^{t} \beta^{j-1} y_{t-j} + \beta^t \mu_0,$$

so the effect of μ_0 decays like β^t and initial values do not matter asymptotically. ■

Exercise 8.4 (INGARCH(1,1) as an ARMA(1,1) model)

Under the assumptions of Exercise 8.3, show that $\{y_t\}$ follows an ARMA(1,1) model. (Hint: consider $\nu_t = y_t - \mu_t$.)

Solution.

Step 1: the noise. μ_t is $\mathcal{F}_{t-1}$-measurable and $y_t \mid \mathcal{F}_{t-1} \sim \text{Poisson}(\mu_t)$, so $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E(\nu_t^2 \mid \mathcal{F}_{t-1}) = \mu_t$. Hence $E\nu_t = 0$ and $E\nu_t^2 = E\mu_t = m = \omega/(1-\alpha-\beta)$, the same for all t. For $s < t$, ν_s is $\mathcal{F}_{t-1}$-measurable and $\nu_s\nu_t$ is integrable (Cauchy–Schwarz), so $E\nu_s\nu_t = E\big(\nu_s E(\nu_t \mid \mathcal{F}_{t-1})\big) = 0$. Thus $\{\nu_t\}$ is white noise with variance m (a martingale difference, not i.i.d.).

Step 2: the ARMA equation. Substituting $\mu_{t-1} = y_{t-1} - \nu_{t-1}$ in $y_t = \mu_t + \nu_t = \omega + \alpha y_{t-1} + \beta\mu_{t-1} + \nu_t$,

$$y_t = \omega + (\alpha+\beta) y_{t-1} + \nu_t - \beta\nu_{t-1}, \qquad \text{i.e.} \qquad (1-(\alpha+\beta)B)(y_t - m) = (1-\beta B)\nu_t,$$

using $\omega = (1-\alpha-\beta)m$. This is an ARMA(1,1) model with AR coefficient $\phi = \alpha+\beta \in (0,1)$ and MA coefficient $\theta = -\beta \in (-1,0)$; it is causal and invertible.

Step 3: causality and a check. Iterating Step 2, $y_t - m = \sum_{j=0}^{K-1} \phi^j(\nu_{t-j} - \beta\nu_{t-j-1}) + \phi^K(y_{t-K} - m)$, and the remainder has second moment $\phi^{2K} \operatorname{Var} y_t \to 0$. Hence $y_t - m$ is a mean-square limit of combinations of ν_s, $s \le t$, and is uncorrelated with $\nu_{t+1}, \nu_{t+2}, \ldots$. These are the only facts used in Example 2.4.3, which therefore gives

$$\operatorname{Var} y_t = \frac{m(1+2\phi\theta+\theta^2)}{1-\phi^2} = m\,\frac{1-2\alpha\beta-\beta^2}{1-(\alpha+\beta)^2} = m\,\frac{1-(\alpha+\beta)^2+\alpha^2}{1-(\alpha+\beta)^2},$$

in agreement with Proposition (2), and $\gamma(h) = (\alpha+\beta)^{h-1}\gamma(1)$ as in Proposition (3). ■